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CONCISE CHEMISTRY

– Middle School

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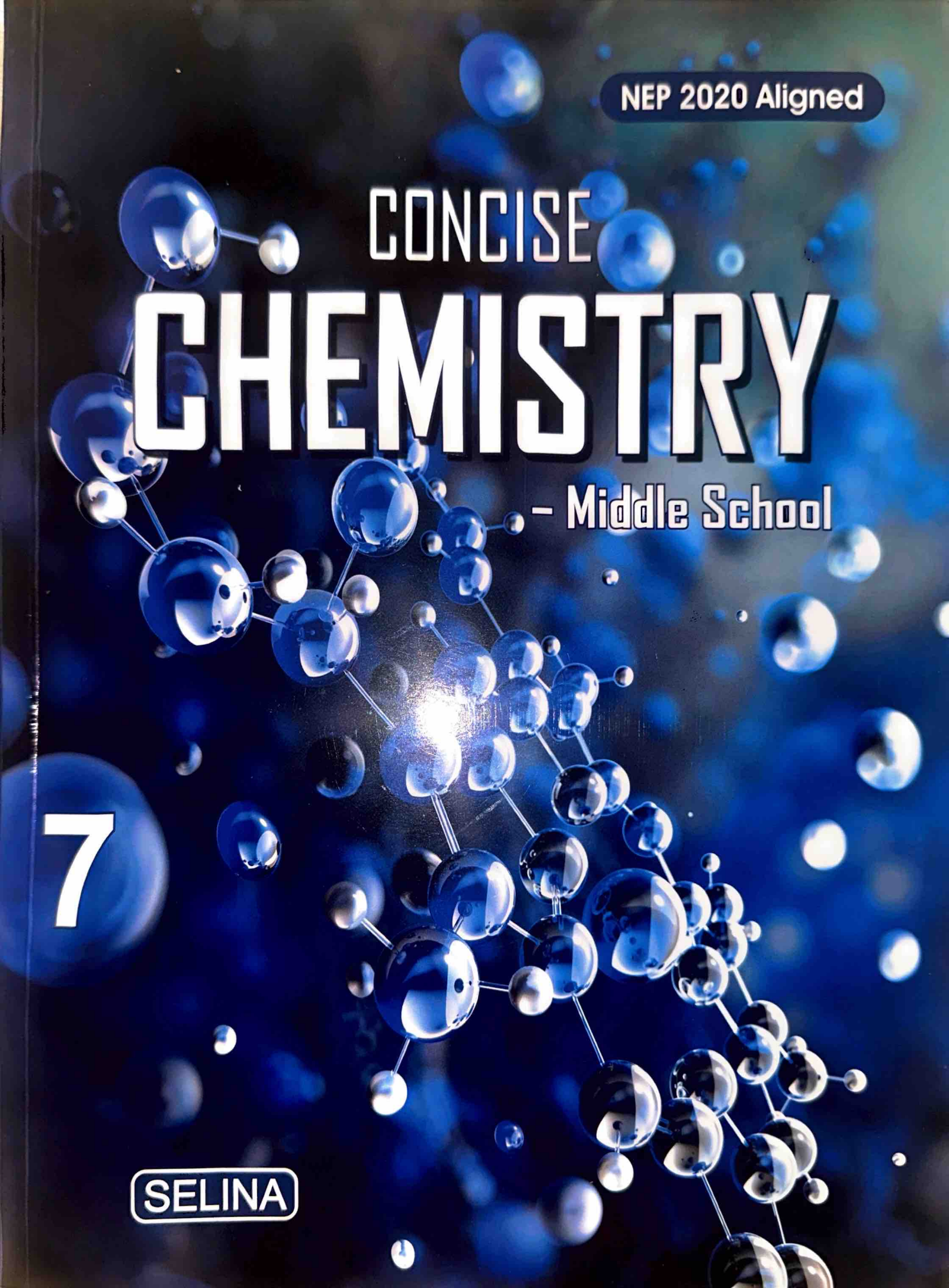

SELINA

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CONCISE CHEMISTRY

– Middle School

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Preface

"*Concise Chemistry Middle School*" meant for classes VI, VII, and VIII, has been carefully designed to cater to the needs of young learners studying under the curriculum prescribed by the Council for the ISC Examinations, New Delhi. This series adheres to the latest guidelines proposed in the National Education Policy (NEP), 2020.

The primary objective of this series is to make the fundamental concepts of Chemistry clear, engaging, and accessible for students. Each book in the series has been structured thoughtfully, employing simple language and illustrations to meet the comprehension level of the students. By using experiments, activities, and diagrams, we aim to foster curiosity, a scientific temper, and an appreciation for chemical processes. To enhance learning and to facilitate quick revision, every chapter includes concise summaries under "**Recapitulation**". **Detailed exercises** are provided at the end of each chapter to test understanding and promote self-assessment. In addition, **Assertion-Reason type and Case Study based questions** have been included to help students connect concepts logically and develop critical thinking. Readers may also visit our website to explore the e-book editions of this series, available for purchase.

We hope this series serves as a valuable resource for both teachers and students in exploring the fascinating world of chemistry. Despite our meticulous efforts, we acknowledge the possibility of overlooked errors or areas needing improvement. We sincerely welcome constructive feedback and suggestions to further refine and enhance this series.

Author

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Key Information on Assertion and Reason Questions

To help students develop critical thinking skills, CISCE has introduced **Assertion-Reason based questions** in class 10 board exams. We are introducing this format of questions in our Middle School series to give students a head start and help them crack these questions from an early age.

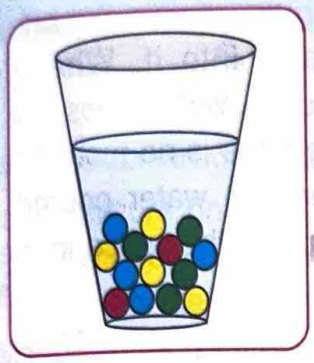
In this question format, two statements are given :

1. **Assertion (A) :** An assertion is a statement of fact, observation or conclusion, which is either true or false.
2. **Reason (R) :** A reason is a statement that explains, defines, justifies or sometimes contradicts the assertion. It can also be either true or false.

To answer these questions, students must consider both statements together, evaluate the truth of each, and basically decide whether the reason correctly explains or supports the assertion.

Matter and Its Composition

1



THEME

This theme focuses on informing and making children aware of the different types of matter/objects found in their surroundings such as stones, water, soil, oil, sugar, air. Some of them have common characteristics in terms of states, some are solids, liquids and some are gases. These states vary in their shape, volume and texture. All these are made up of some materials which have mass and occupy space. Children will also realize that the study of their composition is of great importance in their daily lives.

LEARNING OBJECTIVES

The children will be able to :

- describe matter;
- discuss the constituents (atoms/molecules) of matter;
- explain the forces which keep atoms/molecules in matter together.

INTRODUCTION

There are different types of substances in our surroundings such as water, soil, stones, minerals, sun, moon, stars, plants, animals, etc. We also use innumerable objects in our daily lives like books, pens, bags, shoes, clothes, cars, etc. Some of them are living, like plants and animals, while some are non-living, like water, soil, etc. Some are natural like minerals, air, water and living beings, while some are man-made like books, pencils, papers, houses, etc. All these substances are made up of **matter**.

In fact the food we eat, the water we drink, the air we breathe, the clothes we wear, the houses we live in are all made up of matter. But what is matter ?

Matter is anything that **has mass, occupies space** and can be **perceived by our senses.**

That means matter is not only that, which we can see or touch, but also which we can feel, such as air. Although we cannot see or touch air but we can feel it, especially when the wind blows.

Does matter of only one kind exist ?

No, there are different kinds of matter. Water, sugar, alcohol, milk, gold, coal, hydrogen, oxygen, rocks — all have different kinds of matter in them. A single material* can be used to make different objects, such as wood is used to make chairs, tables, handles, cupboards, etc. Thus, objects made of matter can be of different shapes and sizes.

** The word 'material' is derived from 'matter'.*

Name *two* objects made up of each of the following materials :

1. Metal 2. Wool 3. Cloth
4. Plastic 5. Leather 6. Glass

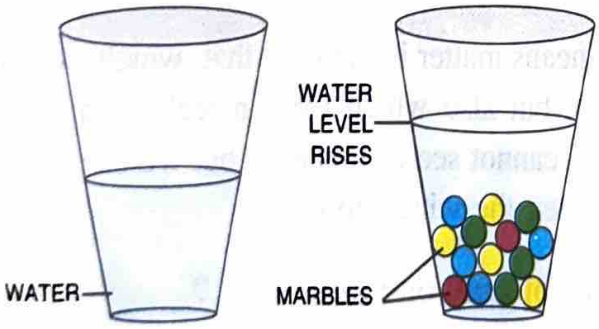
- The quantity of matter that a body contains is known as its **mass**.
- The **space** occupied by a body is called its **volume**.

Matter occupies both mass and space can be proved by the following activities :

Activity 1

Take a glass tumbler. Fill one third of it with water. Put a mark on the glass to measure the level of water. Now weigh this on a balance.

Take some marbles and put them into the glass tumbler one by one. You will notice that the water level crosses the mark and rises. This is because the **marbles occupy space**. Again weigh the glass with the marbles.



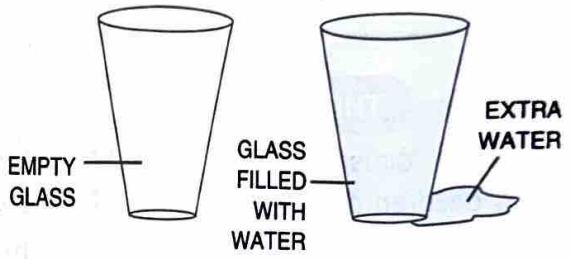
You will find that the second mass is greater than the first one. This proves that **marbles have mass**.

The activity proves that matter has **mass** and occupies **space**.

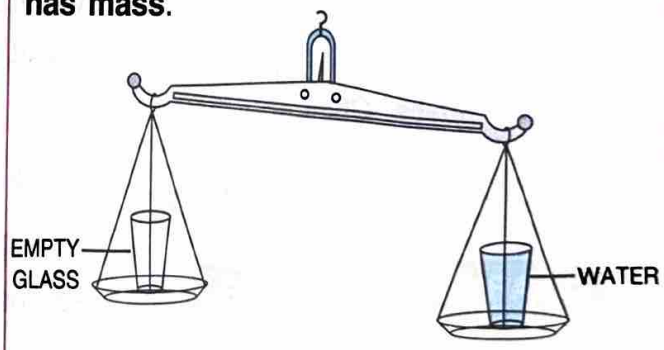
If we hold objects in our hands, then we can feel their mass. Hold a book and you will feel its mass.

Activity 2

Take two identical glass tumblers. Fill one of them with water upto the brim. Pour some more water into it. What do you observe ? The extra water flows out of the tumbler because there is no more space left in it. This is because water poured earlier has already occupied the space in the glass tumbler.



Now place the empty glass tumbler and the one filled with water on the two pans of a weighing balance. The pan with the glass of water tilts to one side. This shows that **water has mass**.



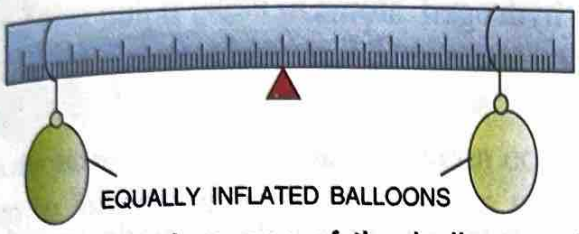
Balance tilted on one side

Activity 3

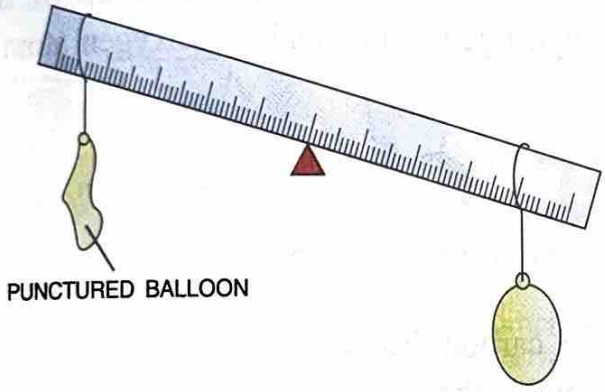
Take two similar balloons. Blow air in them. What do you observe ?

You will observe that the shape of balloons changes, they become inflated because air inside the balloons **occupies space**. Inflate them equally. Suspend one balloon to the left of a metre scale and the other one to the right of it, as shown in the

figure below. Balance the scale in the middle with the help of a peg.



Now puncture one of the balloons with the prick of a needle. You will observe that air escapes from the balloon and the metre scale tilts towards the inflated balloon. This is because of the **mass of air** present in the inflated balloon.

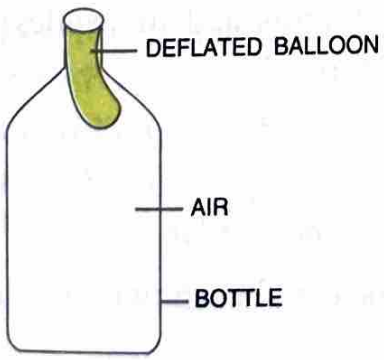


Air occupies space and has mass

This proves that air, which is a kind of matter, **occupies space and has mass**.

Activity 4

Take a plastic bottle, push a deflated balloon into it and stretch the open end of the balloon over the bottle's mouth.



Now, blow and try to inflate the balloon.

What do you observe ? Does the balloon get inflated ?

No, because the space inside the bottle is occupied by air and there is no space left for the balloon to be inflated.

This proves that although air is invisible, yet it occupies space.

Do You Know ?

Mass and weight are different from each other. Mass is the "quantity of matter" and weight is "the force with which the earth pulls a body towards itself". The mass of a body does not change but its weight changes from place to place.

The weight of a body is 1/6 times lighter on the moon than it is on earth, but its mass remains the same.

COMPOSITION OF MATTER

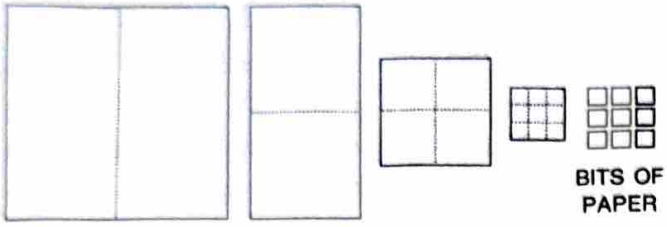
Now you know that all substances are made up of matter, but what is matter made up of ?

Matter is made up of extremely small particles called **atoms**.

To understand this, let us do the following activity.

Activity 5

Take a sheet of paper. Tear it into two half sheets. Fold each half sheet and tear it again. Repeat this until a very small piece of paper is obtained. Now try to get even smaller bits of paper from that small piece. Each time you will notice that there is no change in the appearance and properties of paper. By



looking at the bits of paper you still can clearly understand that it is paper.

Hence we can conclude that paper, a kind of matter, is made up of tiny particles which exhibit all the properties of paper. These particles are atoms.

Note : Earlier it was thought that all kinds of matter are made up of air, water, earth, sky and fire. Later on "Kannada" an Indian philosopher and then "John Dalton", an English scientist, proposed that all kinds of matter are made up of atoms.

"An atom is the smallest possible unit of matter that exhibits all the properties of that matter".

Atoms usually do not have independent existence, therefore they combine with one another to form another minute particle called **molecule**.

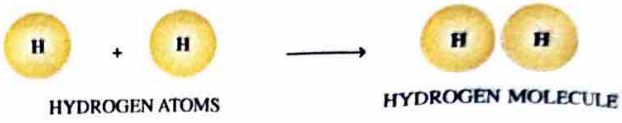
"A molecule is the smallest unit of matter which exhibits all the properties of that kind of matter and also has an independent existence".

Molecules are bigger than atoms. However both the particles are too small to be seen through the naked eye or an ordinary microscope.

A molecule can be formed of atoms of the same kind or of different kinds.

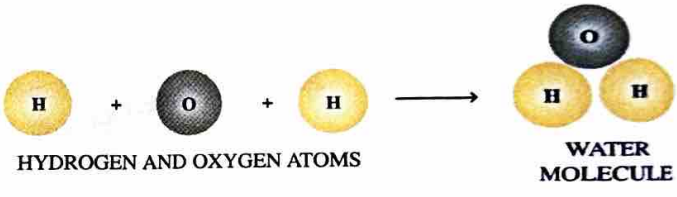
For example :

- A hydrogen molecule is made up of two hydrogen atoms.

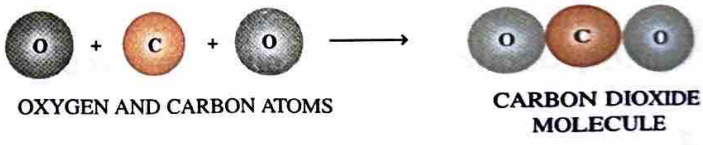


- A nitrogen molecule is made up of two nitrogen atoms.

- A water molecule is made up of two hydrogen atoms and one oxygen atom.



- A carbon dioxide molecule is made up of one carbon atom and two oxygen atoms.



CHARACTERISTICS OF PARTICLES OF MATTER

Properties of matter can be explained on the basis of arrangement and characteristics of particles (atoms and molecules) from which they are formed.

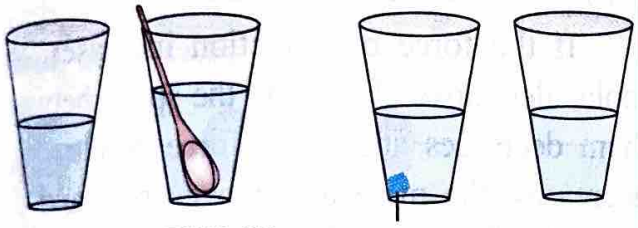
1. *Particles of matter have space between them. This space is called interparticular or intermolecular space.*

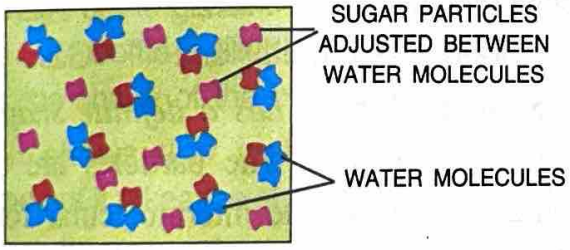
This can be shown by the following activity.

Activity 6

Take half a glass of water. Dip a spoon in it. What do you observe ? The water level rises, indicating that spoon occupies space.

Now remove the spoon, water comes down to its original level. Now add a spoon of sugar to it and stir well. The sugar disappears but the level of water in the glass does not rise, that means the volume of water has not increased. But where did the sugar particles disappear ?





To show intermolecular space between water molecules

Upon the dissolution of sugar in water, the sugar particles being smaller get adjusted between the water molecules. This shows that there are intermolecular spaces in water.

Why does the water level rise on dipping a spoon ? This is because a spoon is a bigger object that cannot be dissolved and adjusted between the intermolecular spaces of water.

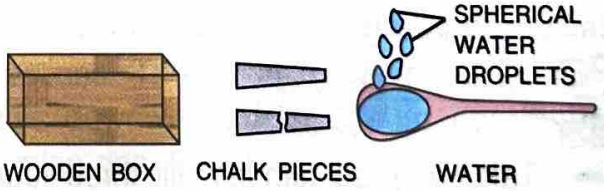
2. *Particles of matter attract each other*

There exists a force of attraction between the particles or molecules of matter, which holds them together. This is known as **intermolecular force of attraction.**

This can be shown by the following activity.

Activity 7

- Take a **wooden block.** Try to break it by applying a force on it. The block does not break. The block is made up of particles which do not separate easily.
- Take a **chalk piece**, apply some force on it and it breaks up into two pieces.
- Why does a chalk break into pieces by applying some force by hand but not a wooden block ? This is because the intermolecular force is weaker in chalk than in the wooden block, as chalk is a soft solid while wood is hard.
- Take some water in a table spoon and throw it up. What do you observe ? Falling droplets of water are spherical in shape. Why are they spherical ? Because water molecules have a tendency to stick together.



To show force of attraction between molecules

All these show that there is intermolecular force of attraction between the particles of wooden block, particles of chalk piece and particles of water.

Hence, it is proved that intermolecular attraction between the particles keeps them together.

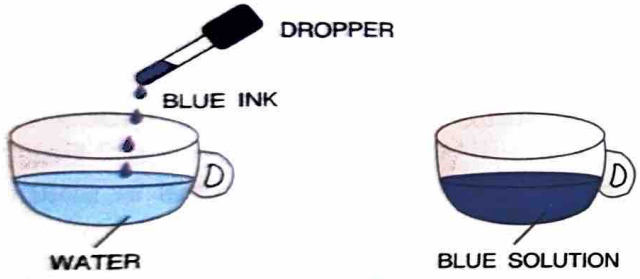
3. *Particles of matter are always in random motion :*

By now you know that matter is made up

of minute particles which have gaps between them. But do these particles stand still in the bulk or do they keep moving ? This can be better understood by the following activity.

Activity 8

Take a cup of water. Add few drops of blue ink into it. What do you observe? You will observe that the ink slowly mixes uniformly in water making the whole solution blue. This is because the water as well as ink particles (molecules) are in continuous **random motion**. Due to motion, the blue coloured particles of the ink spread all over and give blue colour to the water.

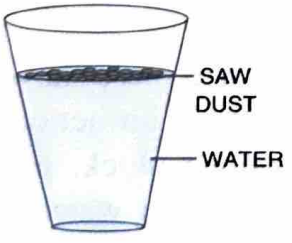


Water in the cup turns into a blue solution due to ink

Activity 9

Take a glass tumbler, fill three fourth of it with water and sprinkle some saw dust on it. Observe it carefully using a magnifying glass.

You will notice that particles of saw dust move randomly on the surface of water.



Randomly moving particles of saw dust

Why do these particles move ? This is because they possess a kind of energy called **kinetic energy**. This energy increases with an increase in temperature and decreases with a decrease in temperature. Hence, the movement of particles increases or decreases with the increase or decrease in temperature.

Kinetic energy is the energy that an object has due to its motion.

STATES OF MATTER

Now you know that intermolecular spaces, intermolecular attraction and random motion are three important characteristics of particles of any kind of matter.

If the force of attraction increases, the molecules move closer and the space between them decreases and if the force of attraction decreases, the molecules move apart and the space between them increases.

This fact accounts for the three different states of matter. Those are

• Solid • Liquid and • Gas

Solid : "*A solid has a definite shape and a definite volume*." The particles are tightly packed in solids. The intermolecular force of attraction is very strong and intermolecular space is almost negligible. The particles are not free to move. They can only vibrate about

their mean positions. Solids are rigid and incompressible. *Examples :* Iron, stone, sand, wood, coal, etc.

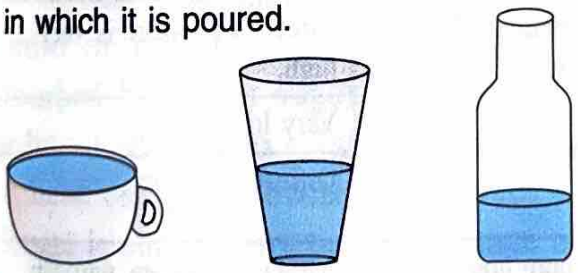
Liquid : *"A liquid has a definite volume but no definite shape."*

The particles in a liquid are less tightly packed in comparison to solids. In liquids, the intermolecular force of attraction is weaker and intermolecular gaps are larger. Thus the particles are free to move within the bulk. Liquids can flow and are therefore also called **fluids**. *Examples :* Water, milk, oil, alcohol, petrol, etc.

Activity 10

Take a cup filled with water. Water acquires the shape of the cup. Pour this water into a glass. It fills half of the glass and takes the shape of the glass. Pour this water into a bottle. The bottle is filled one third and water takes the shape of the bottle.

It shows that water has the same volume but it takes the shape of the container in which it is poured.



Water in three different containers

Gas : *"A gas has neither a definite shape nor a definite volume."*

The particles in a gas are far apart. The intermolecular force of attraction between the particles is very weak and the space between them is large. The particles are free to move in any direction *i.e.* they can flow.

They can be compressed inside a small container and can spread into a large space available to them. *Examples :* Air, hydrogen, oxygen, nitrogen, carbon dioxide, etc.

Gases also take the shape of the container in which they are kept.

Examples : 1. Air when pumped into a football, takes the shape of the football.

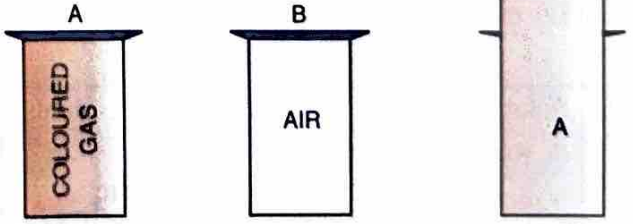
2. The fragrance of a perfume spreads into the whole room because the gaseous particles move rapidly and occupy the whole space.

Activity 11

Take two jars. Fill one of the jars with reddish brown bromine vapours and the other contains air. Invert the empty jar over the jar containing the bromine vapours and remove the lid. You will observe that the coloured gas (bromine) also spreads in the upper jar because it gets more space.

It is to be noted that gases flow in all directions. The air in the inverted jar also flows downwards into the jar containing bromine gas. But since air is colourless, we are unable to see its movement.

This shows that gases have neither a fixed shape nor a fixed volume. They have no free surfaces.



A gas jar containing coloured gas

The coloured gas spreads in both the jars

All substances that can flow are called **fluids**. Liquids and gases are fluids.

(a) Molecules lie closely packed together, with great force of attraction between them

(b) Molecules lie fairly apart from each other, with little force of attraction between them

(c) Molecules lie very far from each other with very little force of attraction between them

Fig. 1.1 : The position of molecules in the (a) solid, (b) liquid and (c) gaseous states.

The phenomenon of *intermixing* of one kind of particles with another kind is called *diffusion*.

INTERCONVERSION OF STATES OF MATTER

In everyday life, we come across substances that change from one state to another when conditions are changed.

Interconversion of states of matter is the process by which matter changes from one state to another and back to its original state without any change in its chemical composition, when conditions are changed.

Table 1.1 Properties of solids, liquids and gases.

S.No.	Property	Solids	Liquids	Gases
1.	Intermolecular space	Molecules are closely packed; have negligible intermolecular space.	Molecules are not closely packed; have more inter-molecular space.	Molecules are at a greater distance from one another; have the maximum inter-molecular space.
2.	Shape	Have fixed shape	Have no fixed shape. Take the shape of the container in which they are kept.	Have no fixed shape. Take the shape of the container in which they are filled.
3.	Volume	Have fixed volume.	Have fixed volume.	Have no fixed volume.
4.	Fluidity	Do not flow.	Flow from a higher level to a lower level.	Flow in all directions.
5.	Effect of pressure *i.e.* compression	Effect of pressure is very low.	Effect of pressure is higher than on a solid.	Effect of pressure is very high.
6.	Density	High	Low	Very low
7.	Diffusion	Do not diffuse under normal conditions	Diffuse slowly	Diffuse very fast
8.	Transparency	Solids are generally opaque *e.g.* wood, stones, *etc.* (glass and diamond are transparent)	Most of the pure liquids are transparent in nature *e.g.* water, alcohol, petrol.	All gases, oxygen, nitrogen carbon dioxide, *etc.* are transparent in nature.
9.	Lustre	Amongst the solids, all the metals have lustre.	Amongst the liquids, mercury is lustrous.	Gases are not lustrous.
10.	Solubility in water	Some solids are insoluble in water. *e.g.* wood, mud, etc. Other solids such as sugar, common salt are soluble in water.	Liquids can be miscible as well as immiscible in water. *e.g.* alcohol and vinegar are soluble (miscible) in water while oil and water, petrol and water are immiscible.	Some gases are insoluble in water. *e.g.* nitrogen, hydrogen. Some gases are soluble in water. *e.g.* oxygen, carbon dioxide, ammonia, etc.

The change in the state of matter is mainly caused by

- **change in temperature**
- **by applying pressure.**

(1) Change of state by changing the temperature :

When temperature increases, a solid changes into liquid and then into vapour and when temperature decreases just the reverse happens.

For example, water is a liquid under ordinary conditions but when it is kept in a refrigerator, it gets cooled and changes into ice at 0°C. Ice when kept at room temperature, again changes back into liquid water.

Water when heated starts changing into steam above 100°C, but when this steam is cooled, it again changes into its liquid form.

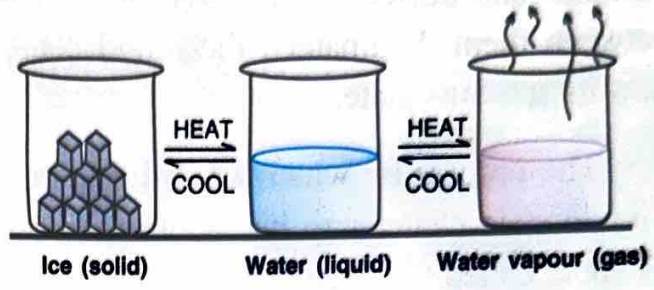
$$\underset{\text{(solid state)}}{\text{Ice}} \underset{\text{cool}}{\overset{\text{heat}}{\rightleftharpoons}} \underset{\text{(liquid state)}}{\text{Water}} \underset{\text{cool}}{\overset{\text{heat}}{\rightleftharpoons}} \underset{\text{(gaseous state)}}{\text{Water vapour}}$$

In all the *three* states, the properties of water remain the same.

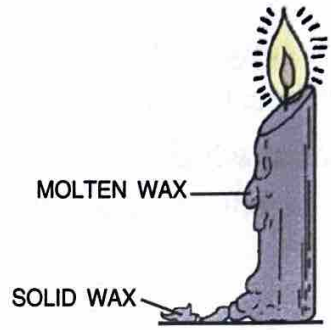
When a substance changes from solid to liquid or liquid to vapour, heat energy is absorbed but when vapour changes to liquid or liquid changes to solid state, same amount of heat energy is evolved. The net energy change is almost zero, because change of state does not change the chemical composition of the substance.

Another example is wax. When you ignite a candle, solid wax melts into liquid wax which rises in the wick. The wax near the glowing wick melts and some part of the molten wax trickles down and solidifies.

$$\text{Solid wax} \underset{\text{cool}}{\overset{\text{heat}}{\rightleftharpoons}} \text{Molten wax}$$



Interconversion of states of water



States of wax

In both the above cases, the change of state of the substance is due to change in temperature.

Effect of increasing temperature on states of matter

When a **solid** is heated, the particles of the solid gain energy and start vibrating more vigorously. A stage is reached where the particles gain enough energy to overcome the force of attraction and they start moving. Thus the solid changes into its **liquid** state.

The process by which a substance in its solid state changes to its liquid state on heating is called **melting** or **fusion**.

The temperature at which a solid starts changing into liquid is called its **melting point**.

As the **liquid** is heated, its particles start gaining energy and move more vigorously which further increases the gaps between the

particles and decreases the force of attraction between them. Ultimately, the liquid changes into its **gaseous** state.

The process by which a substance in its liquid state changes to its vapour (gaseous) state is called **vaporisation** or **evaporation**.

Evaporation can also take place at room temperature but it becomes faster on heating.

When a liquid changes into its gaseous state on heating, it is called **boiling**.

The temperature at which a liquid starts boiling and changes into its gaseous state is called its **boiling point**.

Effect of decreasing temperature on states of matter

When a **gas** is cooled, the energy of the particles decreases and their movement becomes slow. The gaps between the particles decrease, thus the force of attraction between them increases. Ultimately, the gas changes into its **liquid** state.

The process by which a susbtance in its gaseous state changes into its liquid state is called **condensation** or **liquefaction**.

The temperature at which a gas starts changing into its liquid state is called its **condensation point**.

On further cooling a liquid, the particles lose more energy and their movement becomes even slower. The particles come closer, decreasing the intermolecular gaps and increasing the intermolecular force of attraction to an extent that the liquid changes into its solid state.

The process by which a substance in its liquid state changes into its solid state is called **freezing** or **solidification**.

The temperature at which a liquid starts changing into its solid state is called its **freezing point**.

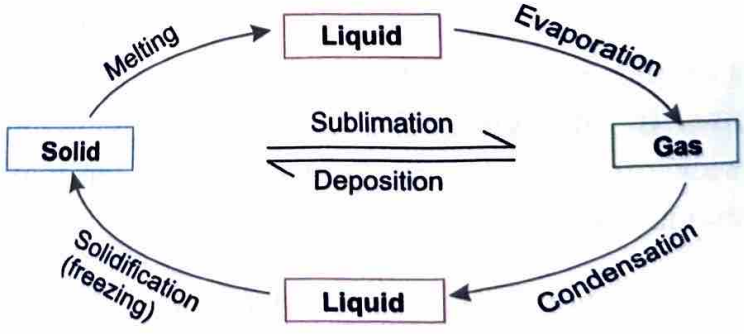
(2) Change of state by changing the pressure :

The state of a substance can also be changed by applying pressure. ***For example,*** gases on increasing pressure change into liquid form. This is how we can obtain liquid oxygen, liquid carbon dioxide, liquid cooking gas, etc.

The cooking gas sold in the gas cylinders is in the form of liquid under high pressure, thus called as Liquified Petroleum Gas (LPG). When the regulator valve is opened, pressure is released and LPG comes out in the form of gas.

Sublimation : There are some substances like camphor, naphthalene, iodine crystals, ammonium chloride, etc. which change directly from solid to gaseous state on heating (this process is called **sublimation**) and vice versa on cooling.

The process by which a substance directly changes from its **solid state** to its **gaseous state** *without passing through liquid state* is called **sublimation**.



RECAPITULATION

- Matter has mass and occupies space.
- Matter is made up of atoms and molecules.
- Atoms are the smallest particles of matter which may or may not have independent existence.
- Molecules are capable of independent existence. They are made up of atoms of same kind or different kinds.
- The atoms and molecules are in random motion.
- There are gaps between the molecules of matter called as intermolecular space.
- There exists a force of attraction between the molecules known as intermolecular force of attraction.
- Matter exists in *three* states : solid, liquid and gas.
- Matter can change from one state to another on changing temperature and pressure.
- The change of state of a matter from one form into another is called interconversion of states of matter.

EXERCISE

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(*i*) The intermolecular forces are weakest in:
(*a*) solids (*b*) liquids
(*c*) gases (*d*) both (a) and (b)

(*ii*) The intermolecular spaces are the least in:
(*a*) aluminium (*b*) water
(*c*) oxygen (*d*) carbon dioxide

(*iii*) The state of matter which has a definite volume but takes the shape of a container is :
(*a*) solids (*b*) liquids
(*c*) gases (*d*) none

(*iv*) Which of the following is transparent ?
(*a*) Diamond (*b*) Graphite
(*c*) Wood (*d*) Coal

(*v*) The process by which a solid changes directly from the solid state to the gaseous state is called :
(*a*) diffusion (*b*) sublimation
(*c*) evaporation (*d*) melting

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

(*i*) **Assertion (A) :** Matter exists in three states, *i.e.* solids, liquids and gases.
Reason (R) : All matter can change from one state to another on changing temperature and pressure.
(*a*) (1) (*b*) (2) (*c*) (3) (*d*) (4)

(*ii*) **Assertion (A) :** When a gas is cooled, the intermolecular gaps decrease.
Reason (R) : The kinetic energy of the gaseous particles decrease upon cooling.
(*a*) (1) (*b*) (2) (*c*) (3) (*d*) (4)

(*iii*) **Assertion (A) :** Most of the liquids have their own shape.

Reason (R) : Liquids can flow from a higher to lower level.

(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(iv) **Assertion (A) :** Particles of matter are always in a random motion.

Reason (R) : Particles of matter possess kinetic energy.

(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(v) **Assertion (A) :** A piece of wood is a hard solid, while that of chalk is a soft solid.

Reason (R) : In solids, the intermolecular forces of attraction are very strong.

(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

3. Fill in the blanks :

(a) Air is a matter because it has ..... and ..... and it can be ..... .

(b) The molecules are made up of .....

(c) The quantity of matter in an object is called its ..... .

(d) The state of matter with definite volume and definite shape is called ..... .

(e) The substances which can flow are called ..... .

4. Write whether the following statements are true or false.

(a) Solids can diffuse very fast. .....

(b) The quantity of matter in an object is called volume. .....

(c) Atoms combine to form molecules. .....

(d) Particles of matter can move because they possess kinetic energy. .....

(e) When a liquid is heated, it solidifies. .....

5. Match the following :

Column A	Column B
(a) Fluids	*(i)* space between the molecules
(b) Melting point	*(ii)* change of state from gas to liquid
(c) Atoms	*(iii)* substances which can flow
(d) Intermolecular space	*(iv)* the smallest particles of matter.
(e) Condensation	*(iv)* the temperature at which a solid changes to a liquid.

6. Classify the following into solid, liquid and gas: Coal, kerosene, wood, oxygen, sugar, blood, water vapour, milk, wax.

7. Give appropriate terms for the following :

(a) The change of a solid into liquid.

(b) The force of attraction between the molecules of matter.

(c) The particles of matter which may or may not have independent existence.

(d) The process due to which a solid directly changes into its vapours.

(e) The change of vapour into a liquid.

B. SHORT ANSWER TYPE QUESTIONS

1. Give two examples for the following :

(a) substances which can flow

(b) substances which can sublime

(c) substances which can diffuse.

2. Define matter.

3. Differentiate between an atom and a molecule.

4. What do you understand by :

(a) Intermolecular force of attraction,

(b) Intermolecular space ?

5. Why are liquids and gases called as fluids ?

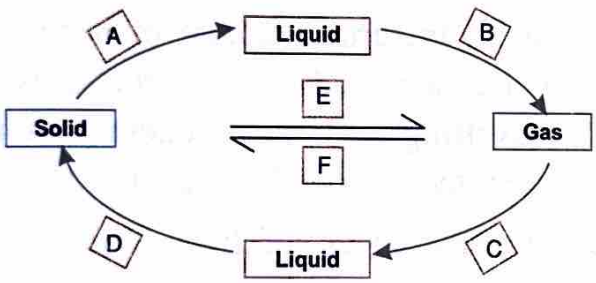
C. LONG ANSWER TYPE QUESTIONS

1. Define :

(a) Solid *(b)* Liquid *(c)* Gas

Give *two* examples of each type.

2. Write your observations and conclusions for the following :
   (a) A few marbles are put in a glass half filled with water.
   (b) Ice is kept at room temperature.
3. State three main characteristics of the particles of matter.
4. *(a)* Define interconversion of states of matter.
   (b) Why do solids, liquids and gases differ in their physical state ?
   (c) Under what conditions do solids, liquids and gases change their state ?
5. Name the processes marked as A, B, C, D, E and F.



6. Give reasons :
   (a) When a stone is dipped in a glass containing some water, the level of water rises but when a spoon of sugar is added to it and stirred, the water level does not rise.
   (b) A drop of ink added to water in a glass turns the whole water blue.

D. CASE STUDY BASED QUESTION

Sapna was making an orange flavoured drink. She poured 3-4 teaspoons of concentrated orange juice into a glass. She then added water until the glass was full and stirred it slowly.

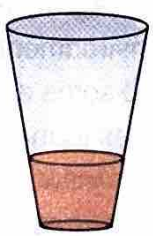
(a) Explain how the orange juice spread into water?

(b) What is the effect of stirring ?

(c) What is this intermixing of particles called ? Name the type of substances which show this phenomenon the fastest.

(d) "There is no more movement in the liquid in the glass once the orange colour of the juice has stopped spreading."

Is this statement correct ? Justify your answer.



Concentrated orange juice

Water + orange juice

Answers

Exercise : I — Objective type questions

1. *(i)–(c), (ii)–(a), (iii)–(b), (iv)–(a), (v)–(b)*
2. *(i)–(c)* : Assertion is true but the reason is false because not all types of matter can change from one state to another.
   (ii)–(a) : The reason correctly explains the assertion. Intermolecular gaps decrease only when kinetic energy of the particles decrease upon cooling the gas.
   (iii)–(d) : Assertion is false as liquids do not have any definite shape. The reason is true.
   (iv)–(a) : The reason correctly explains the assertion. Due to kinetic energy possessed by the particles, they are always in a random motion.
   (v)–(b) : The reason does not explain why wood is a hard solid while chalk is a soft solid.

Project

- Think and try to develop a working model to show the existence of water in all the *three* states.

Physical and Chemical Changes

2

THEME

The theme focuses on informing children and making them aware about the different types of changes physical and chemical that are regularly observed occurring in the environment. Some occur on their own and some are caused due to human activities to meet their requirements. Keeping in view the unending role of these changes, it becomes worthwhile that children learn about them.

LEARNING OBJECTIVES

The children will be able to :

- differentiate between physical and chemical changes;
- perform activities related to physical and chemical changes;
- classify changes such as respiration, preparation of solution of sugar, burning of paper ripening of fruit, spoiling of food materials as physical and chemical changes;
- discuss that in a chemical change, a new substance with different properties is formed.

INTRODUCTION

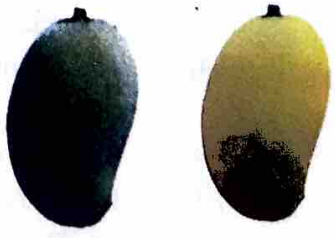
Change is an important feature of nature. Every moment several changes take place around us. These changes are exciting and they generate in us the curiosity to know the reasons behind each of them.

Some common changes which we observe in nature and in our surroundings are changes of day and night, change of climate, rainfall, melting of ice, freezing of water, growth of plants and animals, digestion of food, burning of fuel, ripening of fruits, cooking of rice, formation of curd from milk, rusting of iron, etc.

But what happens to the substance when a change occurs in it ?

Let us try to understand with the help of following examples :

(i) A raw mango is green but it turns yellow on ripening.



Raw mango *Ripe mango*

Thus when a fruit ripens, its ***colour*** changes.

(ii) A sapling is very small in size but it becomes larger with time as it grows into a plant, *i.e.* its shape and size change.

A sapling

A plant

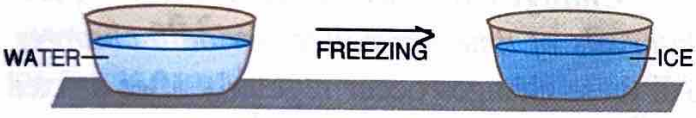
Thus when plants grow, their ***shapes and sizes*** change.

(iii) Molten wax is soft but it becomes hard on cooling.

Raw rice is hard but boiled rice is soft.

Thus, the ***hardness*** of the substance might also change when a change occurs.

(iv) Water changes into ice when kept in the freezer for few hours.



Thus a substance can change its ***state*** when a change occurs.

(v) The earth revolves around the sun which means the ***position*** of the earth with respect to the sun changes continuously with time.

From the above examples it is clear that, when changes take place, the substances may change their colour, shape, size, hardness, state, position, etc.

Thus we see that almost all substances around us undergo changes. But while some changes are easy to detect, others are too small to be noticed. Some of these changes are useful to us while others are not. We always think and try to ensure that useful changes should take place faster while the harmful changes should stop or they should rarely happen.

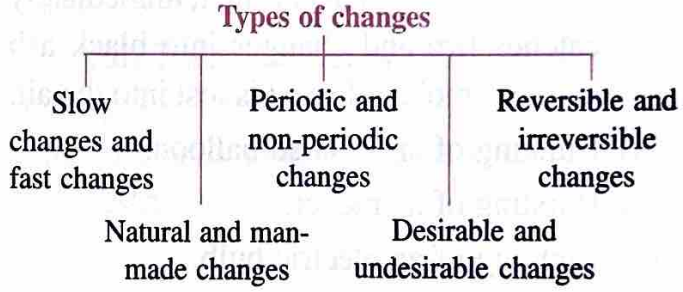
Activity 1

Make a list of five objects which do not show any changes in them with time.

We should remember that **there is always a reason** (*i.e.*, some cause) **behind every change**. Since most of the changes are different in nature, we need to classify them and study them accordingly in detail.

TYPES OF CHANGES

There are different types of changes based on how or on what basis, we try to classify them.



Slow Changes and Fast Changes

*The changes that take longer time to complete are called **slow changes**. These changes may take hours, days, months or even years to occur.*

Examples of slow changes :

(i) A young boy growing old. It takes a few years.

(ii) The evaporation of water from a lake, though happening continuously, takes hours.

(iii) The germination of seeds takes days.

(iv) Formation of fossil fuels from dead plants and animals is a very slow change. They take hundreds and thousands of years to be formed.

(v) Rusting of iron, tooth decay, freezing of water, change of seasons are all slow changes.

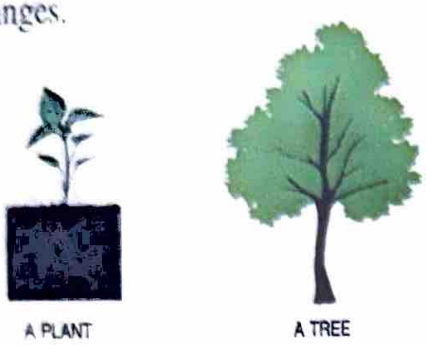


Fig. 2.1 : A plant transforming into a tree is a slow change

The changes that take place in a very short interval of time are called fast changes.

Examples of fast changes :

(i) When a piece of paper is lit, it immediately catches fire and changes into black ash and the smoke released is lost into the air.

(ii) Bursting of an inflated balloon.

(iii) Bursting of a cracker.

(iv) Lighting of an electric bulb.

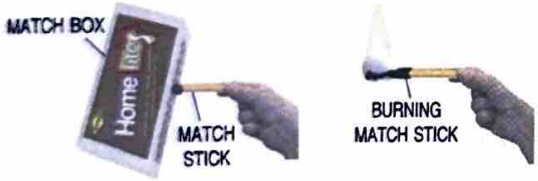


Fig. 2.2 : Burning of a matchstick is a fast change

Activity 2

Classify the following changes into slow or fast change.

Blinking of eyes, change of seasons, explosion of crackers, rusting of iron, growing of nails, formation of curd from milk, cutting of a cloth, cooking of rice.

Natural and Man-made Changes

Some of the changes take place in nature by themselves and are never ending. Such changes are called natural changes. The formation of coal from dead and decaying plants is a natural change.

Examples of natural changes

Change of day and night, change of seasons, earthquakes, growth of all living organisms, growing of a tree, eruption of volcanoes.

Any change that occurs due to the efforts of human beings is called a man-made change.

Examples of man-made changes

The formation of steel from raw iron, making a kite from a piece of paper, preparing flatbread from wheat flour, *etc.* are man-made changes.

Periodic and Non-periodic Changes

Changes that are repeated at regular intervals of time are called periodic changes. All these changes occur regularly after a fixed time period and not randomly.

Examples of periodic changes are :

(i) Change of day and night.

(ii) Change of seasons — spring, summer, autumn and winter.

(iii) Full moon and new moon nights.

(iv) Development of high tides and low tides in the sea.

Fig. 2.3 : Pendulum of a clock showing a periodic change

Changes that are not repeated at regular intervals and occur irregularly are called ***non-periodic changes***. These changes can occur at any time and they do not have any periodicity.

Examples of non-periodic changes

These are landslides, earthquakes*, power breakdown, *etc*. In our day-to-day life, we get fever, sneeze, headache, *etc*. These are some non-periodic changes that occur around, as well as within us.

An epidemic, a sudden shower of rain, a duststorm, thunder and lightning are also examples of non-periodic changes.

Desirable and Undesirable Changes

Changes that are useful to mankind are said to be ***desirable changes***. *For example*, cooking of food, formation of curd, ripening of fruits, production of heat from cooking gas to cook food, glowing of bulb to give light, change of season, photosynthesis, etc.

Any change that is harmful and brings about destruction is an ***undesirable change***. *For example*, food becoming stale, rotting of fruits, vegetables, eggs, etc., rusting of iron, floods, storms, epidemics, etc.

The whole world suffered from the Covid-19 pandemic which caused a lot of destruction to the mankind. This change was completely undesirable.

Activity 3

Classify the following changes into desirable and undesirable change :

Growing up of seedlings into plants, conversion of dung into manure, rusting of iron, souring of milk, breaking of a cup, drying of clothes, melting of ice, earthquake and eruption of volcanoes, ripening of fruits, curdling of milk.

Some changes are desirable at one time and undesirable at some other time.

For examples : (i) Cutting of trees is desirable for those who need wood for cooking or making furniture, but it is undesirable because it damages the environment.

(ii) Burning of a fuel for cooking is a desirable change but it is undesirable when it causes air pollution due to the smoke produced.

(iii) Timely arrival of rains (at the time of sowing and ripening) is desirable to a farmer, otherwise it is undesirable as it causes flood and destruction.

Reversible and Irreversible Changes

When a change in a substance can be reversed by changing the conditions, it is said to be a ***reversible change***. A reversible change is indicated by two opposite arrows ($\rightleftharpoons$ or $\rightleftarrows$).

Water converting into steam at 100°C and steam condensing back into water on cooling is an example of a reversible change.

$$\text{Water} \underset{\text{cooling}}{\overset{\text{heating}}{\rightleftharpoons}} \text{Steam}$$

Examples of reversible changes

Glowing of an electric bulb, stretching of a rubber band, ploughing of a field, melting of wax, dissolving sugar into water, vaporization of water, condensation of steam, melting, sublimation, etc.

* Landslides & earthquakes are categorised under 'natural changes.'

If a substance cannot be brought back to its original state after a change, it is said to be an ***irreversible change***. On burning, a piece of paper changes to ash, it is an irreversible change because paper cannot be obtained back from ash.

Examples of irreversible changes

Grinding of wheat or corn into flour, cooking of food, ripening of fruits, rusting of iron, souring of milk, burning of candle or wood, growth of living organisms and their ageing, etc.

Activity 4

State whether the change is reversible or irreversible.

Change	Reversible/ Irreversible
1. Raw egg to boiled egg.	
2. Batter to idli.	
3. Wet clothes to dry clothes.	
4. Woollen yarn to knitted sweater.	
5. Grain to its flour.	
6. Cold water to hot water.	
7. Straight string to a coiled string.	
8. Bud to flower.	
9. Milk to Cheese.	
10. Cowdung to biogas.	
11. Ice cream to melted ice-cream.	
12. Stretched rubber band to its normal size.	
13. Souring of milk.	
14. Ripening of a mango.	
15. Dissolving sugar in water.	

PHYSICAL AND CHEMICAL CHANGES

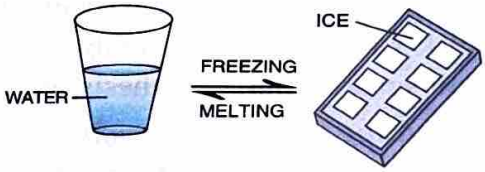
In science, all kinds of changes mentioned above can broadly be classified into *two* types:

(i) physical change (ii) chemical change.

Physical Change

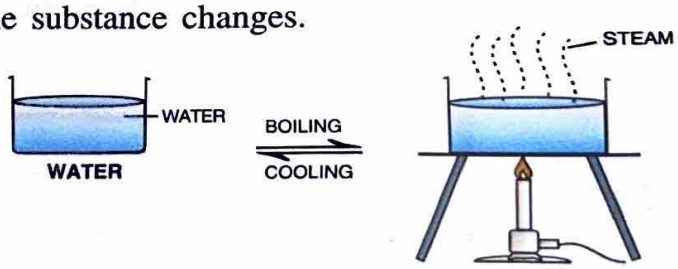
A **physical change** is a temporary change, in which no new substance is formed and the chemical composition of the original substance remains the same, even though some of its physical properties like colour, state, shape, size, etc. may change.

The **interconversion of states of matter** is a physical change. You might have observed that ice when taken out of a refrigerator melts and changes into water. If this water is put back in the refrigerator, it freezes into ice again. This means that the *properties of water and ice are the same. There is* ***no change in chemical compostion.*** *On melting or freezing,* ***no new substance*** *is formed, only the state of the substance changes.*



Water freezing into ice

Similarly, when water is boiled it changes into steam. On cooling, the steam again changes back into water. Again no new substance is formed, only the physical state of the substance changes.

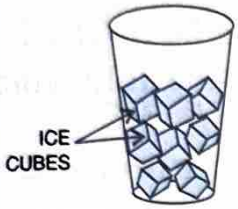


Water turning into steam

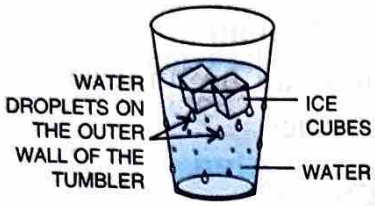
Thus, on removing the cause of change the substance returns to its original state.

Activity 5

Put some ice cubes in a dry glass tumbler and place it on a table for few minutes.



Ice cubes in a dry tumbler



Ice cubes after some time

(i) What do you observe on the outer surface of the tumbler?

(ii) From where do these water droplets have come ?

(iii) What happens to the ice inside the tumbler?

(iv) Are these changes condensation and melting ?

Water droplets are seen on the outer wall of the tumbler. These droplets have come from the water vapour present in the air.

When the surrounding warm air comes in contact with the cold outer wall of the tumbler, the water vapour present in the air get **condensed** to form water droplets on the walls of the tumbler.

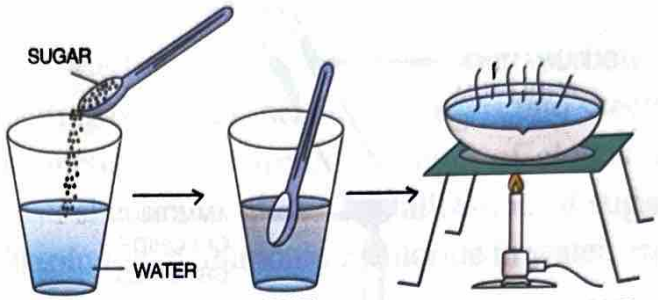
The ice cubes inside the tumbler **melt** into water.

Activity 6

To show that dissolution of salt in water is a physical change.

Take a glass of water. Put some salt in the water and stir it. The salt dissolves and makes the water salty. There is no change in the appearance of the water.

This shows that *salt retains its properties and composition.* This further indicates that



on mixing salt with water, ***no new substance is formed.***

Dissolution of salt in water is a physical change.

Now heat the solution. Water evaporates, leaving behind the salt.

We observe that by applying a simple method, only the physical properties of the original substance change. Therefore, ***the change is a physical change.***

Activity 7

To show that sublimation of ammonium chloride is a physical change.

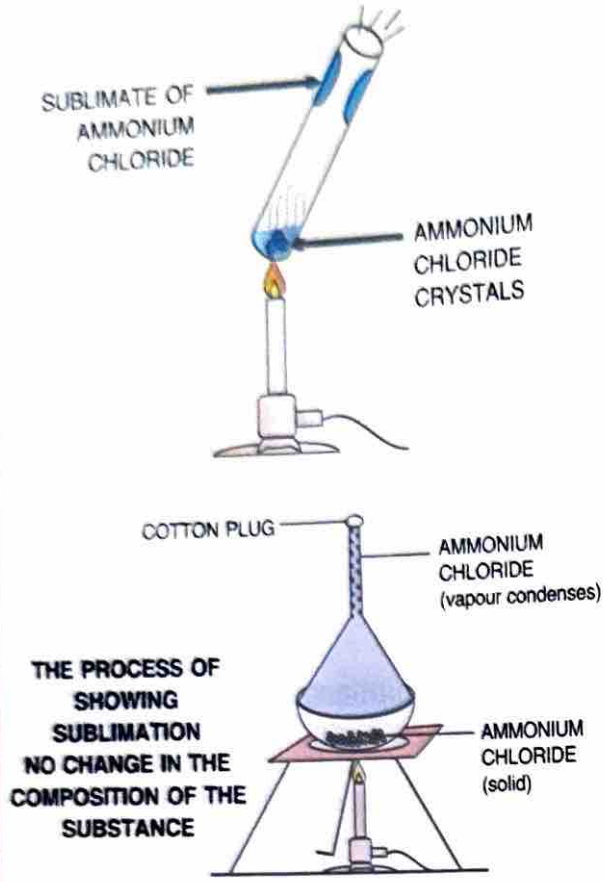
(To be demonstrated by the teacher)

Take some solid ammonium chloride in a test-tube and heat it gently.

You will observe that the solid changes into vapours. When the vapour touches the upper cooler part of the test tube, it changes back into solid ammonium chloride.

This shows that no new substance is formed, only the state changes directly from solid to gas. But when the vapour is cooled, it again changes into the original substance, that means the change is ***temporary*** and ***reversible.***

$$\text{Solid} \underset{\text{cooling}}{\overset{\text{heating}}{\rightleftharpoons}} \text{Gas}$$



Ammonium chloride sublimates on heating

Note : The process of changing a solid directly into a gas on heating and without passing through a liquid state is called **sublimation**. The solid substance formed after cooling of vapours is called **sublimate**. Some other substances which can sublime are camphor, naphthalene balls, iodine, dry ice (solid carbon dioxide), etc.

Characteristics of a Physical Change

1. No new substance is formed.
2. There is no change in the composition of the substance.
3. The properties of the substance remain the same.
4. The change is temporary and can be reversed by changing the conditions in most of the cases.
5. There may be changes in the state, size, shape, colour, and smell of substances during a physical change.
6. There may or may not be any exchange of energy during a physical change.

Can all physical changes be reversed ?

This can be understood by the following activities.

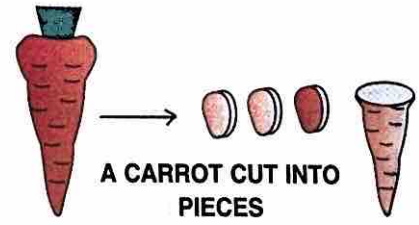
Activity 8

Take a carrot and cut it into pieces. What do you observe ?

The shape and size of the carrot has changed. Now observe the colour of the carrot pieces and taste one of them.

Do you find any difference between the colour or taste of the whole carrot and its pieces ?

No, there is no difference in the colour or taste. That means the properties of carrot are present in its pieces too although the shape and size have changed. Therefore, **it is a physcial change**.



An irreversible physical change

Thus, physical changes only affect the physical form of a substance and the properties of the final product remain identical to the properties of the original substance.

Generally, the change can be reversed but not always. In the example of carrot mentioned above, once the carrot is cut into pieces, the original carrot cannot be obtained.

Physical changes can be understood further by some more examples, explained ahead.

1. **Electric iron used for ironing clothes :** A cold electric iron becomes hot when current is passed through it, which is used for ironing clothes. But when the flow of current is stopped, the hot iron becomes cold again. No other change takes place. Hence, it is a physical change.
2. **Glowing of an electric bulb :** When electric current passes through the filament of a bulb, it starts glowing and gives light. When the bulb is switched off, the current stops flowing and the bulb stops glowing and returns back to its original condition. Hence, it is a physical change.
3. **Blowing of a balloon :** When air is blown into a balloon, it gets inflated. Its shape and size changes. When the air is allowed to escape from the balloon, it becomes deflated and returns back to its original condition. Hence, it is a physical change.
4. **Preparation of sugar solution :** When sugar is dissolved in water, a solution is prepared which is sweet to taste, *i.e.* the property of sugar is retained. No new substance is formed. Sugar can be easily recovered from its solution by the process of crystallisation. Hence, it is a physical change.
5. **Cutting of paper into pieces :** It is a physical change as although the original sheet of paper cannot be obtained, no new substance is formed. Even the smallest bits of paper retain their properties.
6. **Drying of wet clothes :** Wet clothes become dry due to the evaporation of water. No new substance is formed. Hence, it is a physical change.

Some more examples of physical changes

Melting of ice-cream, melting of butter, melting of metals, beating of metals into sheets, magnetisation of iron, formation of clouds or mist, shaping of glass, crystallization of sugar, dissolution of ammonium chloride in water, *etc.*

TERMS INVOLVED IN SOME PHYSICAL CHANGES

Let us now study the meaning of different terms used to explain physical changes.

- **Dissolving :** Dissolving is a process in which a substance known as solute mixes completely with another substance known as solvent to form a homogeneous mixture called solution. *Example :* Sugar dissolves in water.
- **Freezing :** The process in which a substance in a liquid state changes into its solid state on cooling at a particular temperature is called freezing or solidification.
- **Freezing point :** The temperature at which a liquid starts changing into its solid state at a given pressure is called its freezing point. All pure substances have a definite freezing point. *Example :* Pure water freezes at 0°C into ice.
- **Melting :** Melting is a process in which a substance in solid state changes into liquid state at a particular temperature.
- **Melting point :** Melting point of a solid is the temperature at which it starts melting. Solids have definite melting points. *Example:* Melting point of ice is 0°C. Numerically, the melting point and freezing point of a substance are the same.
- **Boiling :** The process in which a liquid on heating changes into its vapour state at a particular temperature is called boiling.

Note : Boiling is an extreme form of evaporation.

- **Boiling point** : The temperature at which a liquid starts changing into its vapours on heating at a given pressure is called its boiling point. All pure liquids have a definite boiling point. *Example* : Boiling point of water is 100°C.
- **Evaporation** : The process due to which a substance in liquid state changes into its vapour state at any temperature below its boiling point is called evaporation or vaporisation. *Example* : Water from rivers, lakes, ponds, etc. changes into vapour due to evaporation.
- **Condensation** : The process in which a substance in vapour or gaseous state changes into liquid state upon cooling is called condensation.
- **Condensation point** : The temperature at which a gaseous substance starts changing into liquid state on cooling is called condensation point. Numerically, the boiling point and condensation point of a liquid are the same.

Why wet clothes dry faster during summers but they take more time to dry in rainy season and winters?

This is because in summers, the temperatures are higher and the humidity (i.e. amount of water vapour in air) is low due to which evaporation is faster.

However during rainy season, humidity is high while during winters temperatures are low and as a result evaporation is slow.

Table 2.1 : Differences between evaporation and boiling

Evaporation	Boiling
1. Evaporation is a slow process.	1. Boiling is a fast process.
2. Evaporation takes place from the surface of the liquid.	2. Boiling takes place from all parts of the liquid.
3. Evaporation takes place at all temperatures below its boiling point.	3. Boiling takes place at a fixed temperature on heating, *i.e.* at its boiling point.

Chemical change

Chemical change is a permanent change in which new substance(s) is/are formed whose chemical composition and physical and chemical properties are completely different from those of the original substance(s).

We cook food everyday from raw grain and raw vegetables. The cooking requires heat energy which is supplied by a kerosene stove, cooking gas or an electric heater. Now, it is interesting to know whether you can get back the raw grain and vegetables from this cooked food.

The answer is NO, because *the composition of the raw substances change due to heat used for cooking and new substances are formed.*

Raw vegetable $\xrightarrow{heat}$ Cooked vegetable

The cooked materials retain their new forms even after the cause of change (heating) is removed. This shows that *the change is permanent and irreversible.* Such changes are chemical changes.

Activity 9

Roll out a flatbread from a ball of dough and bake it on a pan. Can we get the dough back ?

No, because it is a chemical change.

Characteristics of a Chemical Change

1. The change is permanent and cannot be reversed.
2. One or more new substances are formed.
3. Composition and properties of the original substance change.
4. There is an exchange of energy during a chemical change. This means heat or light or both might be given out or consumed.

A chemical change is generally called a ***chemical reaction****. The original substance is called the* ***reactant*** *and the new substance formed is called the* ***product****.*

Reactant/s $\xrightarrow[\text{change}]{\text{chemical}}$ *Product/s*

Examples of chemical changes

Bursting of a cracker, growing of a plant into a tree, souring of milk, preparation of soap from oil, changing of sugar into a black powdery substance on heating, burning of magnesium ribbon, burning of coal, decomposition of organic matter, boiling of egg, burning of a matchstick, respiration, etc.

Chemical changes can be further described by the following processes and activities.

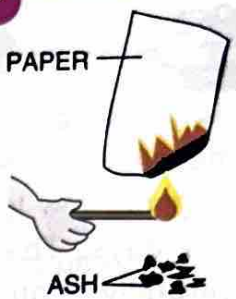
1. Burning : Burning is a chemical change in which a substance reacts with oxygen or air to produce a new substance called oxide with the evolution of energy in the form of heat and light. *Example :* Burning of fuel, candle, etc.

Give examples of two substances other than paper which on burning produce ash.

Activity 10

To show that burning of paper is a chemical change.

Take a piece of paper and burn it. It turns into *ash*. This is a new substance, whose properties are different from that of the paper. Ash cannot be changed into paper again. Therefore, it is a chemical change.



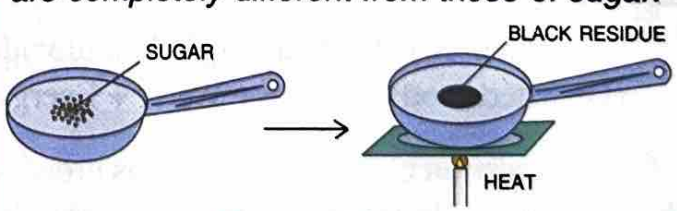
Burning of paper

Activity 11

To show that burning of sugar is a chemical change.

Put a spoonful of sugar in a pan and heat it. You will observe that the sugar first **melts** and then changes its colour to **reddish brown**. Ultimately, it turns **black** and gets **charred**. Now stop heating the sugar and try to get it back. You cannot. On heating the sugar you get a **black residue**, called ***charcoal**** and some gaseous products like carbon-dioxide and water vapour. The gases immediately mix with air and you do not even see them.

Now, taste the black substance, charcoal. It is not sweet. This shows that *its properties are completely different from those of sugar.*



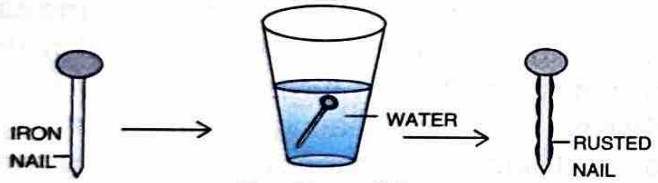
Burning of sugar

* *Charcoal* is a form of carbon, an element.

2. Rusting : Rusting is a chemical process in which iron reacts with oxygen in presence of moisture to produce a reddish brown substance called **rust**.

Activity 12

Take an iron nail and bring it near a magnet. The nail gets attracted towards the magnet. Now leave the nail in tap water for few days. Then take it out and observe carefully. You will see a reddish brown coating on the nail. This coating is called ***rust***#. Again bring the magnet near the nail. But now the nail is not attracted towards the magnet. This shows that rust is a new substance, which is not magnetic in nature. You cannot get iron back from rust. Therefore, the change is irreversible and permanent. Hence, rusting of iron is a chemical change.



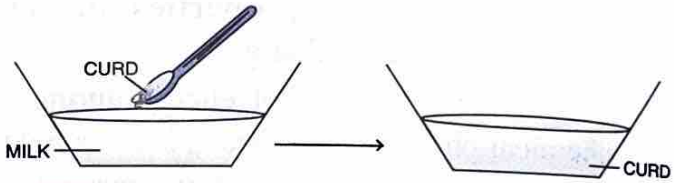
Rusting of iron

3. Respiration : Respiration is a chemical process that takes place in all living beings. In this process, oxygen present in air which is inhaled by the living beings reacts with the digested food to release energy, carbon dioxide and water.

Glucose + Oxygen → Carbon dioxide + Water + Energy
(Digested food)

4. Digestion of food : During this process, the food in our body breaks into new, simpler compounds like proteins, carbohydrates, fats, etc. by the action of enzymes. These compounds are then absorbed by our body to release energy. We can not get back the original food substances. Hence, digestion of food is a chemical change.

5. Formation of curd from milk : It is a chemical change because curd cannot be changed back into milk. The properties of curd are different from that of milk. The formation of curd from milk is a slow, desirable and irreversible change.



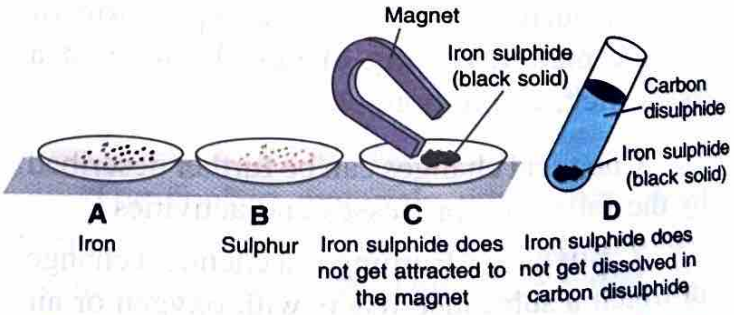
Milk changing into curd (a permanent change)

6. Heating of iron and sulphur : When iron and sulphur are heated together, a new substance iron sulphide is formed.

The properties and colour of this new substance is different from those of iron and sulphur.

Iron is greyish black and sulphur is yellow, but iron sulphide is a black solid.

Iron gets attracted towards a magnet, but iron sulphide does not.



Heating of iron and sulphur is a chemical change

Sulphur dissolves in carbon-disulphide, but iron sulphide does not.

Rust is a hydrated oxide of iron.

Hence, formation of iron sulphide on heating iron and sulphur powder together is a chemical change.

How will you know that a chemical change has taken place ?

We can tell if a chemical change has taken place because it is usually accompanied by :

Change in colour : Ripening of fruit is a chemical change. The colour of the raw fruit changes on ripening. Burning or charring of sugar or burning of a matchstick gives a black substance.

Evolution of a gas : Reaction of a metal like zinc with dilute hydrochloric acid is a chemical change because a salt called zinc chloride is formed and a gas called **hydrogen** is evolved.

Change in energy : Release or absorption of energy in the form of heat, light and sound takes place during a chemical change.

Burning of a cracker is a chemical change. When a cracker explodes, it produces heat and light energy and sound is also heard.

Activity 13

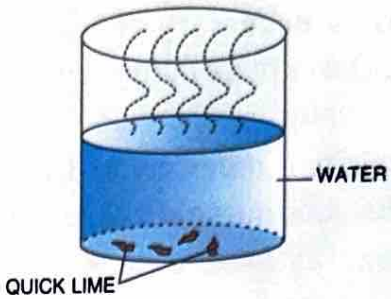
To show that dissolution of quick lime in water is a chemical change.

(To be demonstrated by the teacher)

Take a beaker and put 1-2 pieces of quick lime in it. Add some water to it.

You will observe that the quick lime pieces first crumble and then dissolve in water, evolving large amount of heat energy. If you touch the beaker, you will notice that the beaker feels hot to the touch.

A new white solid substance, calcium hydroxide, is formed which partially dissolves in water to produce a solution called lime water. The remaining white solid, called slaked lime, settles at the bottom of the beaker.



Large amount of energy is evolved during a chemical change

This proves that dissolution of quick lime in water is a chemical change in which energy change also takes place.

Activity 14

To show that on adding zinc to dilute hydrochloric acid, a chemical change takes place.

(To be demonstrated by the teacher)

Take some dilute hydrochloric acid in a test tube. Add some zinc granules to the acid in the test tube.

You will observe some bubbles coming out with some sound. This indicates **evolution of a gas.**

Touch the test tube. It has become hot. This indicates that some heat energy has been released during the reaction.

Bring a lighted match stick near the mouth of the test tube. A pop sound is heard and the flame of the match stick gets extinguished. This confirms that the gas evolved is **hydrogen gas.**

Different types of energies are needed for different chemical changes.

Example : • **Heat** is necessary for the cooking of food, burning of paper, etc.

- **Light** is necessary for photosynthesis (which takes place during daytime) or during photography.
- **Electricity** is necessary to decompose water into its constituent elements, hydrogen and oxygen.

Energy change may also take place during a physical change but the amount of energy involved is less in comparison to a chemical change.

Example : When a solid melts into a liquid, it absorbs some heat energy; while when a liquid boils to form vapour, it also absorbs some heat energy. Similarly when a liquid freezes into solid, heat energy is given out; while when vapour condenses into liquid, energy is given out.

IMPORTANCE OF CHEMICAL CHANGES

Various types of chemical changes take place in our daily life. Some are natural while some are man-made. Most of them are very useful to us. Some are listed below :

1. Raw grains cannot be eaten but when they are cooked, chemical changes take place in them and they become edible.
2. Soaps, detergents, paints, fertilizers, medicines, *etc.* are all produced due to chemical changes.
3. Many useful metals like iron, aluminium, copper, *etc.* are extracted from their combined states by various chemical processes.
4. Burning of fuel is an important chemical change because it produces a lot of heat energy which is used for domestic and industrial purposes.
5. Different life processes taking place in plants and animals are all chemical changes which help to sustain life.

An example of simultaneous physical and chemical change

When a candle is lit, the wax melts and turns into liquid state. As some of the molten wax drops on the floor or table, it solidifies again. Therefore, this is a ***physical change.*** *Simultaneously, most of the molten wax rises up the wick, turns into vapour and burns with the flame. Two new substances are formed: water vapour and carbon dioxide. The candle becomes smaller and smaller. This is a* ***chemical change.***

Thus, we see that the melting of wax is a physical change and the burning of the candle is a chemical change.

Candle wax $\xrightarrow{\text{heat}}$ Carbon dioxide + Water vapour

Table 2.2 : Differences between physical and chemical changes

Physical change	*Chemical change*
1. Change is temporary.	1. Change is permanent.
2. No new substance is formed. There is change only in physical properties.	2. New substances are formed, with entirely different properties.
3. Change can be reversed by simple methods.	3. Change cannot be reversed by simple methods.
4. Heat or light, may or may not be given out or consumed.	4. Heat or light, or both, are given out or consumed.
5. The original form of the substance can be obtained easily by simple physical methods.	5. The original substance cannot be obtained by simple physical methods.

RECAPITULATION

- Changes in matter take place around us all the time.
- The changes that we see around us can be classified as natural or man-made, slow or fast, desirable or undesirable, periodic or non-periodic, reversible or irreversible, physical or chemical, etc.
- All changes can broadly be classified into two types (i) physical and (ii) chemical changes.
- A change in which no new substance is formed and where the composition and the properties of the substance involved remain the same is known as a physical change.
- A change in which a new substance is formed whose composition and properties are entirely different from those of the original substance is known as a chemical change.
- Energy is either released or absorbed during these changes. Physical changes may or may not involve energy exchange but chemical changes always do so, in the form of heat and light.
- Rusting is a slow chemical process while burning is a fast chemical process.
- The process of a liquid changing into a gas at a particular temperature is called **boiling**. It is a fast process.
- The process of a liquid changing into a gas at room temperature is called **evaporation**. It takes place from the surface of a liquid and is a slow process.

EXERCISE

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(i) A substance which cannot sublime is :

(a) iodine *(b)* camphor
(c) sugar *(d)* dry ice

(ii) When you put some ice cubes in a glass, droplets of water are formed on the outer wall of the glass. This explains the phenomenon of :

(a) melting *(b)* freezing
(c) condensation *(d)* evaporation

(iii) Burning is a :

(a) slow process *(b)* fast process
(c) natural process
(d) none of the above

(iv) Which one of the following is volatile in nature ?

(a) common salt *(b)* petrol
(c) water *(d)* milk

(v) An example of both physical and chemical change is :

(a) burning of candle
(b) melting of ice
(c) cooking of food
(d) glowing of bulb

(vi) The compound rust is a hydrated oxide of :

(a) copper *(b)* aluminium
(c) iron *(d)* gold

(vii) When sugar is heated, its colour changes into :

(a) red *(b)* blue
(c) black *(d)* grey

(viii) A pop sound is heard when a lighted match stick is brought near the mouth of a jar. This indicates the release of :

(a) oxygen gas *(b)* hydrogen gas
(c) nitrogen gas *(d)* water vapour

(ix) When we add water to the following substances, which one will show a chemical change ?

(a) salt (b) sugar

(c) oil (d) quick lime

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

(i) **Assertion (A) :** The dissolution of salt in water is a physical change.

Reason (R) : Salt retains its composition and properties in solution.

(a) (1) (b) (2) (c) (3) (d) (4)

(ii) **Assertion (A) :** Naphthalene balls disappear from the cupboard after sometime.

Reason (R) : Naphthalene balls undergo evaporation.

(a) (1) (b) (2) (c) (3) (d) (4)

(iii) **Assertion (A) :** During a chemical change, new substances with entirely different properties are formed.

Reason (R) : Bursting of a cracker is a chemical change.

(a) (1) (b) (2) (c) (3) (d) (4)

(iv) **Assertion (A) :** Heating of iron with sulphur is a physical change.

Reason (R) : A new substance, iron sulphide is formed, when iron and sulphur are heated together.

(a) (1) (b) (2) (c) (3) (d) (4)

3. Fill in the blanks.

(a) A new substance is always formed in a ..... change.

(b) Melting of ice is a ..... change.

(c) When a candle burns, wax melts. Melting of wax is a ..... change.

(d) Chemical change occurs as a result of ..... between two substances.

(e) Burning of a fuel is a ..... change.

(f) Revolution of the earth around the sun is a ..... change.

(g) Growing of a seedling into a plant is a ..... change.

4. Write *true* or *false* against the following statements :

(a) Cutting of paper into pieces is a chemical change. .....

(b) Rusting of iron is a chemical change. .....

(c) Earthquake is a desirable change. .....

(d) Melting of ice is a physical change. .....

(e) Burning of sugar is a temporary change. .....

5. Match the following.

Column A	Column B
(a) Evaporation of water	(i) Non-periodic change
(b) Milk turning sour	(ii) Periodic change
(c) Earthquake	(iii) Chemical change
(d) Change of seasons	(iv) Physical change
(e) Glowing of bulb	(v) Reversible change

6. State whether the following are physical or chemical changes.

(a) glowing of a bulb (b) burning of sugar

(c) heating of water

(d) growing of a piglet into a pig

(e) burning of wood

(f) passing electric current through a heater's rod

(g) water cycle in nature

(h) respiration in living beings

(i) shaping a piece of glass

(j) lightning

(k) dissolving sugar in water

(*l*) heating a mixture of iron filings and sulphur
(*m*) mixing oil with water
(*n*) cutting wood into small pieces.
(*o*) photosynthesis
(*p*) Digestion of food
(*q*) Melting of wax
(*r*) Boiling of an egg
(*s*) Slaking of lime

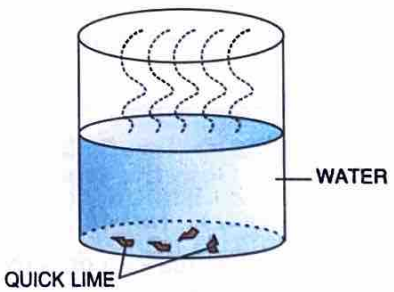
7. Find the odd one out from the following :
   (*a*) Burning of wood, cooking of food, glowing of bulb, formation of curd.
   (*b*) Water turning into ice, cutting a glass into pieces, dissolving sugar into water, boiling an egg.
   (*c*) Drying of clothes, ripening of fruits, photosynthesis, rusting of iron.

B. SHORT ANSWER TYPE QUESTIONS

1. Mention a change which is always
   (*a*) desirable (*b*) undesirable (*c*) periodic
2. Name a chemical change which takes place in the presence of :
   (*a*) Heat (*b*) Light (*c*) Electricity
3. What do you observe when :
   (*a*) water is boiled,
   (*b*) a piece of paper is burnt,
   (*c*) some ice cubes are kept in a glass tumbler,
   (*d*) solid ammonium chloride is heated,
   (*e*) an iron nail is kept in tap water for few days,
   (*f*) a spoon of sugar is heated in a pan,
   (*g*) a lighted match stick is brought near the mouth, of the test tube containing hydrogen gas,
   (*h*) quick lime is dissolved in water,
   (*i*) little amount of curd is added to a bowl containing warm milk and kept for five hours ?
4. Is burning a physical change or a chemical change ? Why ?

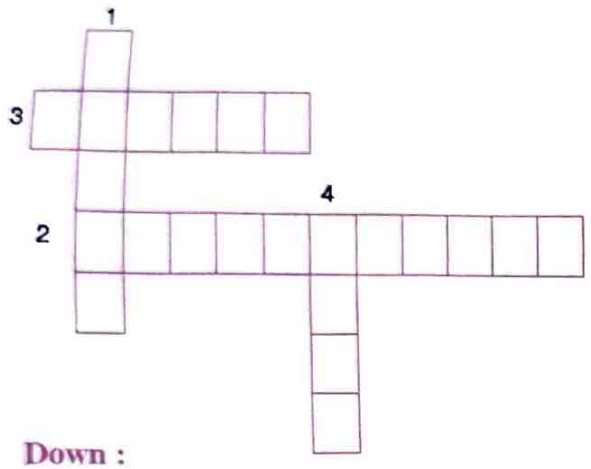
C. LONG ANSWER TYPE QUESTIONS

1. (*a*) Define :
   (*i*) a physical change
   (*ii*) a chemical change
   (*b*) Give two examples for each of the above two changes.
2. What are reversible and irreversible changes? Give one example of each.
3. A burning candle shows both physical and chemical changes. Explain.
4. State three differences between evaporation and boiling.
5. State four differences between physical and chemical changes.
6. Give reasons :
   (*a*) Burning is a chemical change.
   (*b*) Cutting of a mango into pieces is a physical change.
   (*c*) Rusting of iron is an undesirable change.
7. The diagram given ahead shows an experiment in which a few pieces of quicklime are added to water.



   (*a*) What do you observe when quicklime is added to water ?
   (*b*) Why does the beaker become hot ?
   (*c*) What type of change is depicted here ?
   (*d*) What conclusion can be drawn on the basis of your observation ?
8. (*a*) Draw a well-labelled diagram to show the process of sublimation.
   (*b*) Is sublimation a physical change or a chemical change ? Explain.
   (*c*) Name two substances which can sublime.

9. Solve the crossword puzzle given below with the help of the following clues :



Down :

1. A large amount of energy is given out when quicklime is added to this substance.
4. A reddish-brown substance formed when iron is exposed to moist air

Across :

2. It is a physical change in which a liquid changes into vapour.
3. An object which attracts iron fillings.

D. CASE STUDY BASED QUESTION

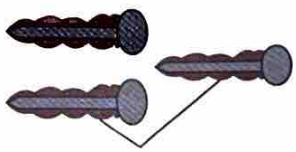
Ravi had brought some iron nails to fix pictures on the walls of his room. However, he got busy and he left those nails in the courtyard of his house to work with later. Upon returning to the courtyard after a few days, he saw that a reddish-brown substance was deposited on the nails. His father told him that this is a common phenomenon on iron articles, especially during rainy season.

(a) Name the reddish-brown substance formed on the iron nails.

(b) Is it a physical change ? Justify your answer.

(c) What will you observe when a magnet is brought near the nails ?

(d) State two conditions under which the above phenomenon takes place. Why is this faster during rainy season ?



Reddish-brown coating on iron nails

Answers

Exercise : 2 — Objective type questions

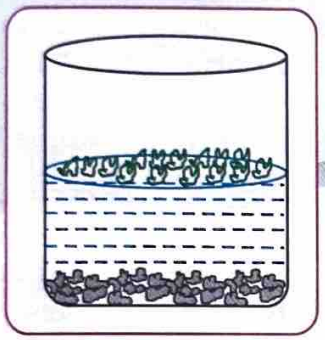
1. *(i)–(c), (ii)–(c), (iii)–(b), (iv)–(b), (v)–(a), (vi)–(c), (vii)–(c), (viii)–(b), (ix)–(d)*
2. *(i)–(a)* : The reason correctly explains the assertion. Salt retains its properties when dissolved in water, that is why it is a physical change.
   (ii)–(c) : Assertion is true but the reason is false. Naphthalene balls disappear after sometime due to *sublimation*, not due to evaporation.
   (iii)–(b) : The reason does not explain the process of chemical change, it only gives an example.
   (iv)–(d) : Assertion is false as heating of iron with sulphur is a chemical change.

Project

- Maintain a record for one year of the seasonal changes in vegetables, clothing and natural events around you. Identify the changes that can or cannot be reversed.
- Observe preparation of dishes at your home. Identify two changes in the process that can be reversed.

Elements, Compounds And Mixtures

3



THEME

This theme will enable children to understand that the earth mainly consists of mixtures containing elements and compounds. These are of different types and often the separation of components of mixtures is required for practical utility. We will also discuss the different techniques for separation of the components of a mixture to get the pure components.

LEARNING OBJECTIVES

The children will be able to :

- identify elements and compounds on the basis of their properties and the type of atoms present in them;
- differentiate between mixtures and compounds on the basis of their properties and composition of constituents;
- provide examples of elements, compounds and mixtures from daily life;
- discuss different techniques for separation of components of mixtures;
- justify the reason for the use of a particular technique in separation of a mixture;
- explain chromatography and its importance.

A. Elements and Compounds

INTRODUCTION

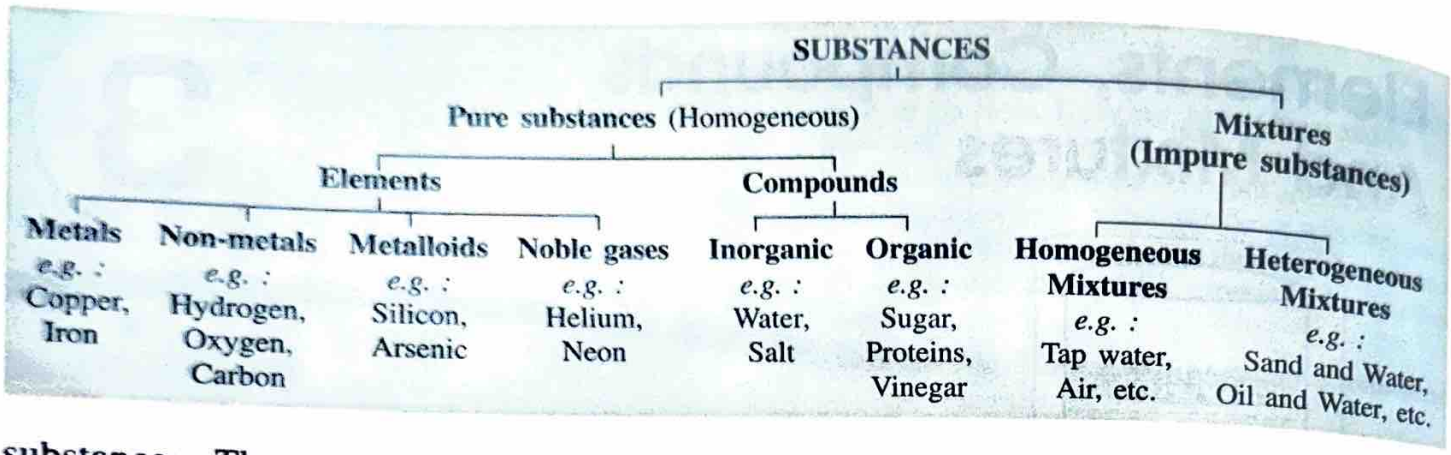
There are millions of substances in this world such as iron, aluminium, water, air, etc. They are all different from one another in their composition, properties and uses.

- They are made up of different kinds of matter.
- They may be pure or impure.

To study these substances accurately and conveniently, they have been classified on the basis of similarities and dissimilarities into three main classes : elements, compounds and mixtures. (You have already studied about these in class VI).

Pure substances : Pure substances have a definite composition and a definite set of properties such as boiling point, melting point, density, etc. They are all homogeneous *i.e.*, their composition is uniform throughout the bulk. **Both elements and compounds are pure substances.**

Impure substances : Impure substances are made up of two or more pure substances mixed together in any proportion. They do not have any definite set of properties. They retain the properties of their constituent



substances. They may be homogeneous or heterogeneous. They are all mixtures. *e.g.* : air, sugar solution, sand and stone, etc.

Activity 1

Make a list of five pure and five impure substances which you use in your daily life.

Let us first recall the two main classes of pure substances *i.e.* elements and compounds with the following activities.

Activity 2

Classify the following list of substances as elements and compounds and form two separate groups.

Water, iron, sodium chloride, copper, plaster of paris, calcium oxide, aluminium, silicon, sodium sulphate, helium and sodium hydrogen carbonate.

Activity 3

The following table shows the symbols and formulae of some substances. Identify them as elements or compounds and write their chemical names.

Symbol/ Formula	Element/ Compound	Name of the substance
Na		
H_2	Element	Hydrogen
CO_2		
NH_3		
O_2		
Fe		
He		
HCl		
N_2		
H_2O		
FeS	Compound	Iron sulphide
Cl_2		
NaCl		
CaO		

Now let us revise elements and compounds in more detail.

ELEMENTS

An **element** is a pure substance that cannot be converted further into anything simpler than itself by any physical or chemical process. Thus, each element has its own unique properties.

These elements are made up of extremely small particles called atoms.

Atoms are the smallest units of an element.

All atoms in an element are identical, but they differ from the atoms of other elements.

Examples : A piece of aluminium sheet contains only aluminium atoms and oxygen molecules contain only atoms of oxygen. Atoms of oxygen and aluminium are different from each other.

Do You Know ?

Robert Boyle was the first scientist to use the term element in 1661.

Antoine Laurent Lavoisier (1743-94) was the first to establish experimentally useful definition of an element.

At present 118 *elements are known. Of these,* 92 *are natural elements while the rest* 26 *have been artificially created.* Some elements are solids, some are liquids and some are gases. In fact, elements are the basic pure substances from which millions of substances are made.

Do You Know ?

Of the 118 elements known to us, some are radioactive in nature, because they emit radiations which may be harmful.

CLASSIFICATION OF ELEMENTS

Based on their properties, elements have been classifed into (i) metals (ii) non-metals (iii) metalloids and (iv) noble (or inert) gases.

Metals : Most of the elements known to us are metals. *Examples :* Gold, silver, copper, aluminium, iron, zinc, tin, lead, etc.

Do You Know ?

- Chalk, milk and our bones contain a common metal **calcium**.
- Chlorophyll contains **magnesium** which helps to capture the energy from sunlight for photosynthesis by plants.
- In mammals, **iron** is found in red blood cells in haemoglobin which carries oxygen in the body.
- Chocolate wrappers are made of **aluminium** (metal).

Non-metals : Non-metals are very less in number in comparison to metals. Excluding inert gases, there are only twelve non-metals known to us. *Examples :* Hydrogen, oxygen, nitrogen, carbon, chlorine, sulphur, phosphorus, fluorine, selenium bromine, iodine and astatine.

Metalloids : These elements show some properties of metals and some properties of non-metals. They are hard solids. *Example :* Boron, silicon, germanium, arsenic, antimony tellurium and polonium.

Inert or noble gases : These elements do not react chemically with other elements or compounds, so they are known as noble or inert gases. They are found in air, in traces. They are only six in number — helium, neon, argon, krypton, xenon and radon.

Activity 4

Following is a list of elements. Select metals, non-metals, metalloids and noble gases from the list : gold, graphite, chlorine, sodium, arsenic, helium, sulphur, xenon, antimony.

SYMBOLS OF ELEMENTS

Now let us revise the method by which each element was given a symbol (studied in class VI).

Each element has a name and a unique chemical symbol approved by the International Union of Pure and Applied Chemistry (IUPAC).

The symbol of an element is an abbreviated form or short hand notation of its name in English mostly, but sometimes in Latin, Greek, etc.

The symbol of an element also represents an atom of that element.

Following are the guidelines to write the symbols of elements :

1. Each element is denoted by a symbol, which is usually the first letter of its name, always written in capital.

Example : Oxygen is an element. It is denoted by the symbol 'O'. Similarly, hydrogen is denoted by the symbol 'H'.

2. However, when the first letter of more than one element is the same, the symbol is denoted by two letters, first letter is written in capital while the second is written in small letter.

Example : (i) Hydrogen and Helium are elements whose first letter is H. Hence symbol of hydrogen is H, while that of helium is He (not HE). (ii) Boron is represented by symbol 'B', while bromine is denoted by 'Br' and barium by 'Ba'.

3. The first letter of the symbol of an element is always the first letter of its name but the second letter of the symbol can be any other letter in its name.

Example : Carbon, calcium and chlorine all start with letter C. Hence, symbol of carbon is C (first letter only), symbol of Calcium is Ca (first two letters) and symbol of chlorine is Cl (first and third letter).

4. For some elements, when the first two letters of their English name are the same, the symbol of one of the elements is derived from its Latin or Greek name.

Example : The first two letters of cobalt and copper are the same. Hence, symbol of cobalt is Co (derived from its English name), while symbol of copper is Cu (derived from its Latin name — Cuprum).

Table 3.1 : Names and symbols of some elements derived from their English names.

Name in English	*Symbol*	*Name in English*	*Symbol*
Hydrogen	H	Argon	Ar
Helium	He	Calcium	Ca
Lithium	Li	Chromium	Cr
Beryllium	Be	Manganese	Mn
Boron	B	Cobalt	Co
Carbon	C	Nickel	Ni
Nitrogen	N	Zinc	Zn
Oxygen	O	Arsenic	As
Fluorine	F	Bromine	Br
Neon	Ne	Iodine	I
Magnesium	Mg	Platinum	Pt
Aluminium	Al	Krypton	Kr
Silicon	Si	Barium	Ba
Phosphorus	P	Radon	Rn
Sulphur	S	Xenon	Xe
Chlorine	Cl	Radium	Ra
		Uranium	U

Table 3.2 : Names and symbols of some elements derived from their Latin and Greek names.

Name in English	*Name in Latin/Greek*	*Symbol*
Sodium	Natrium	Na
Potassium	Kalium	K
Iron	Ferrum	Fe
Copper	Cuprum	Cu
Silver	Argentum	Ag
Gold	Aurum	Au
Mercury	Hydrargyrum	Hg
Lead	Plumbum	Pb
Tin	Stannum	Sn
Antimony	Stibium	Sb

Activity 5

Write the names and symbols of the first twenty elements you have studied in class VI.

Compounds : A compound is a pure substance formed by the chemical combination of two or more elements in a fixed ratio by mass.

Examples of compounds are common salt, water, sugar, chalk, sand, carbon dioxide, washing soda, alcohol, etc.

The smallest unit of a compound is molecule.

Properties of molecules of one compound differ from those of molecules of other compounds.

CHARACTERISTICS OF COMPOUNDS

- The properties of compounds are entirely different from those of its constituent elements.

Example : Sodium is a highly reactive poisonous metal while chlorine is a greenish yellow gas but the compound formed by chemical combination of these two elements *i.e.* sodium chloride, known as common salt, is added to food and is completely safe to eat.

- Compounds can be broken down into their constituent elements only by chemical means, not by physical means.

3.3 Table showing some common compounds, elements present in them and their formulae

Compounds	Elements present	Formulae
1. Water	Hydrogen and oxygen	H_2O
2. Sodium chloride (common salt)	Sodium and chlorine	NaCl
3. Magnesium oxide	Magnesium and oxygen	MgO
4. Calcium oxide (quick lime)	Calcium and oxygen	CaO
5. Carbon dioxide	Carbon and oxygen	CO_2
6. Sodium carbonate	Sodium, carbon and oxygen	Na_2CO_3
7. Sodium bicarbonate (baking soda)	Sodium, hydrogen, carbon and oxygen	$NaHCO_3$
8. Hydrated calcium sulphate (plaster of paris)	Calcium, sulphur, hydrogen and oxygen	$CaSO_4 \cdot 1/2H_2O$
9. Cane sugar	Carbon, hydrogen, oxygen	$C_{12}H_{22}O_{11}$
10. Silica (sand)	Silicon and oxygen	SiO_2
11. Glucose	Carbon, hydrogen, oxygen	$C_6H_{12}O_6$
12. Sodium sulphate	Sodium, sulphur, oxygen	Na_2SO_4

Note : Hydrated sodium carbonate ($Na_2CO_3 \cdot 10H_2O$) is called washing soda.

Example : Common salt is a compound chemically known as sodium chloride, that can be broken down into its constituent elements sodium and chlorine only by chemical means.

Similarly water, a compound, can be broken down into more simpler substances, hydrogen and oxygen, from which it is formed by the process of **electrolysis** (a chemical method).

- Compounds have a fixed composition of their own.
- Energy is either absorbed or liberated during the formation of a compound.
- A compound is represented by a definite chemical formula.

EXERCISE — I

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

   (i) Atoms of different kinds combine to form molecules of :

   (a) an element *(b)* a compound
   (c) a mixture *(d)* none of the above

   (ii) Which of the following is an inert gas ?

   (a) H_2 *(b)* Cl_2
   (c) He *(d)* S

   (iii) The molecular formula of common salt is :

   (a) MgO *(b)* NaCl
   (c) CO_2 *(d)* Na_2SO_4

   (iv) Which of the following is a metalloid ?

   (a) Argon *(b)* Silicon
   (c) Berrylium *(d)* Phosphorus

   (v) Quicklime is a compound of :

   (a) Calcium and oxygen
   (b) Magnesium and oxygen
   (c) Calcium and chlorine
   (d) Magnesium and sulphur

2. Select elements and compounds from the following list :
   Iron, plaster of paris, chalk, common salt, copper, aluminium, calcium oxide, cane sugar, carbon, silica, sodium sulphate, uranium, potassium carbonate, silver, carbon dioxide.
3. Write the symbols of : Helium, Silver, Krypton, Antimony, Barium.

B. SHORT ANSWER TYPE QUESTIONS

1. Write the names of the following elements : Na, C, Kr, U, Ra, Fe, Co.
2. Name the main metal present in the following:

   (a) Haemoglobin *(b)* Chalk
   (c) Chlorophyll *(d)* Chocolate wrapper
3. Name the particles of which :

   (a) an element is made up of
   (b) a compound is made up of
4. Define :

   (a) Elements *(b)* Compounds

C. LONG ANSWER TYPE QUESTIONS

1. What do you understand by :

   (a) Metalloids *(b)* Noble gases

   Give *two* examples of each.
2. Give *four* examples of non-metallic elements.

B. Mixtures and separation of the components of mixtures

Most of the substances that we come across in our daily life are mixtures. Mixtures are made up of elements, compounds or both elements and compounds. The substances which form mixtures are called **components** or **constituents** of mixtures. Components of a mixture are mixed in any proportion without any chemical combination.

Mixtures can be defined as impure substances which are formed by mixing two or more pure substances (elements and/or compounds) in any proportion such that they do not undergo any chemical change and retain their individual properties.

Example : Air is a mixture. Its constituents are oxygen, nitrogen, carbon dioxide, water vapour, dust particles, etc. All these constituents retain their individual properties.

Activity 6

Identify the mixtures from the following list : Air, water, sugar, salt, milk, tea, alcohol, honey, soil, glucose.

Characteristics of mixtures

1. In mixtures, components are loosely held together without any chemical force acting on them or between them. Hence components retain their individual properties as no new substance is formed.

   Examples : **(1)** In a mixture of salt and chilli powder, salt particles retain their salty taste and chilli powder particles retain their hot and bitter taste.

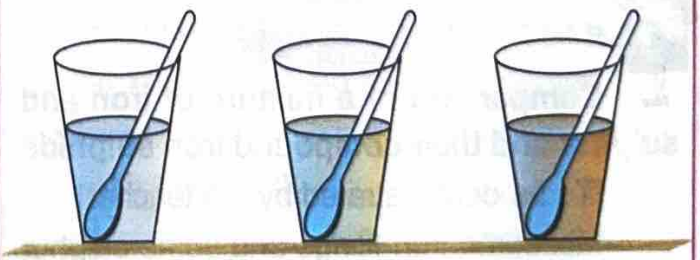
   (2) In a mixture of salt and sugar, both the components retain their salty and sweet tastes respectively.

2. Mixtures do not have any fixed amount of components *i.e.* they can have their components in varying proportions.

   Example : A mixture of sugar and water can be formed by mixing varying proportions of these two substances.

Activity 7

Take three glasses. Fill three-fourth of each glass with water. Add a spoonful of lime juice to the first glass and stir it properly so that lime juice mixes well with water. To the second glass, add two spoons of lime juice and stir it. To the third glass, add three spoons of lime juice and stir it. Now taste all the *three* lime solutions you have prepared. You will find that all samples are sour to taste but the degree of sourness is different. In each case, we are getting a homogeneous mixture of lime juice and water but they differ in their tastes due to the difference in the ratio of water and lime juice in them.



I
A glass with one spoon of lime juice in water (sour)

II
A glass with two spoons of lime juice in water (more sour than I)

III
A glass with three spoons of lime juice in water (more sour than both I and II)

3. Mixtures do not have any specific set of properties.

Example : In a mixture of rice and wheat, both the components can be easily recognized because they retain their properties.

4. Components of mixtures can be separated by simple physical methods.
   Example : Salt solution is a mixture of salt and water. When this solution is heated, water gets evaporated leaving behind salt.
5. *The melting point and the boiling point of a mixture is not fixed. It depends on the proportions of its components present in it, e.g. pure water boils at 100°C but salty water boils at a higher temperature than 100°C.*
6. *Mixtures can be homogeneous or heterogeneous.*
7. Formation of mixtures *does not involve any energy exchange.*
8. Mixtures cannot be represented by any symbol or chemical formula.

DIFFERENCES BETWEEN COMPOUNDS AND MIXTURES

To understand the differences between mixtures and compounds, let us do the following activity.

Activity 8

Comparison of a mixture of iron and sulphur and their compound iron sulphide

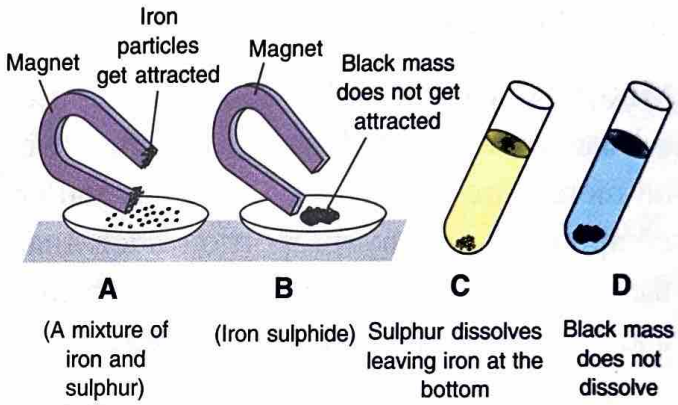
[To be demonstrated by the teacher]

Mix some iron filings and some sulphur thoroughly in a mortar and place some portion of it on a watch glass. We shall call it Sample **A**. Now take the remaining portion of the sample in a test tube and heat the bottom of the test tube. The black mass formed is put on another watch glass. We shall call it sample **B**.

(a) Bring a magnet over sample A and sample B. What do you observe ?

In sample A, iron is attracted by the magnet and sulphur remains on the watch glass, whereas in sample B the magnet has no effect on iron.

(b) Put portions of sample A and sample B in carbon disulphide taken in two separate test tubes. What do you notice ?



In sample A, sulphur dissolves in carbon disulphide but iron does not; while the black mass (sample B) settles down in the test tube without dissolving.

Therefore it is clear that in sample A, iron and sulphur particles retain their individual properties. However in sample B, on heating, iron chemically combines with sulphur forming iron sulphide. Therefore in iron sulphide, iron and sulphur particles do not exist separately as such. They lose their individual properties.

The above experiment confirms that there are two ways in which different kinds of matter can be made to combine.

(i) They can be merely brought together in any proportion and mixed to form a **mixture**.

(ii) They may be heated or allowed to react chemically to form a **compound**.

Types of Mixtures

• On the basis of distribution of components

We come across different kinds of mixtures in our day-to-day life. Some of them quite obviously look like mixtures, such as sand and water.

But there are also mixtures in which we cannot see the components, such as a sugar solution.

Thus mixtures are divided into *two* main types :

1. Homogeneous mixtures
2. Heterogeneous mixtures.

Homogeneous Mixtures

A mixture in which its constituents are uniformly distributed throughout its volume and cannot be recognized separately is called a homogeneous mixture.

Example : A salt solution is a mixture made up of salt and water, but we cannot see salt particles and water separately. If we add one, two or three spoonfuls of salt in a glass of water and stir it, in each case, the solution formed is homogeneous but the proportion of salt and water is not the same.

Some other examples of homogeneous mixtures are sugar solution, water and alcohol, vinegar, air, etc.

A mixture of two or more gases is always considered a homogeneous mixture because all gases mix with each other in all proportions.

Fig. 3.1 : Mixtures that look like pure substances

Table 3.4 Differences between compounds and mixtures

Compound	*Mixture*
1. A compound is a pure substance.	1. A mixture is an impure substance.
2. Compounds are always homogeneous.	2. Mixtures may be homogeneous or heterogeneous.
3. A compound has a fixed composition, *i.e.* it is formed when two or more pure substances chemically combine in a definite ratio by mass.	3. A mixture has no fixed composition, *i.e.* it is formed by mixing two or more substances in any ratio without any chemical reaction.
4. Formation of a compound involves change in energy.	4. Formation of a mixture does not involve any change in energy.
5. Compounds have specific properties which differ from the properties of their constituent elements.	5. Mixtures do not have any specific set of properties. They exhibit properties of their components.
6. Elements of compounds can be separated only by complex chemical processes.	6. Components of mixtures can be separated by simple physical methods.
7. Compounds have definite molecular formulae.	7. Mixtures have no definite formulae. *e.g.* air.

Heterogeneous Mixtures

A mixture in which the components or constituents are not uniformly distributed throughout its volume and can be easily recognized separately is called a heterogeneous mixture.

Example : Soil is a heterogeneous mixture of many elements and compounds. Its composition changes from place to place. That is why we find different substances in the soil at different places.

Other examples of heterogeneous mixtures are water and petrol, sand and stones, sulphur and iron filings, rice and lentils, chalk powder and water, kerosene and water, etc.

Most of the mixtures of two or more solids is considered as a heterogeneous mixture, whether you mix and grind them thoroughly or not [exception : alloys].

*Many substances which appear to be pure are actually mixtures. **Examples** : tap water, milk, honey, **fruit juice**, ice cream, ink, medicines, butter, **brass, bronze**, etc.*

Do You Know ?

1. **Tap water** is not **pure** because it **contains** small amounts of dissolved salts and air in it which we cannot see. **They add** taste to water. (*Pure water has no taste*).
2. **Ink** is a mixture of many **dyes** depending on required shades and applications.
2. **Milk** is made of fats, **carbohydrates**, proteins, salts, vitamins and water **present** in different proportions. Cream **floats** on milk when it is cooled after boiling.
3. **Honey** contains sugar and a **number** of other substances.
4. **Fruit juice** is a mixture of sugar, **salts** and other organic compounds.
5. **Medicines** are made by mixing **different** pure substances in different **proportions.**
6. **Alloys** are homogeneous mixtures of two or more metals or **non-metals** prepared by mixing them in **molten state**, *e.g.*
   (a) Brass is an alloy of copper and zinc.
   (b) Bronze is an alloy of copper, **tin** and zinc.

Table 3.5 Types of mixtures and their examples

S. No.	States of components	Types of mixtures	Examples
(i)	Solid + solid	Heterogeneous	Sand and sugar, sand and salt, sand and stone, *etc.*
		Homogeneous	Brass, bronze, stainless steel, (all alloys), *etc.*
(ii)	Solid + liquid	Heterogeneous	Sand and water, charcoal and water, *etc.*
		Homogeneous	Sugar in water, salt in water, iodine in alcohol, *etc.*
(iii)	Liquid + liquid	Heterogeneous	Oil in water.
		Homogeneous	Alcohol and water, acetone and water, *etc.*
(iv)	Gas + liquid	Homogeneous	Aerated drinks like cold drinks, beer, *etc.*
(v)	Gas + gas	Homogeneous	Pure air
(vi)	Solid + gas	Heterogeneous	Smoke (contains soot particles) in air.

- **On the basis of states of components**

Mixtures may exist in any of the three states of matter, i.e. solid, liquid or gas, depending upon the physical states of its components.

NEED FOR THE SEPARATION OF COMPONENTS OF MIXTURES

We need many substances for purposeful uses in our daily life. But most of these substances are available in the form of mixtures. These mixtures contain unwanted substances which degrade their properties.

Examples : **(1)** It is necessary to remove small stones and husk, etc. from cereals like rice, wheat or pulses before cooking because they are harmful.

(2) Common salt needs to be separated from sea water, for its purposeful uses, as it is an important nutrient and adds taste to our food.

The purpose of separating the constituents of a mixture are to :

(i) get useful substances.

(ii) remove undesirable and harmful substances.

(iii) get completely pure substances for preparing other useful substances. *e.g.* Water is required in its purest form to prepare medicines, in laboratories for preparing solutions, in car batteries, etc. Therefore, all the impurities present in water need to be removed.

METHOD OF SEPARATION

The process by which constituents of a mixture are set apart from one another to get pure substances is called separation.

The principle of separation depends upon the

- type of mixture
- characteristic properties of mixture, such as size, shape, colour, density, melting point, boiling point, solubility, ability to sublime, volatility, magnetic nature, etc.

Thus for different types of mixtures, different methods are applied to separate the components. Some of these methods are described below :

(A) Separation of solid-solid mixtures

On the basis of their types, solid-solid mixtures can be separated by handpicking, winnowing, magnetic separation, gravity separation, sublimation, solvent extraction methods, etc.

Hand-picking : This method of separation can be used when the quantity of a mixture is small and the substance to be separated forms a small portion of the mixture. The substance should be large enough in size to be recognized by naked eyes and picked out by hand. Small stones are picked out from rice, pulses and spices by this method.

Winnowing : The process of separating light solids from heavier ones with the help of wind is called winnowing. This technique is generally used by farmers.

Example : In a mixture of rice and husk, rice is separated from husk by winnowing.

Magnetic separation : This method is used when one of the components of the mixture is magnetic in nature *i.e.* gets attracted towards a magnet. Iron, being magnetic,

gets attracted towards a magnet and hence can be separated from other non-magnetic substances.

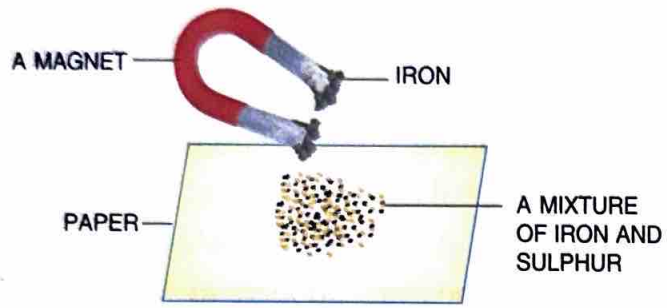


Fig. 3.2 : Separation by a magnet

Example : Mixtures of iron and sulphur, iron and sand, *etc.* are separated by this method.

Gravity separation method : This method is used for the mixtures in which an insoluble solid component is much heavier than water and the other insoluble component is much lighter than water.

Example : If a mixture of sand and sawdust is put in water, sawdust being lighter floats while sand settles down. Now sawdust with water is slowly decanted and transferred to another container leaving the sand at the bottom and then filtered to separate the sawdust.

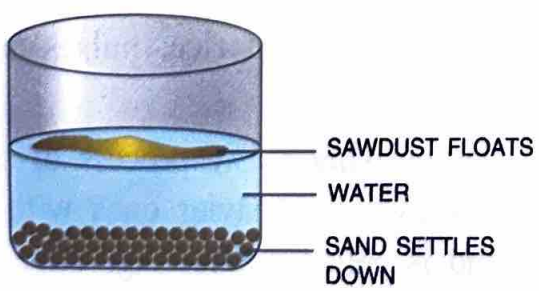


Fig. 3.3 : Separation by gravitational method

Sublimation : *The process in which a solid substance directly changes into its vapour state on heating without passing through its liquid state is called* ***sublimation.*** This method is used for solid mixtures in which one of the components can *sublime* on heating. The solid which sublimes changes into vapour, while the other one is left behind.

Camphor, naphthalene, iodine, and ammonium chloride are some solids that undergo sublimation.

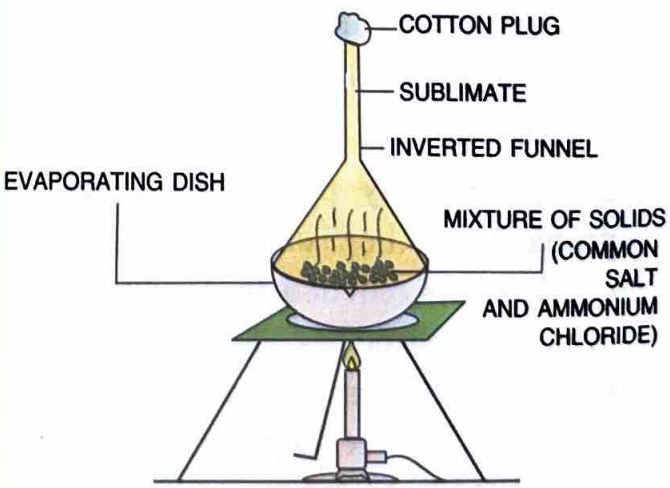
However on cooling, the vapour again turns into a solid.

Example : Mixture of sand and iodine, common salt and ammonium chloride, *etc.* are separated by sublimation.

Activity 9

To separate common salt and ammonium chloride

Take a mixture of common salt and ammonium chloride and place it in a dish and cover with an inverted funnel and heat it. On heating, ammonium chloride changes into vapour, which condenses into a solid along the neck of the funnel (from where it may be scraped off), whereas common salt is left behind in the dish.



Sublimation : Separating a mixture of common salt and ammonium chloride

Note : The stem of the inverted funnel is plugged with cotton to prevent the ammonium chloride vapour from escaping.

Solvent Extraction Method : The process by which a soluble solid component of a mixture is separated from an insoluble solid component by using a suitable solvent is called solvent extraction method.

The soluble solid gets dissolved in the solvent, while the insoluble solid is left behind. The soluble component is then recovered by evaporation.

Example : A mixture of sand and salt can be separated by this method. Salt gets dissolved in water while sand settles down in the container. The salt solution is then filtered. Salt is recovered from the solution by evaporation.

(B) Separation of solid-liquid mixtures

These mixtures can be **homogeneous** (a sugar solution) or **heterogeneous** (a mixture of sand and water). Different methods are used depending upon the type of mixture.

- *The methods used to separate components of heterogeneous mixtures, i.e. in which the solid component is **insoluble** in the liquid component are :*

1. Sedimentation and decantation : *The settling down of suspended, insoluble, heavy solid particles in a solid-liquid mixture when left undisturbed is called **sedimentation**.*

*The solid which settles at the bottom is called the **sediment** while the clear liquid above it is called the **supernatant liquid**.*

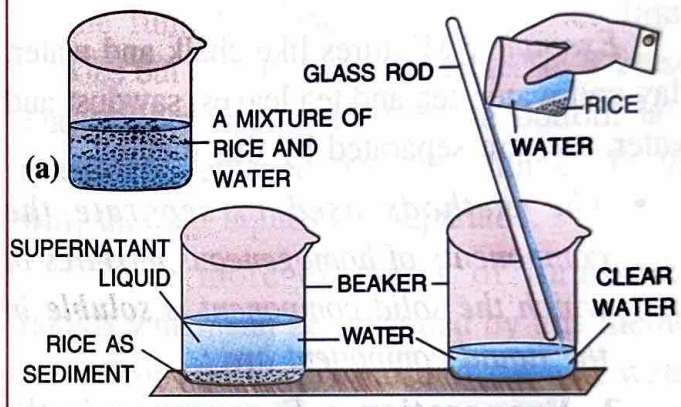
*The process of pouring out the clear liquid, without disturbing the **sediment**, is called **decantation**.*

This method is used when the solid component is insoluble and heavier than the liquid component.

Example : A mixture of sand and water, pulses and water, etc.

Activity 10

Take some rice and water in a beaker and stir it. Now allow the mixture to stand for some time. You will see that the rice settles at the bottom of the beaker. This is called *sedimentation*. Now pour out water gently into another vessel without disturbing the rice. This is called decantation.



2. Filtration : *The process of separating insoluble solid particles from a liquid by allowing it to pass through a filter is called filtration.*

The filter allows the liquid to pass through it but not the solid. The insoluble solid left on the filter is called the ***residue***, while the liquid which passes through the filter is called the *filtrate*.

The substances that can be used as filters are a layer of sand, charcoal, cotton, glass wool, unglazed porcelain, filter paper, *etc*. Even the strainer that we use to separate liquid tea from tea leaves is a filter.

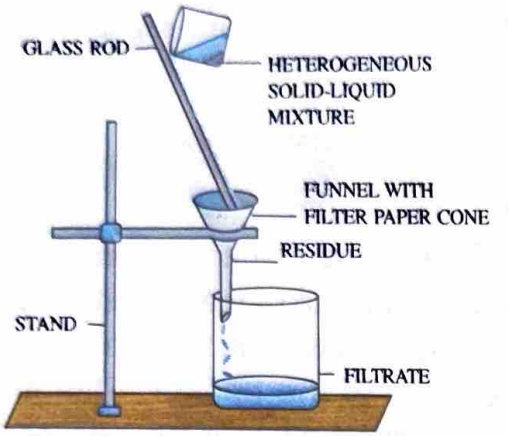


Fig. 3.4 : Process of filtration to separate insoluble suspension from water (on small scale)

Example : Mixtures like chalk and water, clay and water, tea and tea leaves, sawdust and water, *etc.* are separated by this method.

- *The methods used to separate the components of homogeneous mixtures in which the solid component is* ***soluble*** *in the liquid component are :*

3. Evaporation : *Evaporation is the process of converting a liquid into its vapour state, either by exposing it to air or by heating.*

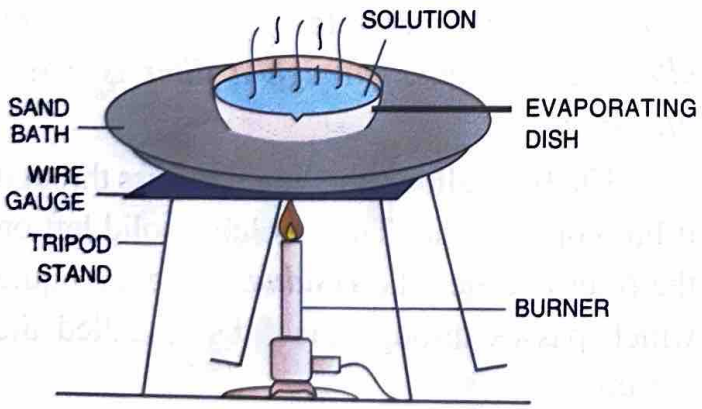


Fig. 3.5 : Evaporation

In this method, the liquid component of the mixture escapes in the form of vapour leaving behind the solid component.

This method is used to separate common salt from sea water. Sea water is collected in shallow beds and allowed to evaporate in the sun. As a result, only salt is left behind.

4. Distillation : *Distillation is the method of getting a pure liquid from a solution by evaporating and then condensing the vapours.*

When the solution is heated, the liquid component of the mixture evaporates in the form of vapour. The vapours are then condensed back into the liquid form which is very pure and is called as the **distillate.**

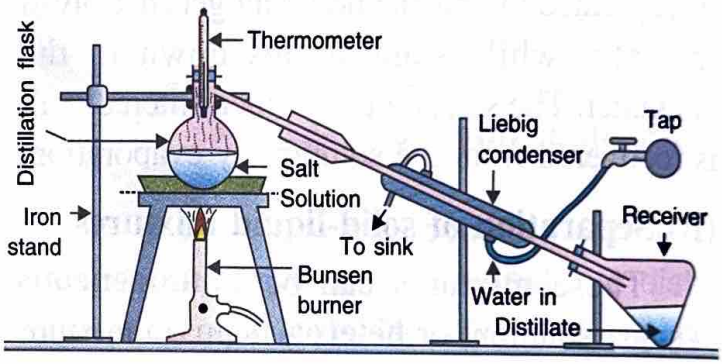


Fig. 3.6 : Distillation

Tap water, which is a mixture containing dissolved salts, is purified by *distillation*. The pure water so obtained is called the *distilled water.* It is used by doctors for preparing medicines, by chemists for making solutions and in industries for various purposes.

The advantage of this process is that both the components of the solid-liquid mixture are obtained.

(C) Separation of liquid-liquid mixtures

Liquid-liquid mixtures are of two types :

(1) Homogeneous liquid-liquid mixtures : Liquids which dissolve in each other completely in all proportions are called **miscible liquids**. *Example* – alcohol is miscible with water.

(2) Heterogeneous liquid-liquid mixtures : Liquids which do not dissolve in each other are called **immiscible liquids**. *Example* – oils are immiscible with water.

5. Fractional distillation : The process of distillation used to separate the components of a homogeneous liquid-liquid mixture on the basis of the difference in their boiling points is called fractional distillation. The difference in boiling points of the different liquids must be 25°C or more.

Example : A mixture of water and alcohol. Alcohol boils at a lower temperature than water. The vapours of alcohol are collected and cooled while water is left behind in the flask.

For the separation of components, a fractionating column is fixed over the distilling flask, so that if water vapours move up along with alcohol, it should condense and come back to the flask.

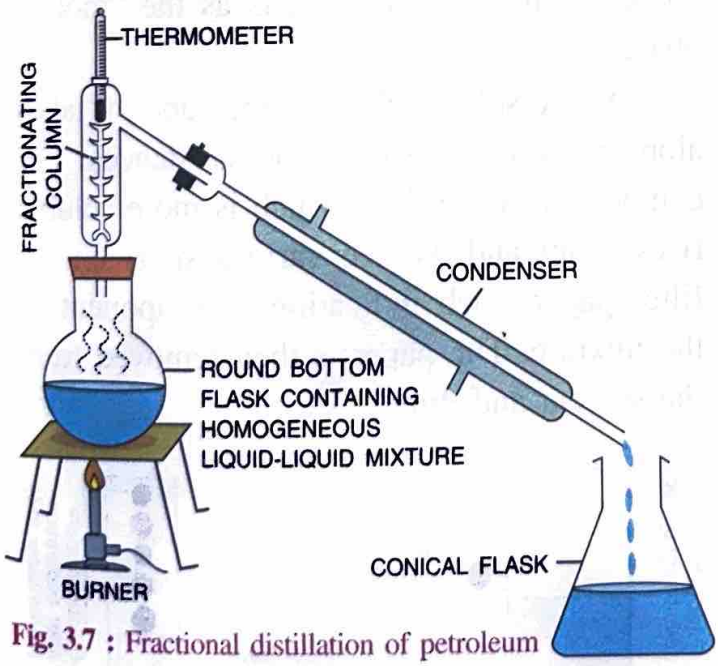


Fig. 3.7 : Fractional distillation of petroleum

Petrol, kerosene, diesel, *etc.* are obtained from crude petroleum oil in a similar way.

6. By separating funnel : It is a simple device used to separate the components of a liquid-liquid heterogeneous mixture, *i.e.* immiscible liquids, when one of the liquid components is heavier than the other.

Example : Kerosene oil and water. The mixture is placed in a separating funnel and allowed to stand for sometime. The components form two clear layers. Water being heavier forms the lower layer and oil being lighter forms the upper layer. When the stopper of the funnel is opened, the heavier liquid trickles out slowly and is collected in a vessel. The stopper is closed when the bottom layer is entirely removed from the funnel. In this way, the two liquids are separated.

Some more examples of immiscible liquids which can be separated by this method are petrol and water; mustard oil and water; carbon tetrachloride and water, etc.

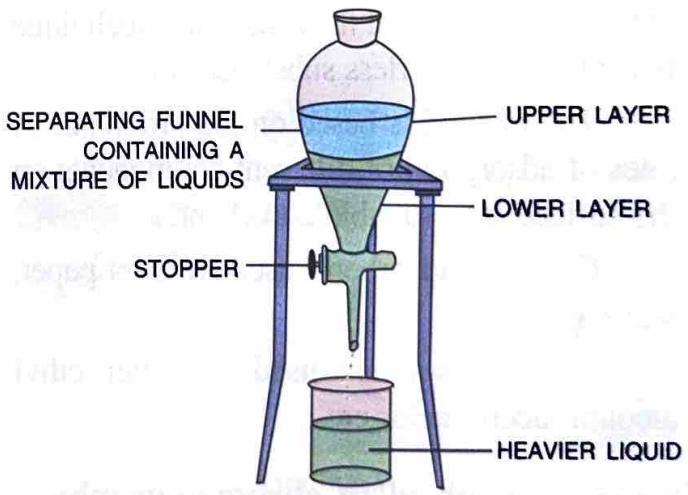


Fig. 3.8 : Separation of immiscible liquids using separating funnel

(D) Separation of a gas - liquid mixture:

A mixture of gas in liquid can be separated by heating. Dissolved gas escapes from the liquid on heating.

Example : Drinking water contains air dissolved in it. When it is boiled, air escapes and so the boiled water becomes tasteless.

(E) Modern techniques :

Chromatography : This is one of the latest techniques to separate the coloured components of a complex mixture when all the components are very similar in their properties.

Example : Components of ink are separated by this method. Ink is a mixture of different dyes, which are separated by chromatography because some of the dyes are less soluble and some are more soluble in a solvent.

The process of separating different dissolved constituents of a mixture by their adsorption* on an appropriate material is called *chromatography*.

The name "chromatography" means colour writing. It is named so, because earlier it was used to separate mixtures containing coloured components only, but these days this technique is applied to colourless substances too.

The method is based on the difference in rates of adsorption of different components on the surface of a suitable adsorbent.

Common adsorbents used are filter paper, silica gel, etc.

Common solvents used are water, ethyl alcohol, acetic acid, etc.

Principle involved in chromatography

Chromatography separates the components of a mixture on the basis of differences between two phases, one of which is stationary while the other is mobile.

The stationary phase is made up of common adsorbents like filter paper, silica gel, etc.

The mobile phase is made up of common solvents like water, acetic acid, ethyl alcohol, etc.

The components of the mixture dissolved in the mobile phase get adsorbed on the stationary phase at different speeds, hence they get separated.

The simplest type of chromatography is "**Paper chromatography**".

In this method, a special type of paper called chromatographic paper (called Whatman filter paper) or ordinary filter paper is taken. A line is drawn with the pencil near the bottom edge of the paper. A drop of the mixture is placed on the filter paper above this line. The paper is then dipped in a solvent, taken in a beaker, such that the line drawn on the paper is above the level of the solvent.

The filter paper acts as the "stationary phase" while the solvent acts as the "mobile phase".

As the solvent rises on the paper, it takes along with it the drop of the substances. The component of the drop which is more soluble rises faster and we see various spots on the filter paper, each indicating a component of the mixture. The paper is then removed from the solvent and dried.

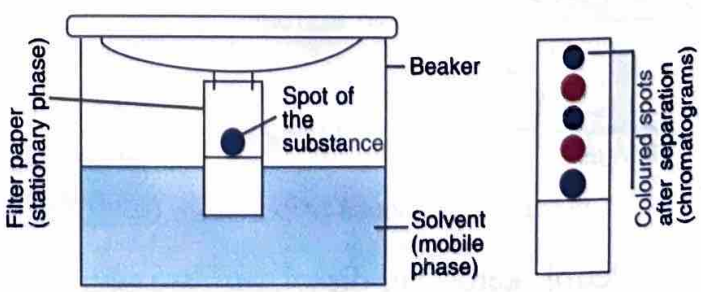


Fig. 3.9 : Paper chromatography

** Adsorption – It is a process by which particles of a substance get adhered (attached) to the surface of a solid material.*

Advantages of chromatography

- A very small quantity of the substance can be separated.
- Components with very similar physical and chemical properties can be separated.
- It identifies the different constituents of a mixture.

Uses of chromatography

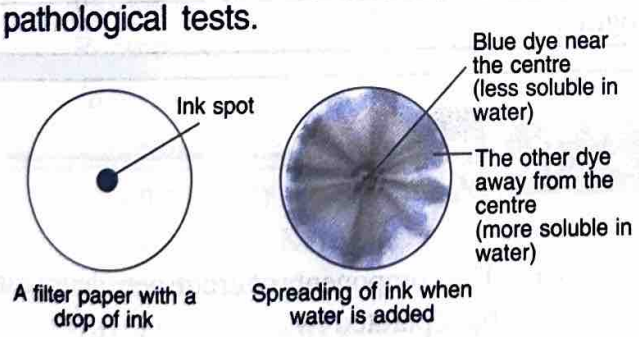
It can be used to separate :

- pigments from natural colours
- drugs from blood (pathological tests)
- colours from a dye.

Activity 11

Take a filter paper. Put a drop of ink on the centre of the paper and then slowly add some drops of water on the same point with the help of a dropper. You will observe that the blot of ink spreads out into different coloured rings. Each ring corresponds to a dye. The more soluble dye moves farthest while less soluble remains near the centre.

Chemists often use this technique for testing purity and doctors use it for pathological tests.



Separation of constituents of the mixtures with more than two constituents :

Sand, sawdust and salt : The mixture is taken in a glass beaker and water is added to it. The beaker is then allowed to stand undisturbed for some time. Salt dissolves in water, forming a salt solution, because it is soluble in water. Sawdust being lighter floats on the surface of water while sand being heavier settles down. Now salt solution along with sawdust is decanted slowly on the filter paper fixed in a funnel such that sand is left in the beaker as sediment. Salt solution passes through the filter paper while sawdust remains on it. The salt solution is evaporated to get salt from the water. In this way, all the components get separated.

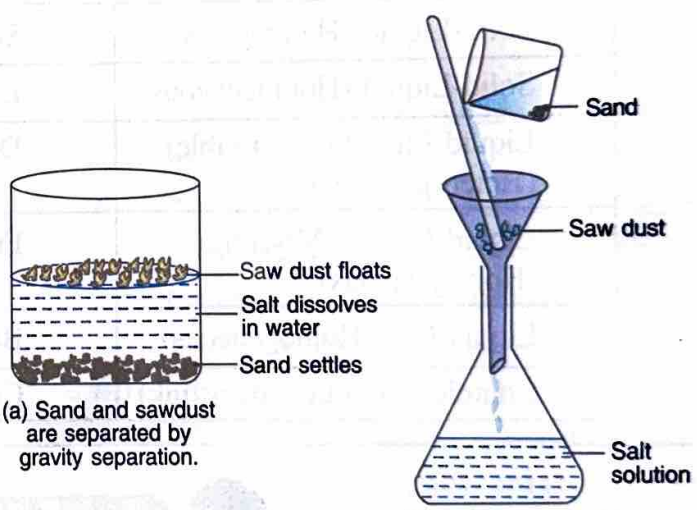


Fig. 3.10 : Separating a mixture of sand, sawdust and salt

RECAPITULATION

- Pure substances are broadly classified into elements and compounds. Mixtures are impure substances.
- Elements are made up of the same kind of atoms.
- Compounds are made up of different kinds of atoms combined chemically together in a fixed ratio.
- Atoms of elements are represented by symbols.
- Molecules of elements and compounds are represented by formulae formed by symbols of atoms. The formulae indicate the number and types of atoms present in the molecules.
- Of 118 elements known to mankind, millions of compounds have been formed.
- A mixture is an impure substance made up of different kinds of elements and compounds held loosely.
- The constituents of mixtures retain their individual properties and can be separated by physical methods.
- The method of separation of constituents of a mixture depends upon the characteristic properties of its constituent components.
- The table below summarizes the different methods used for separating the various types of mixtures :

Types of mixtures	*Methods of separation*
Solid-Solid (Heterogeneous)	Hand-picking, winnowing, sieving, sublimation, magnetic separation, gravity separation, solvent extraction method.
Solid-Liquid (Heterogeneous)	Sedimentation and decantation, filtration.
Solid-Liquid (Homogeneous)	Evaporation, distillation.
Liquid-Liquid (Immiscible) (Heterogeneous)	Decantation, using a separating funnel.
Liquid-Liquid (Miscible) (Homogeneous)	Fractional distillation
Liquid-Gas (Homogeneous)	Boiling
Complex mixtures (dyes, ink)	Chromatography

EXERCISE — II

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(*i*) A pure liquid is obtained from a solution by :

(*a*) evaporation (*b*) distillation
(*c*) filteration (*d*) sublimation

(*ii*) The components of crude petroleum can be separated by :

(*a*) distillation (*b*) evaporation
(*c*) filtration
(*d*) fractional distillation

(*iii*) An example of a homogeneous mixture is:

(*a*) tap water (*b*) distilled water
(*c*) sand and water (*d*) water and oil

(iv) In chromatography, the filter paper is :
(a) stationary phase (b) mobile phase
(c) mixture (d) none of the above

(v) A set of mixtures is :
(a) ink, honey, ice cream, milk
(b) tap water, gold, common salt, alloy
(c) milk, brass, silver, honey
(d) butter, petroleum, tap water, iron

(vi) The substance which can be separated by sublimation from a mixture is :
(a) salt (b) camphor
(c) petrol (d) sand

(vii) A solid-solid mixture is :
(a) Salt and water
(b) Mustard oil and water
(c) Honey (d) Alloys

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

(i) **Assertion (A) :** The molecules of elements and compounds are represented by formulae, formed using atomic symbols.
Reason (R) : Mixtures are impure substances.
(a) (1) (b) (2) (c) (3) (d) (4)

(ii) **Assertion (A) :** Kerosene oil and water can be separated by a separating funnel.
Reason (R) : Kerosene oil and water are immiscible liquids.
(a) (1) (b) (2) (c) (3) (d) (4)

(iii) **Assertion (A) :** Mixtures are either homogeneous or heterogeneous in nature.
Reason (R) : A salt solution in water is a heterogeneous mixture.
(a) (1) (b) (2) (c) (3) (d) (4)

(iv) **Assertion (A) :** When the first letter of more than one element is the same, their symbol is denoted by the first two letters of their name.
Reason (R) : The symbol of calcium is Ca and that of chlorine is Cl.
(a) (1) (b) (2) (c) (3) (d) (4)

3. Fill in the blanks :
(a) ..... are made up of same kind of atoms.
(b) ..... and ..... are pure substances.
(c) In a ....., the substances are not combined chemically.
(d) Clay is separated from water by a method called ..... .
(e) ..... is a process to obtain a very pure form of a solid dissolved in a liquid.
(f) Camphor and ammonium chloride can ..... .

4. Write true or false.
(a) The components of mixtures retain their individual properties. .....
(b) All elements are made up of the same kind of atoms. .....
(c) Compounds can be homogeneous or heterogeneous. .....
(d) Air can be represented by a chemical formula. .....
(e) In a homogeneous mixture, the components are uniformly distributed throughout its bulk. .....

5. Match the following :

(a) Ink	(i) Sublimation
(b) Crude petroleum	(ii) Distillation
(c) Carbon tetra-chloride and water	(iii) Fractional distillation
(d) Tap water	(iv) Chromatography
(e) Sand and ammonium chloride	(vi) Separating funnel

6. Give an appropriate term for the following :
   (a) The solid particles which remain on the filter paper after filtration.
   (b) The liquid which evaporates and then condenses during the process of distillation.
   (c) The process of transferring the clean liquid after the solid settles at the bottom of the container.
   (d) The process by which two miscible liquids are separated.

7. Name the process by which the components of the following mixtures can be separated.
   (a) Iron and sulphur
   (b) Ammonium chloride and sand
   (c) Common salt from sea water
   (d) Chaff and grain
   (e) Water and mustard oil
   (f) Sugar and water

B. SHORT ANSWER TYPE QUESTIONS

1. Give *one* example for each of the following :
   (a) A solid – solid heterogeneous mixture
   (b) A solid – liquid heterogeneous mixture
   (c) A liquid – liquid heterogeneous mixture
   (d) A solid – solid homogeneous mixture
   (e) A solid – liquid homogeneous mixture
   (f) A liquid – liquid homogeneous mixture

2. Define :
   (a) Evaporation (b) Filtration
   (c) Sublimation (d) Distillation
   (e) Miscible liquids (f) Immiscible liquids

3. How is distillation more advantageous than evaporation ?

4. Differentiate between a homogeneous and a heterogeneous mixture.

C. LONG ANSWER TYPE QUESTIONS

1. State four differences between compounds and mixtures.

2. What are the characteristic properties of a pure substance ? Why do we need them ?

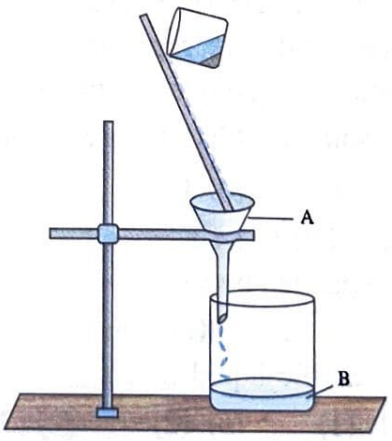
3. How will you separate a mixture of common salt and chalk powder ? Explain.

4. (a) What is chromatography ?
   (b) Why is it named so ?
   (c) What are the advantages of chromatography ?
   (d) Name the simplest type of chromatography?
   (e) On what principle is this method based ?
   (f) What is meant by stationary phase and mobile phase in chromatography ?

5. On what principle are the following methods of separation based ? Give one example of a mixture for each of the methods mentioned in which they are used.
   (a) Sublimation (b) Filtration
   (c) Sedimentation and decantation
   (d) Solvent extraction method
   (e) Magnetic separation
   (f) By using a separating funnel
   (g) Fractional distillation

6. Draw a well-labelled diagram to show the separation of a mixture of kerosene oil and water.

7. Study the figure given below and answer the following questions :



   (a) Label A and B.
   (b) Name the process shown.
   (c) For what type of mixtures is this method used ?
   (d) What is the substance left on the filter paper called ?

D. CASE STUDY BASED QUESTION

A chemistry teacher had taught the concept of elements, compounds and mixtures to her students in the class.

Aaliya, one of the students, was watching her father cook food one day after she was taught about homogeneous and heterogeneous mixtures.He mixed flour, water and common salt and kneaded them into a dough to prepare breads, all of which tasted the same throughout. She also observed her mother serving fruit cream after the dinner which had different kinds of fruits mixed with cream.

(a) What kind of substance is the dough ?

(b) Give the chemical name of common salt. Name a method by which common salt is separated from its salt solution.

(c) Why is fruit cream a heterogeneous mixture ?

(d) Define an 'element'. Name the elements from which sugar is made up of ?

Answers

Exercise : 3 (I) — Objective type questions

1. *(i)–(b), (ii)–(c), (iii)–(b), (iv)–(b), (v)–(a)*

Exercise : 3 (II) — Objective type questions

1. *(i)–(b), (ii)–(d), (iii)–(a), (iv)–(a), (v)–(a), (vi)–(b), (vii)–(d)*

2. *(i)–(b)* : Both statements are true, but the reason does not explain the assertion.

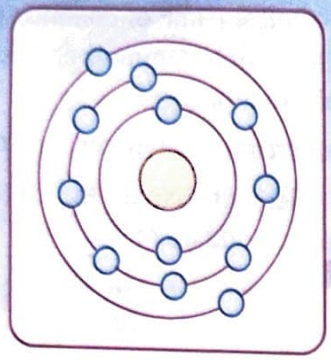
(ii)–(a) : The reason correctly explains the assertion. Kerosene oil and water are immiscible, therefore it is possible to separate them using a separating funnel.

(iii)–(c) : Assertion is true but the reason is false. Salt solution in water is a homogeneous mixture.

(iv)–(d) : Assertion is false because the symbol of an element can be represented by *any* two letters, taken either from the English or Latin name. The reason is true.

Atomic Structure

4



THEME

This theme will enable children to understand that every matter is made up of tiny particles known as atoms and molecules. Molecules are also made up of atoms. Hence atoms are the building blocks of matter. The physical and chemical properties of matter are governed by atoms. Therefore, the knowledge of the concepts of atoms of elements, molecules of elements and compounds and radicals of compounds is necessary to understand different processes and principles of Chemistry.

LEARNING OBJECTIVES

The children will be able to :

- define atom, molecule and radical;
- discuss the significance of valency of elements and radicals;
- define valency in terms of number of hydrogen atoms combined or replaced by one atom of the element;
- apply the definition based on hydrogen atom to find out the valency of other elements and radicals;
- correlate the valency of the elements with group number of periodic table.

INTRODUCTION

Now you know that all substances are made up of matter. There are different kinds of matter and every matter is made up of very tiny particles called **atoms**. Atoms combine to form molecules. Atoms and molecules are too small to be seen by naked eye. They can only be seen through a powerful microscope. Let us know about atoms and molecules in detail.

AN ATOM

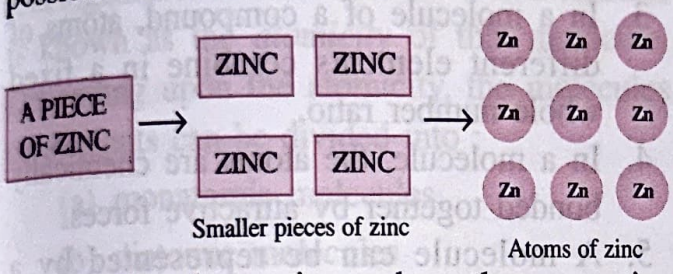
The word **atom** comes from the word **'atomos'** meaning 'indivisible' coined by a Greek philosopher Democritus (460-361 B.C.). He forwarded the idea that the universe was made up of tiny indivisible particles called atoms. Also, according to Maharishi Kannada, an Indian philosopher, matter consists of indestructible particles called "Paramanus". In 1808 John Dalton, an English scientist, suggested that an atom is the basic unit of matter.

*An atom is the smallest particle of an element that exhibits all the properties of that element. It **may or may not exist** independently but takes part in every chemical reaction.*

Example : Take a small piece of zinc and crush it into smaller pieces. All these pieces show properties of

zinc. On grinding crushed pieces further, they break up into very fine particles which still show the properties of zinc. But, there comes a stage when the particles cannot be further subdivided into particles exhibiting properties of zinc. These indivisible particles are the atoms of zinc.

"In other words, atom is the smallest possible unit of an element".



All the above pieces show the properties of element zinc.

Do You Know ?

Atoms are so small that it would take millions of them, just to cover a full stop.

The main characteristics of atoms as suggested by John Dalton are :

- An atom is the smallest particle of matter which cannot be divided further into smaller particles.
- Atoms of the same element are all identical but they differ from the atoms of other elements.
- An atom of an element exhibits all the properties of that element. Different elements differ in their properties.
- Atoms can neither be created nor be destroyed.
- Atoms may or may not have independent existence but they can take part in chemical reactions.

Earlier it was thought that atoms of an element cannot be divided further, but studies in the early twentieth century showed that an atom is itself made up of even smaller particles called *fundamental or subatomic particles*. They are:

• Electrons, • Protons and • Neutrons.

Electrons : *Electrons are the negatively charged particles in an atom with one unit negative charge and negligible mass. The mass of an electron is 1/1837 times the mass of one hydrogen atom.*

Protons : *Protons are the positively charged particles present in an atom with one unit positive charge and one unit mass.*

Neutrons : *Neutrons are the particles with no electrical charge, so they are neutral but have one unit mass.*

Table 4.1 : Symbol, charge and mass of the three fundamental particles of an atom.

Particle	Symbol	Charge	Charge in Coulombs	Mass in amu	Mass in kg	Location in an atom
Electron	${}_{-1}e^0$ or e^-	–1 unit	1.602×10^{-19} C	negligible	9.1×10^{-31} kg	In orbits outside the nucleus
Proton	${}_{+1}p^1$ or p^+	+1 unit	1.602×10^{-19} C	1 amu	1.6×10^{-27} kg	Nucleus
Neutron	${}_0n^1$ or n^0	0	0	1 amu	1.6×10^{-27} kg	Nucleus

The subscripts on e, p and n as -1, $+1$ and 0 represent the charge while the superscript 0, 1, 1 represent the mass. The mass of an electron is very less hence considered to be negligible.

Atoms of different elements have different numbers of electrons, protons and neutrons. The number of protons and electrons are always equal in a neutral atom.

In an atom, the protons and neutrons are present in the centrally located part called the **nucleus**, while electrons are present outside the nucleus in circular paths called **orbits** or **shells**.

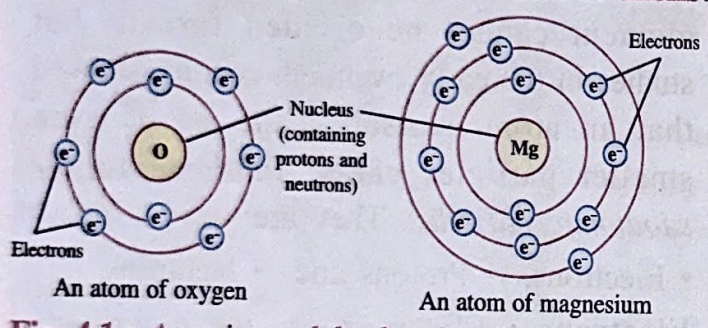


Fig. 4.1 : Atomic models showing the arrangement of electrons, protons and neutrons

Note : • The number of protons in the nucleus of an atom of an element is called its **atomic number**.

• The sum of the number of protons and neutrons in the nucleus of an atom of an element is called its **mass number**.

[Atomic structure will be taught in detail in higher classes].

A MOLECULE

"A molecule is the smallest particle of a pure substance (element or compound) which has independent existence. It is formed when two or more atoms of same element or different elements chemically combine together.

For example : • A molecule of hydrogen is formed when two atoms of hydrogen combine with each other. It is represented by the formula H_2.

• A molecule of water (a compound) is formed when two atoms of hydrogen combine with one atom of oxygen. It is represented by the formula H_2O.

Characteristics of Molecules

1. A molecule is formed by the chemical combination of two or more atoms of the same element or different elements.
2. A molecule exhibits all the properties of that element or compound.
3. In a molecule of a compound, atoms of different elements combine in a fixed whole number ratio.
4. In a molecule, the atoms are chemically bonded together by attractive forces.
5. A molecule can be represented by a chemical formula.

A molecule is the smallest particle of an element or a compund with independent existence, exhibiting all of their properties.

Molecules are of *two* types :

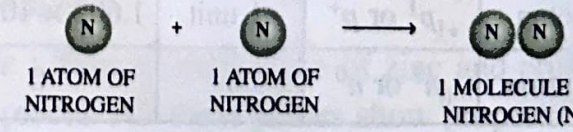
1. Molecules of an element.
2. Molecules of a compound.

Molecules of an element

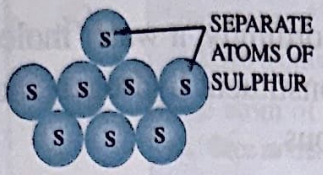
These molecules are made up of two or more atoms of the same element. Molecules of elements like Helium (He), Neon (Ne) are made up of only one atom but atoms of certain elements, like hydrogen (H), oxygen (O), nitrogen (N), chlorine (Cl), *etc.*, cannot exist independently. So they join to form molecules that have independent existence. To form molecules, atoms always join in whole numbers.

Examples :

1. Two atoms of nitrogen join to form one molecule of the element nitrogen.



2. Eight atoms of sulphur join to form a molecule of sulphur.



A SULPHUR MOLECULE (S_8)

Atomicity

The number of atoms of an element that join together to form a molecule of that element is known as the *atomicity* of that molecule. Depending upon the atomicity, the molecules of elements can be divided into :

(a) monatomic molecules

(b) diatomic molecules

(c) triatomic molecules

(d) polyatomic molecules.

Monatomic molecules : The molecules of elements which contain only one atom are said to be **monatomic molecules**. Their atomicity is one. Molecules of inert gases Helium, Neon, Argon, etc. are monatomic and their atoms exist freely under all conditions. Molecules of metals and metalloids are regarded as monatomic.

Examples : Na, Zn, Mg, *etc.*, noble gases : He, Ne, Ar, Xe, *etc.*

Diatomic molecules : The molecules of elements which contain two atoms of the same type are called **diatomic molecules**. Their atomicity is two.

Examples : H_2, O_2, N_2, Cl_2, F_2, *etc.*

Triatomic molecules : The molecules of elements which contain three atoms of the same type are called **triatomic molecules**. Their atomicity is three.

Examples : Ozone (O_3).

Polyatomic molecules : The molecules of elements which contain more than three atoms of the same type are called **polyatomic molecules.**

Examples **:** Phosphorus (P_4), sulphur (S_8), *etc.*

Note : Atomicity refers to the sum total of atoms of same or different elements present in a molecule.

Molecular formula of an element

The molecular formula of an element is the symbolic representation of its molecule. It indicates the number of atoms present in it.

Example : A molecule of chlorine is represented by 'Cl_2' which indicates that two atoms of chlorine join to form one molecule of chlorine. It also shows that the atomicity of chlorine is 2.

Table 4.2 : Names, symbols/formulae, atomicity and state of the molecules of common elements

Name of element	*Symbols/ formulae of molecules*	*Atomicity [Number of atoms in one molecule]*	*State*
Helium	He	1	Gas
Neon	Ne	1	Gas
Sodium	Na	1	Solid
Potassium	K	1	Solid
Hydrogen	H_2	2	Gas
Nitrogen	N_2	2	Gas
Oxygen	O_2	2	Gas
Fluorine	F_2	2	Gas
Chlorine	Cl_2	2	Gas
Bromine	Br_2	2	Liquid
Iodine	I_2	2	Solid
Ozone	O_3	3	Gas
Phosphorus	P_4	4	Solid
Sulphur	S_8	8	Solid

From the above it is clear that :

(i) 'H' represents one atom of hydrogen and 'H_2' represents a molecule of hydrogen.

(ii) '2H' represents two atoms of hydrogen and $2H_2$ represents two molecules of hydrogen.

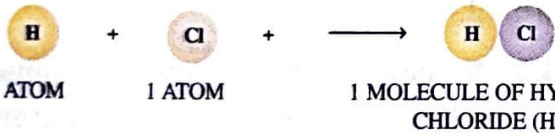
If a numeral is written on the left hand side of a symbol, it represents the number of atoms or molecules.

MOLECULES OF COMPOUNDS

When atoms of two or more elements join together in a fixed ratio by mass, a molecule of a compound is formed.

Examples :

1. One atom of hydrogen and one atom of chlorine combine to form a molecule of hydrogen chloride.



1 ATOM 1 ATOM 1 MOLECULE OF HYDROGEN CHLORIDE (HCL)

2. One atom of carbon combines with two atoms of oxygen to form a carbon dioxide molecule.

1 ATOM 1 ATOM 1 ATOM 1 MOLECULE OF CARBON DIOXIDE (CO_2)

The smallest unit of a compound is its molecule. It exhibits all the properties of that compound.

Every compound has its own specific molecules, which are same in all respects but differ in their properties from the molecules of other compounds. *Example :* Water and sugar molecules are different from each other.

The molecules of a compound are completely different in their properties from the atoms of which they are made.

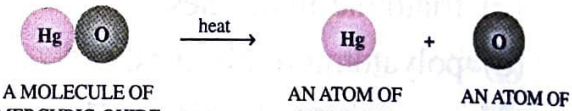
Accordingly, a water molecule is a liquid, but its constituent hydrogen and oxygen atoms are gaseous.

Similarly, carbon dioxide is a gaseous compound but carbon is a solid element.

A molecule of a compound can be broken into its constituent elements using chemical methods.

Examples :

1. Mercuric oxide is a solid compound. When it is heated, it decomposes to give mercury and oxygen, which are elements.



A MOLECULE OF MERCURIC OXIDE AN ATOM OF MERCURY AN ATOM OF OXYGEN

2. When electric current is passed through acidulated water, it ionises to give hydrogen and oxygen gases in the ratio of 2 : 1 [by volume].

From whatever source water is taken and ionised, the ratio of gases hydrogen and oxygen is found to be the same.

[1 MOLECULE OF WATER] [TWO ATOMS OF HYDROGEN AND 1 ATOM OF OXYGEN]

Table 4.3 : Molecules of some compounds and the number of atoms of elements present in them

Name	*Formula*	*No. of atoms of elements*
Sodium chloride (common salt)	NaCl	One atom of sodium and one atom of chlorine
Water	H_2O	Two atoms of hydrogen and one atom of oxygen

Calcium oxide	CaO	One atom of calcium and one atom of oxygen
Sulphur dioxide	SO_2	One atom of sulphur and two atoms of oxygen
Ammonia	NH_3	One atom of nitrogen and three atoms of hydrogen
Sugar (cane sugar)	$C_{12}H_{22}O_{11}$	Twelve atoms of carbon, twenty two atoms of hydrogen and eleven atoms of oxygen

RADICALS

"A radical is an atom of an element or a group of atoms of different elements that behaves as a single unit with a positive or negative charge on it". They are also called ions.

Radicals are of two types :

(i) **Basic radical :** They have positive charge and are also called cations. They are formed by losing electrons. All metallic ions and ammonium ion are basic radicals.

(ii) **Acid radical :** They have negative charge and are also called anions. They are formed by gaining electrons. Most of the non-metallic ions and groups of non-metallic atoms with negative charge are acid radicals.

Ions : Electrically charged atoms or groups of atoms formed by losing or gaining electrons are called ions.

A molecule of a compound is usually made up of two parts. *Example :* Sodium chloride molecule is made up of sodium ion and chloride ion. Sodium ion formed by losing an electron is positively charged while chloride ion formed by gaining an electron is negatively charged represented by the symbols Na^+ and Cl^- respectively. The charge on these radicals show their combining capacity *i.e.* **valency**. These radicals keep their identity in many reactions.

Naming an ion or a radical

The names of simple positive ions (radicals) are derived from the names of the parent element.

Example : A sodium atom (Na) on losing an electron becomes a sodium ion (Na^+).

The names of simple monoatomic negative ions (radicals) are derived from the parent elements by replacing the last few letters of the name by a suffix-ide.

Example : An atom of chlorine (Cl) upon gaining an electron becomes a chloride ion (Cl^-).

$$Cl \longrightarrow Cl^-$$

$$\text{(chlorine atom)} \xrightarrow[-\text{ne}]{+\text{ide}} \text{(chloride ion)}$$

The names of polyatomic anions are derived by adding the suffix – ite, ate, ide, etc.

Table 4.4 : Representation of some positive ions (radicals) and their charge (valency)

Name of radical	*Representation*	*Charge/Valency*
Hydrogen	H^+	+1
Sodium	Na^+	+1
Potassium	K^+	+1
Silver	Ag^+	+1
Magnesium	Mg^{2+}	+2
Calcium	Ca^{2+}	+2
Zinc	Zn^{2+}	+2
Iron (II)	Fe^{2+}	+2
Copper (II)	Cu^{2+}	+2
Iron (III)	Fe^{3+}	+3
Aluminium	Al^{3+}	+3
Gold	Au^{3+}	+3
Tin (IV)	Sn^{4+}	+4
Platinum (IV)	Pt^{4+}	+4

Table 4.5 : Representation of some negative ions (radicals) and their charge (valency)

Name of radical	*Representation*	*Charge/Valency*
Fluoride	F^-	–1
Chloride	Cl^-	–1
Bromide	Br^-	–1
Iodide	I^-	–1
Hydride	H^-	–1
Oxide	O^{2-}	–2
Sulphide	S^{2-}	–2
Nitride	N^{3-}	–3
Phosphide	P^{3-}	–3

Table 4.6 : Representation of some polyatomic ions (radicals) and their charge (valency)

Name of radical	*Representation*	*Charge/Valency*
Ammonium ion	NH_4^+	+1
Hydroxide ion	OH^-	–1
Nitrate ion	NO_3^-	–1
Nitrite ion	NO_2^-	–1
Bisulphate ion	HSO_4^-	–1
Bisulphite ion	HSO_3^-	–1
Bicarbonate ion	HCO_3^-	–1
Acetate ion	CH_3COO^-	–1
Carbonate ion	CO_3^{2-}	–2
Sulphate ion	SO_4^{2-}	–2
Sulphite ion	SO_3^{2-}	–2
Dichromate ion	$Cr_2O_7^{2-}$	–2
Phosphate ion	PO_4^{3-}	–3

Table 4.7 : Differences between atoms, molecules and radicals

Atoms	Molecules	Radicals
1. An atom is the smallest particle of an element (matter) which may or may not have independent existence.	A molecule is the smallest particle of an element or a compound capable of independent existence, made up of one or more than one atom of same or different types.	A radical is a single atom of an element or a group of atoms of different elements behaving as a single unit with positive or negative charge on it.
2. An atom represents all the properties of that element.	A molecule represents all the properties of that element or compound.	Oppositely charged radicals combine to form molecules of compounds
3. *E.g.* : Na and Cl represent atoms of sodium and chlorine elements respectively.	*E.g.* : H_2O and CO_2 are the molecules of compounds water and carbon dioxide respectively. H_2 and O_2 are the molecules of elements hydrogen and oxygen respectively.	*E.g.* : Cl^-, NH_4^+, SO_4^{2-} ions are the radicals of chloride ion, ammonium ion and sulphate ion respectively. (NH_4^+, and Cl^- are oppositely charged ions which can combine to form a compound ammonium chloride [NH_4Cl]).

Valency

Valency is the combining capacity of an element or of a radical.

The valency of an element or of a radical can be defined as, the number of hydrogen atoms that will combine with or displace one atom of that element or radical. This concept of valency with respect to hydrogen can be applied to find the valency of other elements and radicals.

Examples : The valency of hydrogen is taken as 1.

(a) In hydrogen chloride molecule (HCl), one atom of chlorine combines with one atom of hydrogen; hence valency of chlorine is 1.

(b) In water (H_2O), one atom of oxygen combines with two atoms of hydrogen; hence valency of oxygen is 2.

(c) In ammonia (NH_3) gas, one atom of nitrogen combines with three atoms of hydorgen; hence valency of nitrogen is 3.

(d) In a methane (CH_4) molecule, one carbon atom combines with four atoms of hydrogen; hence valency of carbon is 4.

The valency of an element or an ion is very useful in writing the formula of a compound.

Variable valency

Certain elements exhibit more than one valency, that means they show variable valency.

If an element exhibits two different positive valencies, then suffix ***ous*** is used for lower valency and suffix ***ic*** is used for higher valency or their valency is represented in Roman numerals beside their names.

Table 4.8 : Variable positive valency

Metal	*Radicals*	*Valency*
Iron	Ferrous [Iron (II)]	2
	Ferric [Iron (III)]	3
Copper	Cuprous [Copper (I)]	1
	Cupric [Copper (II)]	2

Non-metals like nitrogen, phosphorus and sulphur also show variable valency. Nitrogen and phosphorus show valencies of 3 and 5 while sulphur exhibits valency of 2, 4 and 6.

RELATIONSHIP BETWEEN VALENCY OF ELEMENTS AND PERIODIC TABLE

You know that valency is the combining capacity of an atom of an element with the atoms of other elements.

Let us study the table given below in which symbols and valency of first twenty elements of the periodic table are given :

Table 4.9 : Name, symbol and valency of first twenty elements

Name of the elements	*Symbol*	*Valency*
1. Hydrogen	H	1
2. Helium	He	0
3. Lithium	Li	1
4. Beryllium	Be	2
5. Boron	B	3
6. Carbon	C	4
7. Nitrogen	N	3
8. Oxygen	O	2
9. Fluorine	F	1
10. Neon	Ne	0
11. Sodium	Na	1
12. Magnesium	Mg	2
13. Aluminium	Al	3
14. Silicon	Si	4
15. Phosphorus	P	3
16. Sulphur	S	2
17. Chlorine	Cl	1
18. Argon	Ar	0
19. Potassium	K	1
20. Calcium	Ca	2

It is noticed that valency [combining capacity] of elements increases from 1 to 4 and then again decreases to 1. For some of the elements like Helium, Neon and Argon, the combining capacity is zero because they are inert gases.

Why do the atoms of some elements have the same valency ? Is there any relationship between them? Do they show similar properties?

Examples :

- Both lithium and sodium have the same valency one. Both are metals.
- Sodium and chlorine also have valency one. But sodium is a metal while chlorine is a non-metal.

To understand this more clearly it is necessary to know about the **Periodic Table**.

"Periodic Table is a tabular arrangement of elements in vertical columns and horizontal rows indicating the regular trends in the properties of elements."

There are 118 elements known to us. In order to study these elements in an organised manner, they need to be classified.

The first attempt was made on the basis of some of their physical and chemical properties due to which elements could be classified into metals and non-metals.

Groups (vertical columns) → / Periods → (horizontal rows) ↓	IA	IIA	IIIA	IVA	VA	VIA	VIIA	Zero
1	H							He
2	Li	Be	B	C	N	O	F	Ne
3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca						

Fig. 4.2 : A part of periodic table showing the first 20 elements

Later on more fundamental properties were taken into consideration and elements were arranged systematically in a tabular form.

Based on the fundamental properties of atoms, Dmitri Mendeleev, a Russian Chemist, created the Periodic table in 1869. However it was Henry Moseley's discovery which helped to refine Mendeleev's table to establish the Modern Periodic Table as we know it now.

- The horizontal rows of the table are called **periods**.
- The vertical columns are called **groups**.

Let us see the first 20 elements in the periodic table given in Table 4.9.

The groups are numbered in Roman numerals as IA, IIA, IIIA, IVA, VA, VIA, VIIA and zero group. The periods are numbered in Arabic numerals as 1, 2, 3, 4, 5, 6, 7.

You will notice that elements present in the same group have same valency (combining capacity) and it also corresponds to the group number upto IV.

Examples : Group IA has lithium, sodium and potassium. They are all reactive metals with valency one. Group IVA has carbon and silicon with valency 4.

Valency of elements present in groups V, VI and VII are 3, 2, 1 respectively. Group zero contains inert gases with zero combining capacity.

The above arrangement makes it very clear that metals with the same valency show similar properties and so do the non-metals.

You will study the details of the periodic table in higher classes.

Note : According to IUPAC system (International Union of Pure and Applied Chemistry), now the group number is also represented as 1, 2, 3, 4..... upto 18.

MOLECULAR FORMULA OF COMPOUNDS

Each compound is represented by a chemical formula with the help of symbols and valency of different atoms present in it. Since the chemical formula represents the molecule of a compound, it is more commonly known as its molecular formula.

A molecular formula of a compound is the symbolic representation of its (one) molecule. It shows the number of atoms of each element present in it. These atoms combine in whole numbers to form the molecule.

For example : A molecule of sulphur dioxide gas is represented as SO_2. It indicates that one molecule of SO_2 is formed by one atom of element sulphur and two atoms of element oxygen.

WRITING THE CHEMICAL FORMULA OF A COMPOUND

To write the chemical formula of a compound, the following information should be available:

(i) Symbols of the elements or the radicals that constitute the compound.

(ii) Valencies (combining capacity) of the elements or the radicals.

Step-by-step method for writing the formulae of chemical compounds.

1. *Example* – Calcium chloride

I. Write the symbols

On the left hand side	*On the right hand side*
Calcium Ca	Chloride Cl

II. Write the valency of the symbols

(Ignore the (+) and (–) signs)

At the top right corner	*At the top right corner*
Ca^2	Cl^1

III. Interchange the valency number

Ca^2 Cl^1

1 2

IV. Write the interchanged numbers at the base

Ca_1 Cl_2

V. Write the formula of the compound

(ignore base number if it is 1)

$CaCl_2$

Note : The number 1 with Ca is not written in the formula because the symbol itself represents one atom.

One more example of writing a molecular formula is in the manner where valency number of positive and negative ions are divided by a common number.

2. *Example* – Magnesium oxide

Step I : Write the symbols and valencies.

Symbols	
Magnesium Mg	Oxide O

Valencies	
Magnesium 2	Oxide 2

Step II : Mg^2 O^2

Step III : Interchange the valencies of the radicals.

Mg^2 O^2

2 2

Step IV : Write the interchanged numbers at the base.

Mg_2 O_2

Step V : Reduce the valency to the lowest ratio, if divisible by a common factor.

$Mg_{2\div 1}$ $O_{2\div 1}$

Step VI : (Ignore the base number, as it is one) → Mg_1 O_1

Therefore, the formula is **MgO**.

3. *Example* – Calcium nitride

Step I : Write the symbols and valencies.

Symbols	
Calcium	Nitride
Ca	N

Valencies	
Calcium	Nitride
2	3

Step II : Ca^2 N^3

Step III : Interchange the valencies of radicals.

Ca^2 N^3

3 2

Write the interchanged numbers at the base. (There is no common factor)

Ca_3 N_2 Therefore, the formula is Ca_3N_2.

4. *Example* – Zinc hydroxide

Step I : Write the symbols and valencies.

Symbols	
Zinc	Hydroxide
Zn	OH

Valencies	
Zinc	Hydroxide
2	1

Step II : Zn^2 OH^1

Step III : Interchange the valency number. When the radical itself bears any number in its formula or in its symbol, or if it is OH radical and more than 1 number comes after exchanging the valency number, put the parenthesis (round brackets) sign enclosing the radical symbol.

Zn^2 OH^1

1 2

Step IV : Zn_1 $(OH)_2$

Step V : (Ignore the base number of Zn as it is one). Therefore, the formula is $Zn(OH)_2$.

5. *Example* – Ammonium carbonate

Step I : Write the symbols and valencies.

Symbols	
Ammonium	Carbonate
NH_4	CO_3

Valencies	
Ammonium	Carbonate
1	2

Step II : NH_4^1 CO_3^2

Step III : Interchange the valency number

NH_4^1 CO_3^2

2 1

Step IV : Radicals already bear some numbers in their formulae. So parenthesis (round brackets) are required to be put around both the radical formulae.

$(NH_4)_2$ $(CO_3)_1$

Step V : (Ignore the base number of carbonate, as it is one). Therefore, the formula is $(NH_4)_2CO_3$.

Activity

Write the molecular formulae of chlorides, oxides and sulphides of following bivalent metals (valency-2)

1. Copper 2. Iron 3. Lead 4. Mercury

Significance of molecular formula

The molecular formula of a compound other than saving time, space and energy also gives us the following information :

(i) It represents one molecule of a compound.

(ii) The number of each kind of atoms present, *i.e.*, the ratio in which the atoms are present in one molecule.

(iii) The mass of one molecule of the compound can be calculated. Molecular mass is the algebric sum of the masses of all the atoms present in a given molecule.

Example :

- A molecule of sulphuric acid is represented by the formula H_2SO_4.
- The elements present in it are hydrogen, sulphur and oxygen.
- One molecule of sulphuric acid has two atoms of hydrogen, one atom of sulphur and four atoms of oxygen. The ratio in which atoms of hydrogen, sulphur and oxygen are present is 2 : 1 : 4.
- If the masses of all the atoms present in the molecule are added, the molecular mass of sulphuric acid is obtained.

Note : We can represent the number of molecules of a compound by writing a number before the formula. *For example:* $2H_2SO_4$ represents two molecules of sulphuric acid.

5NaCl represents five molecules of sodium chloride.

Table 4.10 : Molecular formulae, the common names and the state of some common compounds

Name of Compounds	*Formula*	*State*
1. Sodium chloride [common salt]	NaCl	Solid
2. Sugar	$C_{12}H_{22}O_{11}$	Solid
3. Glucose	$C_6H_{12}O_6$	Solid
4. Sodium bicarbonate [baking soda]	$NaHCO_3$	Solid
5. Washing soda [sodium carbonate, decahydrate]	$Na_2CO_3 \cdot 10H_2O$	Solid
6. Marble & chalk [calcium carbonate]	$CaCO_3$	Solid
7. Sand [silica]	SiO_2	Solid
8. Calcium hydroxide [slaked lime]	$Ca(OH)_2$	Solid
9. Sodium hydroxide [caustic soda]	NaOH	Solid
10. Copper sulphate [blue vitriol]	$CuSO_4 \cdot 5H_2O$	Solid
11. Water	H_2O	Liquid
12. Acetic acid [vinegar]	CH_3COOH	Liquid
13. Hydrochloric acid	HCl	Liquid
14. Sulphuric acid	H_2SO_4	Liquid
15. Nitric acid	HNO_3	Liquid
16. Carbon dioxide	CO_2	Gas
17. Carbon monoxide	CO	Gas
18. Sulphur dioxide	SO_2	Gas
19. Sulphur trioxide	SO_3	Gas
20. Ammonia	NH_3	Gas
21. Hydrogen sulphide	H_2S	Gas
22. Nitrogen dioxide	NO_2	Gas
23. Nitric oxide [nitrogen monoxide]	NO	Gas
24. Nitrous oxide [laughing gas]	N_2O	Gas
25. Phosphorous pentoxide	P_2O_5	Solid

RECAPITULATION

- Matter is made up of tiny particles called atoms and molecules.
- Atoms are the smallest particles of elements.
- Atoms may or may not have independent existence.
- Molecules have independent existence. They are the smallest particles of compounds.
- Radicals are charged particles consisting of an atom or group of atoms of different elements. Radicals can be positive or negative. They are also called as ions.
- Valency is the combining capacity of an atom of an element. The number of hydrogen atoms that combines with an atom of another element determines the valency of that element.
- Some elements exhibit variable valency like iron, copper, etc.
- Periodic table represents the tabular arrangement of elements in horizontal rows called periods and vertical columns called groups in order to classify the elements and their systematic study.
- The valency of elements corresponds to the group number of the table upto 4 and then decreases to one and finally zero. This shows the similar behaviour of the elements present in the same group.
- Molecular formula is the representation of molecules with the help of symbols and the valencies of its constituting atoms.

EXERCISE

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(i) The smallest particle of matter which may or may not have independent existence is :

(a) Molecule *(b)* Atom
(c) Radical *(d)* Element

(ii) The number of atoms of the same type which combine to form a molecule of an element is called its :

(a) Atomicity *(b)* Valency
(c) Molecular formula
(d) Compound

(iii) The atomicity of phosphorus is :

(a) 1 *(b)* 2
(c) 3 *(d)* 4

(iv) The number of atoms present in a molecule of water is :

(a) 2 *(b)* 3
(c) 4 *(d)* 5

(v) What does 2H represent ?

(a) two atoms of hydrogen
(b) two molecules of hydrogen
(c) one molecule of hydrogen
(d) none of the above

(vi) The symbol for calcium ion is :

(a) Ca *(b)* Ca^{+}
(c) Ca^{2+} *(d)* Mg^{2+}

(vii) Which of the following sets of elements are present in the same group of the Periodic Table ?

(a) sodium, magnesium
(b) beryllium, boron
(c) oxygen, sulphur
(d) carbon, nitrogen

(viii) Which of the following is a non-metallic polyatomic cation ?

(a) nitrate ion *(b)* ammonium ion
(c) aluminium ion *(d)* chloride ion

(ix) The valency of iron in Fe_2O_3 is :

(a) 1 (b) 2

(c) 3 (d) 6

(x) Which of the following has valency 4?

(a) Aluminium (b) Oxygen

(c) Carbon (d) Phosphorus

(xi) The sulphate radical is written as SO_4^{2-}. What will be the formula of calcium sulphate?

(a) $Ca(SO_4)_2$ (b) $Ca_2(SO_4)$

(c) $Ca(SO_4)_3$ (d) $CaSO_4$

(xii) Which of the following exhibits variable valency ?

(a) Calcium (b) Copper

(c) Carbon (d) Chlorine

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

(i) **Assertion (A) :** Atoms of an element exhibit all the properties of that element.

Reason (R) : Atoms of an element have the same atomic number.

(a) (1) (b) (2) (c) (3) (d) (4)

(ii) **Assertion (A) :** A hydrogen molecule is monoatomic.

Reason (R) : The number of atoms of an element that combine to form a molecule is known as its atomicity.

(a) (1) (b) (2) (c) (3) (d) (4)

(iii) **Assertion (A) :** Radicals or ions are electrically charged atoms or a group of atoms formed by losing or gaining electrons.

Reason (R) : Basic radicals, also called as cations are formed by gaining electrons.

(a) (1) (b) (2) (c) (3) (d) (4)

(iv) **Assertion (A) :** Periodic table is a tabular arrangement of elements in vertical columns called groups and horizontal rows called periods.

Reason (R) : The arrangement of elements is done for their systematic and convenient study.

(a) (1) (b) (2) (c) (3) (d) (4)

(v) **Assertion (A) :** An atom consists of three fundamental particles called electrons, protons and neutrons.

Reason (R) : The sum of the number of protons and neutrons is called the mass number of an atom.

(a) (1) (b) (2) (c) (3) (d) (4)

3. Fill in the blanks :

(i) Atoms are ..... .

(ii) An ion with positive charge is called .....

(iii) An ion with negative charge is called .....

(iv) $2H_2$ means two ..... of hydrogen.

(v) ..... is a triatomic molecule.

(vi) Some metals have ..... valency.

(vii) Chemical name of caustic soda is .....

4. Write true or false for the following statements:

(a) An atom is the smallest particle of an element that exhibits all its properties. .....

(b) The molecules of compounds are formed when atoms of the same type combine chemically. .....

(c) A basic radical is negatively charged. .....

(d) The valency of oxygen is three. .....

(e) An atom is electrically neutral. .....

5. Match the following :

(a) The combining power of an element.	*(i)* Inert gas
(b) The negatively charged particle of an atom.	*(ii)* Period
(c) Helium	*(iii)* Valency
(d) Positively charged ion	*(iv)* Electron
(e) Horizontal rows in a periodic table.	*(vi)* Cation

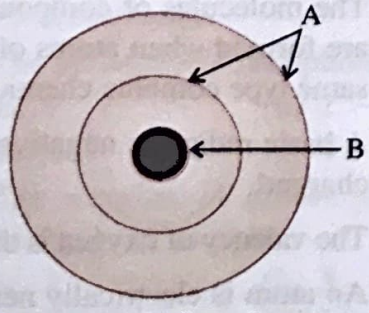
6. Name or give appropriate terms for the following :
   - *(a)* The smallest particle of an element.
   - *(b)* The smallest particle of a molecule.
   - *(c)* The positively charged subatomic particle of an atom.
   - *(d)* The centrally located part of an atom.
   - *(e)* The subatomic particle of an atom which has no charge.
   - *(f)* The number of atoms present in a molecule of an element..
   - *(g)* The symbolic representation of a molecule.
   - *(h)* A group of atoms that react as a single unit..
   - *(i)* The combining capacity of an element.
   - *(j)* The tabular arrangement of elements in horizontal rows and vertical columns.

7. Name the following radicals :

(a) SO_4^{2-}	*(b)* HCO_3^-
(c) OH^-	*(d)* $Cr_2O_7^{2-}$

B. SHORT ANSWER TYPE QUESTIONS

1. Look at the diagram given below carefully and answer the questions that follow :



Atomic model of an element

   - *(a)* Label A and B.
   - *(b)* Name the particles present in *(i)* A and *(ii)* B.
   - *(c)* Which of the particles mentioned in A and B are :
     - *(i)* negatively charged
     - *(ii)* positively charged
     - *(iii)* neutral

2. Write the names of the following compounds:

(a) $(NH_4)_2SO_4$	*(b)* $Ca(NO_3)_2$
(c) FeS	*(d)* Na_3PO_4
(e) NH_4OH	*(f)* $CuCO_3$
(g) HgO	*(h)* $ZnCl_2$
(i) ZnS	*(j)* H_2S.

3. Write the names of the elements present in the following compounds.

(a) Common salt	*(b)* Ammonia
(c) Sulphuric acid	*(d)* Glucose
(e) Sodium hydroxide	*(f)* Acetic acid

4. Write the symbols and valencies of the following radicals :

(a) Magnesium ion	*(b)* Ammonium
(c) Carbonate	*(d)* Nitrate
(e) Oxide	*(f)* Bisulphate
(g) Aluminium ion	

5. Define the following terms :

(a) Atoms	*(b)* Molecules
(c) Radicals	*(d)* Valency
(e) Periodic table	

6. Give the group numbers of the following elements present in the periodic table :

(a) Magnesium	*(b)* Carbon
(c) Sulphur	*(d)* Neon

7. What does each of the following represent ?

(a) $2CO_2$	*(b)* $2H_2S$
(c) $5H_2SO_4$	*(d)* $6NaNO_3$

8. The valency of calcium is 2. Write the valencies of other radicals in the following compounds :

(a) CaO	*(b)* $Ca(OH)_2$
(c) $CaCO_3$	*(d)* $CaCl_2$

9. Write the molecular formulae of :
   (c) Calcium hydroxide
   (d) Hydrogen chloride
   (e) Sulphuric acid (f) Iron (II) sulphide
   (g) Iron (III) sulphate (h) Nitric acid
   (k) Magnesium nitride

10. Answer the following questions in brief :
   (a) Give two examples of polyatomic molecules.
   (b) Give the name and symbol of an ion each with the valency of +1, +2 and +3.
   (c) Give the name and symbol of an ion each with the valency of –1, –2 and –3.
   (d) What are the particles present in the nucleus of an atom ?
   (e) Give one difference between the molecules of an element and those of a compound.
   (f) How is a basic radical (cation) formed ?

C. LONG ANSWER TYPE QUESTIONS

1. State five characteristics of an atom.
2. State five characteristics of a molecule.
3. Give two differences between molecules and radicals.
4. (a) Name the three fundamental particles present in an atom.
   (b) What are the two main parts of an atom?
   (c) Give the location of the three fundamental particles in an atom.

5. What is variable valency ? Give two examples of elements showing variable valency.
6. (a) What is a Periodic Table ?
   (b) What are Periods and Groups of the Periodic Table ?
   (c) A part of the periodic table is shown below in which symbols of some elements are given, while some are missing. (Only the first 20 elements are given).
      (i) Give the names and symbols of the missing elements.
      (ii) Also mention whether they are metals, non-metals, metalloids or inert gases.

Groups (vertical columns) ↓ / Periods → (horizontal rows)	IA	IIA	IIIA	IVA	VA	VIA	VIIA	Zero
1	H							—
2	Li	—	B	C	—	—	F	Ne
3	—	Mg	—	—	P	S	—	Ar
4	K	Ca	×	×	×	×	×	×

7. The symbols and valencies of some metal ions (cations) are given below. Write the molecular formulae of their oxides, sulphides and chlorides, provided that the valencies of oxygen, sulphur and chloride are –2, –2 and –1 respectively.

Cation	Valency	Oxide	Sulphide	Chloride
Na	1	.....	Na_2S	.....
Mg	1	MgO	.....	.....
Fe	3	.....	.....	$FeCl_3$

8. The valency and symbols of some cations and anions are given below.

Cations : Ca^{2+}, Fe^{2+}, Al^{3+} Anions : SO_4^{2-}, HCO_3^-, OH^-, PO_4^{3-}

Write the formula of :

(a) Calcium phosphate (b) Aluminium sulphate (c) Iron (II) hydroxide

(d) Calcium bicarbonate (e) Aluminium hydroxide

D. CASE STUDY BASED QUESTION

Read the given passage carefully and answer the questions that follow :

An atom is the basic unit of matter.

Following are the two statements about atoms :

(a) An atom is the smallest particle of matter which cannot be divided further into smaller particles.

(b) An atom of an element exhibits all the properties of that element.

(i) Name the scientist who proposed the above two characteristics of an atom.

(ii) Which of the above two statements is true ?

(iii) Name the particles formed when two or more atoms combine chemically.

Answers

Exercise 4 — Objective type questions

1. (i)–(b), (ii)–(a), (iii)–(d), (iv)–(b), (v)–(a), (vi)–(a), (vii)–(c), (viii)–(b), (ix)–(c), (x)–(d), (xi)–(d), (xii)–(b)

2. (i)–(a) : The reason correctly explains the assertion. Since atoms of an element have the same atomic number, they exhibit all the properties of that element.

(ii)–(d) : Assertion is false as a molecule of hydrogen is diatomic. The reason is true.

(iii)–(c) : Assertion is true but the reason is false as basic radicals / cations are formed by losing electrons.

(iv)–(a) : The reason correctly explains the assertion. For a systematic and convenient study, elements are arranged in a Periodic Table.

(v)–(b) : Both assertion and reason are true but the reason does not explain or justify the assertion.

Project

- Play a game for writing the formulae of compounds.

Example : Make placards with symbols and valencies of the elements separately. Each student should hold two placards, one with the symbol in the right hand and the other with the valency in the left hand. Keeping the symbols in place, students should criss-cross their valencies to form the formula of a compound.

Language of Chemistry

5

THEME

Chemistry involves the study of a large number of elements and compounds that also have been learnt earlier with their representation by their short hand notations *i. e.* symbols and formulae. This theme will enable children to understand that it is not convenient to write the full names of the elements and compounds, and the use of symbols has made the job of the chemists much easier. In addition, they will further realize that Chemistry also involves the occurrence of a large number of chemical reactions that are written in the form of equations known as chemical equations. The writing of chemical equations involves writing of reactants and products as their symbols and formulae. Thus symbols and formulae have also made writing of chemical equations in Chemistry very convenient.

LEARNING OBJECTIVES

The children will be able to :

- identify the names of reactants and products of different chemical reactions;
- write a chemical reaction in the form of a chemical word equation;
- recognize the usefulness of a word equation.

CHEMICAL REACTIONS

Any chemical change in matter which involves its transformation into one or more new substances is called a ***chemical reaction****. It also involves transfer of energy.*

An example of a chemical reaction is burning of a candle. Candle wax burns in air to give carbon dioxide and water vapour along with heat and light energy.

The substances that undergo chemical change are called ***reactants****, and the new substances thus formed are called* ***products****.*

$$\text{Reactants} \xrightarrow[\text{Change}]{\text{Chemical}} \text{Products}$$

Both reactants and products are pure substances, *i.e.*, elements and compounds.

What happens to the reactants during a chemical reaction ?

During a chemical reaction, atoms of reacting molecules rearrange themselves to form one or more products. These atoms are neither destroyed nor new atoms are formed.

***Example* 1 :**

Elements hydrogen and oxygen react with each other to form a new substance water, a compound which

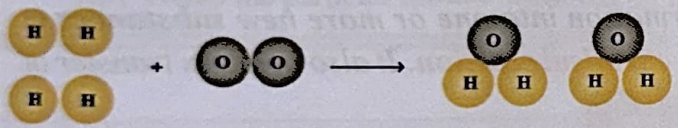
is completely different from hydrogen and oxygen in its properties.

This reaction can be represented in the form of a word equation given below.

Hydrogen + Oxygen ⟶ Water
[Reactants] [Product]

When we write the names of reactants and products to express a chemical reaction, it is called a 'word equation.'

The atoms of oxygen and hydrogen in their molecules rearrange themselves to form water molecules. One molecule of water contains two atoms of hydrogen and one atom of oxygen.



HYDROGEN MOLECULES | OXYGEN MOLECULE | WATER MOLECULES

Example 2 :

Two compounds, sodium oxide and water, react chemically to give a new compound sodium hydroxide.

Sodium oxide + Water ⟶ Sodium hydroxide

Example 3 :

Upon heating, the compound mercuric oxide changes into two elements mercury and oxygen.

Mercuric oxide $\xrightarrow{\text{heat}}$ Mercury + Oxygen

Example 4 :

Potassium chlorate, a compound, changes into two new substances on heating; a compound potassium chloride and an element oxygen.

Potassium chlorate $\xrightarrow[MnO_2]{\text{heat}}$ Potassium chloride + Oxygen

Note : This reaction takes place faster in presence of another compound, manganese dioxide (MnO_2).

Thus, a chemical reaction may involve a combination of two or more elements or compounds reacting with each other to form new compound(s) or it may involve splitting of a compound into two or more elements or compounds. It also involves reaction between an element and a compound or two compounds to give more than one product.

So whenever a chemical change occurs, we can say that a **chemical reaction** has taken place.

CHARACTERISTICS OF OCCURRENCE OF CHEMICAL REACTIONS

How will you recognize that a chemical reaction has taken place ?

Let us do an activity.

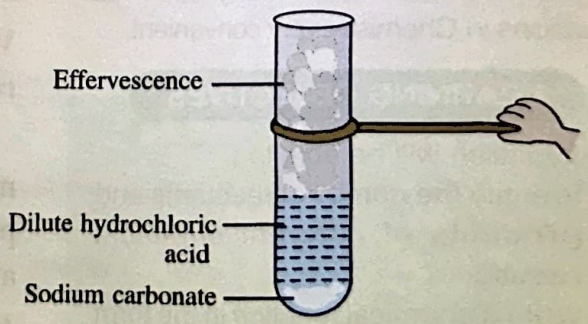
Activity 1

(To be demonstrated by the teacher.)

Take a small amount of sodium carbonate, a white solid in a dry test tube. Add 2-3 ml of dilute hydrochloric acid to it.

What do you observe ?

A brisk effervescence* indicating evolution of a gas, which is a new substance carbon dioxide. At the same time heat energy is also given out as the test tube becomes warm.



* *Effervescence : It is the bubble formation due to evolution of a gas during a chemical reaction.*

This indicates that a chemical reaction has taken place between sodium carbonate and dilute hydrochloric acid to produce new substances.

Sodium carbonate + Hydrochloric acid (dil.) → Sodium chloride + Water + Carbon dioxide

Hence, chemical reactions are characterized by certain changes that can be easily observed. They help to recognize the changes in the reactants and formation of new products. Some of these are as follows :

1. **Change of colour :** In some chemical reactions, change of colour takes place when reactants form the products.

   Example **:** When a few pieces of iron are dropped into a blue solution of copper sulphate, its **blue** colour slowly changes into **light green** and also a **red** substance is formed. The reaction can be represented as,

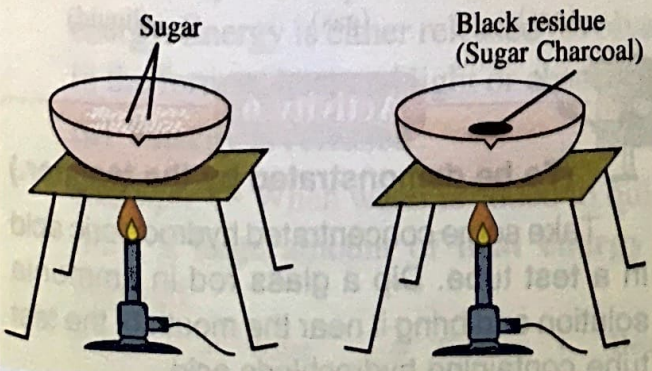
Iron + Copper sulphate (blue solution) → Iron sulphate (light green solution) + Copper (reddish brown solid)

Activity 2

(To be demonstrated by the teacher.)

Take some sugar in a porcelain dish. Keep it over a tripod stand and heat it for some time.

What do you observe ?



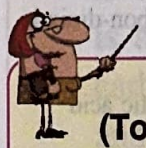
Sugar first melts and then gets charred into a **black** residue which is a new substance charcoal. It no more tastes sweet. Some water vapour is also formed which escapes in air.

This shows that a chemical reaction may involve a change in colour.

This reaction can be represented as,

Sugar (white solid) $\xrightarrow{\text{heat}}$ Charcoal (black solid mass) + Water vapour

Let us do one more activity.



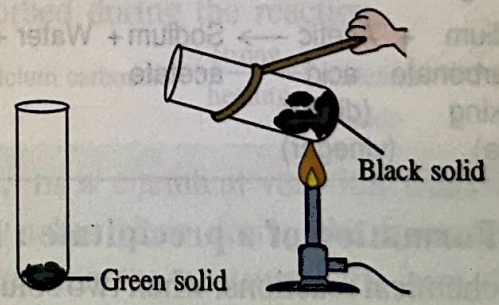
Activity 3

(To be demonstrated by the teacher.)

Take a small amount of a compound copper carbonate, which is a green solid, in a dry test tube. Heat it for some time.

What do you observe ?

The **green** solid changes into a **black** solid and carbon dioxide gas is also evolved.



Thus, change in colour may take place during a chemical reaction.

Copper carbonate (green solid) $\xrightarrow{\text{heat}}$ Copper oxide (black solid) + Carbon dioxide

2. **Evolution of a gas :** Some chemical reactions are characterised by the evolution of a gas, which can be recognised by the effervescence (bubbles), smell or colour.

   Example **:** When zinc granules react with dilute hydrochloric acid, an effervescence

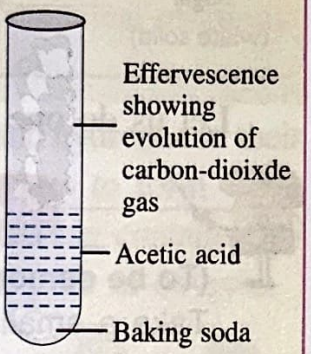
is observed indicating the evolution of a colourless, odourless gas hydrogen. The reaction is represented as,

Zinc + Hydrochloric acid (dilute) ⟶ Zinc chloride + Hydrogen↑ (gas)

Activity 4

(To be demonstrated by the teacher.)

Take some baking soda (sodium bicarbonate) in a test tube. Add some dilute vinegar (acetic acid) to it.



What do you observe?

A strong effervescence (bubbles) is observed indicating the evolution of carbon dioxide gas.

This shows that a gas may be evolved during a chemical reaction.

Sodium bicarbonate (baking soda) + Acetic acid (dilute) (vinegar) ⟶ Sodium acetate + Water + Carbon dioxide

3. Formation of a precipitate : In some chemical reactions, when two solutions are mixed, an insoluble solid known as a precipitate is formed.

Example **:** When barium chloride solution is added to sodium sulphate solution, a **white** precipitate of barium sulphate is formed.

Barium chloride (solution) + Sodium sulphate (solution) ⟶ Barium sulphate↓ (white precipitate) + Sodium chloride (solution)

Activity 5

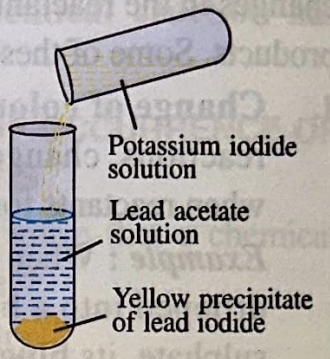
(To be demonstrated by the teacher.)

Take aqueous solutions of potassium iodide and lead acetate in two separate test tubes. Add potassium iodide solution to lead acetate solution.

What do you observe ?

A **yellow** precipitate is formed immediately which is a new substance lead iodide.

This shows that, during a chemical reaction an insoluble solid *i.e.* a **precipitate** can be formed and also a colour change may take place.



Potassium iodide (solution) + Lead acetate (solution) → Lead iodide↓ (yellow precipitate) + Potassium acetate (solution)

4. Change of state : Change of state is observed in many chemical reactions. The reactant may be a solid or a liquid which changes into a gaseous product or vice versa.

Example **:** When hydrogen gas is burnt in oxygen gas it results in the formation of water which is a liquid under normal conditions.

Hydrogen (gas) + Oxygen (gas) $\xrightarrow{\text{heat}}$ Water (liquid)

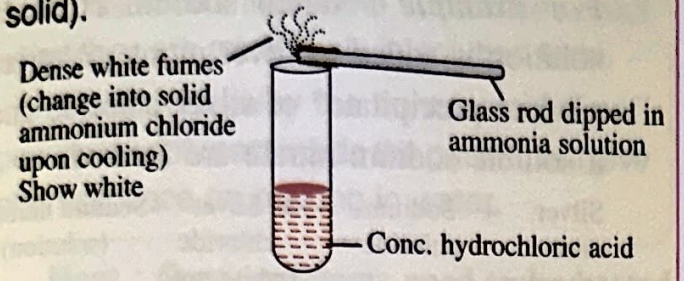
Activity 6

(To be demonstrated by the teacher.)

Take some concentrated hydrochloric acid in a test tube. Dip a glass rod in ammonia solution and bring it near the mouth of the test tube containing hydrochloric acid.

What do you observe ?

Dense white fumes are observed due to the production of ammonium chloride (which is a solid).



Ammonia solution (liquid) + Conc. hydrochloric acid (liquid) → Ammonium chloride (solid)

5. **Change of smell :** During some chemical reactions a strong smell is experienced.

Example **:** • When solid ammonium chloride is heated with sodium hydroxide solution, a gas ammonia is evolved which is recognized by its *strong pungent smell.*

Ammonium chloride (solid) + Sodium hydroxide (solution) $\xrightarrow{\text{heat}}$ Sodium chloride (solution) + Water + Ammonia↑ (gas) (pungent smell)

- When dilute hydrochloric acid is heated with iron sulphide, a gas hydrogen sulphide with a *rotten egg smell* is evolved.

Iron sulphide (solid) + Hydrochloric acid (dil.) (solution) → Iron chloride (solution) + Hydrogen sulphide (gas)↑ (rotten egg smell)

6. **Change of energy :** Chemical reactions are mostly accompanied by a change in energy. Energy is either released (evolved) in the form of heat and light or absorbed.

(a) Energy is released

Example **:** • When water is added to quick lime, a large amount of **heat energy** is evolved.

Calcium oxide (solid) + Water → Calcium hydroxide + Heat

- Magnesium burns in air or oxygen giving a **white dazzling light,** to produce a compound magnesium oxide.

Magnesium + Oxygen $\xrightarrow{\text{heat}}$ Magnesium oxide

- When coal or charcoal burns in air both heat and light are produced, along with a gas carbon dioxide.

Coal + Oxygen $\xrightarrow{\text{heat}}$ Carbon dioxide + Heat + Light

(b) Energy is absorbed :

Example **:** • In photosynthesis, **sunlight is absorbed** by green plants to prepare carbohydrates (food) from carbon dioxide and water.

Carbon dioxide + Water $\xrightarrow{\text{sunlight}}$ Glucose (Carbohydrates) + Oxygen

- Calcium carbonate on heating dissociates to give two new products, calcium oxide and carbon dioxide. Heat energy is absorbed during the reaction.

Calcium carbonate $\xrightarrow[\text{heating}]{\text{strong}}$ Calcium oxide + Carbon dioxide

Note : In a chemical reaction transfer of energy takes place during the formation of products from the reactants in the form of heat and light and sometimes in the form of sound and electricity too.

Exothermic reactions : The chemical reactions in which energy is released are called exothermic reactions.

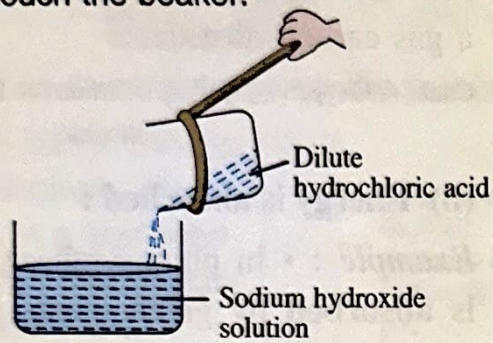
Endothermic reactions : The chemical reactions in which energy is absorbed are called endothermic reactions.

Energy is mostly released or absorbed in the form of heat and light energy, but sometimes in the form of sound and electrical energy as well.

Activity 7

Take some sodium hydroxide solution in a beaker. To this, add some dilute hydrochloric acid and stir properly.

Now touch the beaker.



What do you observe ?

The beaker is slightly warm as some heat is released upon the reaction of these two substances.

Sodium hydroxide + Hydrochloric acid (dil.) → Sodium chloride + Water + Heat

CONDITIONS NECESSARY FOR CHEMICAL REACTIONS

A chemical reaction takes place when one or more of the following conditions are fulfilled.

1. **Close contact :** For a chemical reaction to take place, the reactants must come into close contact, *i.e.,* they should be mixed.

   For example : Sodium reacts with water violently when they come in contact with each other to produce two new substances, sodium hydroxide and hydrogen.

   Sodium + Water ⟶ Sodium hydroxide + Hydrogen

   Without contact, no reaction takes place. Therefore in the laboratory, sodium is stored in kerosene oil to prevent its contact even with moisture.

2. **Solution form :** Some substances react with each other only when they are mixed in the solution form.

   For example : When sodium chloride solution is added to silver nitrate solution, a white precipitate* of silver chloride and a soluble sodium nitrate are formed.

   Silver nitrate (aqueous solution) + Sodium chloride ⟶ Silver chloride (white ppt.) + Sodium nitrate (solution)

   If solid sodium chloride and silver nitrate are mixed together, no change takes place.

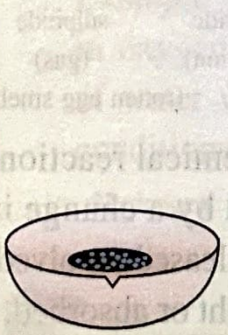
Activity 8

(To be demonstrated by the teacher.)

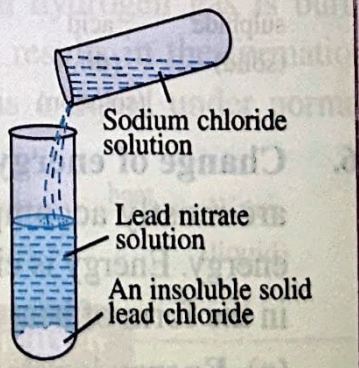
Take some solid sodium chloride and solid lead nitrate and mix them. They are now in contact of each other. Keep the mixture undisturbed for some time.

Do you observe any change ? **No.**

Now take aqueous solutions of sodium chloride and lead nitrate in two separate test tubes. Add the solution of sodium chloride to lead nitrate solution. What do you observe ?



A mixture of solid sodium chloride and lead nitrate



Chemical reaction takes place when the reactants are mixed in solution form

* *Precipitate : An insoluble solid formed on reaction of two solutions is called a precipitate.*

A white insoluble solid (precipitate) is formed immediately which is a new substance known as lead chloride.

This shows that some substances react when they are mixed only in solution form.

Now think, why the contents of 'Eno*' powder do not react inside the packet, but give effervescence on pouring in water.

3. **Heat :** Some reactants need to be heated to undergo a chemical change.

For example : Iron and sulphur when heated together react to produce iron sulphide. Without heating, they do not react even if they are in contact with each other.

$$\text{Iron} + \text{Sulphur} \xrightarrow{\text{heat}} \text{Iron sulphide}$$

4. **Light :** Some reactions take place in the presence of light.

For example : In photosynthesis, carbon dioxide and water react in presence of chlorophyll and light to produce glucose and oxygen.

$$\text{Carbon dioxide} + \text{water} \xrightarrow[\text{sunlight}]{\text{chlorophyll}} \text{glucose} + \text{oxygen}$$

5. **Electricity :** Some chemical reactions require electricity to take place.

For example : When electric current is passed through water (containing some acid), it dissociates to produce hydrogen and oxygen gases.

$$\text{Water (acidulated)} \xrightarrow{\text{electric current}} \text{Hydrogen} + \text{Oxygen}$$

6. **Catalyst :** Some reactions need a catalyst to speed up or slow down the reaction.

For example : Manganese dioxide acts as a catalyst for the decomposition of potassium chlorate into potassium chloride and oxygen at a lower temperature.

$$\text{Potassium chlorate} \longrightarrow \text{Potassium chloride} + \text{Oxygen}$$

Manganese dioxide does not undergo any change in its chemical composition during the reaction.

CATALYST : ***A catalyst is a substance which changes the rate of a chemical reaction without itself undergoing any chemical change.*** Different chemical substances are used as catalysts for different chemical reactions.

CHEMICAL EQUATIONS

There are a large number of elements and compounds. An element can be represented by a symbol, while a compound can be represented by a formula.

Similarly, a chemical reaction can be written in the form of an equation known as **chemical equation.**

The steps involved in writing a chemical equation :

- The reactants are written on the left hand side, while the products are written on the right hand side.
- An arrow (→) sign is put in between the reactants and products with its head pointing towards the products.
- If two or more reactants and products are represented in the equation, each one is separated by a (+) sign.

$$\underset{\text{(Reactants)}}{A + B} \longrightarrow \underset{\text{(Products)}}{C + D}$$

* *Eno : It is a mixture of sodium bicarbonate, sodium carbonate and citric acid. It is used as a medicine to get relief from acidity.*

Chemical equations can be written in the following two ways :

(1) **As word equations :** When names of the reactants and products of a chemical reaction are written in an equation, it is called as a word equation.

For example :

- Carbon + Oxygen $\xrightarrow{\text{heat}}$ Carbon dioxide
  (Reactants) (Product)

Carbon burns in oxygen to form carbon dioxide.

- Sodium hydroxide + Sulphuric acid ⟶ Sodium sulphate + Water
  (Reactants) (Products)

Sulphuric acid when added to sodium hydroxide forms sodium sulphate and water as products.

[*Note :* Many more word equations were described as examples and activities while discussing the characteristics and conditions of chemical reactions].

(2) **Chemical equations using symbols and formulae :** When the reactants and products of a chemical reaction are represented symbolically using their symbols and formulae, it is called as a chemical equations.

For example : Carbon burns in oxygen to form carbon dioxide. This can be written using symbols and formulae for carbon, oxygen and carbon dioxide.

$$\underset{\text{(Reactants)}}{C + O_2} \longrightarrow \underset{\text{(Products)}}{CO_2}$$

In fact, a chemical equation is a short hand description of a chemical reaction. It saves space, time and energy.

In symbolic representation, the molecular formulae of the substances are written because atoms are usually neither stable nor capable of independent existence.

Now consider the following chemical reactions :

(i) Reaction of zinc oxide with carbon to form zinc and carbon monoxide.

Zinc oxide + Carbon ⟶ Zinc + Carbon monoxide
[word equation]

$$ZnO + C \longrightarrow Zn + CO$$
[symbolic equation]

The total number of atoms on the reactant side as well as product side are equal in this case. Such an equation is a **balanced equation.**

(ii) Reaction between hydrogen and chlorine to form hydrogen chloride.

Hydrogen + Chlorine ⟶ Hydrogen chloride
[word equation]

$$H_2 + Cl_2 \longrightarrow HCl$$
[skeletal equation]

Note : We cannot write the equation as H + Cl → HCl, because H and Cl represent the atoms of hydrogen and chlorine, which have no independent existence. The two reactants should be written as H_2 and Cl_2, *i.e.*, in their molecular form. Molecules have independent existence.

In this reaction, the numbers of hydrogen and chlorine atoms on the left hand side are not equal to their numbers on the right hand side.

Such an equation is an unbalanced one and it is known as a ***skeletal equation.***

Some more examples of skeletal equations are given here. Identify the reactants and products in the given equations and also write their word equations.

$H_2 + O_2 \longrightarrow H_2O$

$Mg + O_2 \longrightarrow MgO$

$N_2 + H_2 \longrightarrow NH_3$

$Na_2O + H_2O \longrightarrow NaOH$

$HgO \longrightarrow Hg + O_2$

$Na + H_2O \longrightarrow NaOH + H_2$

$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

$Fe + S \longrightarrow FeS$

$Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

$Fe + HCl \longrightarrow FeCl_2 + H_2$

$CaO + H_2O \longrightarrow Ca(OH)_2$

Remember, all chemical equations must be balanced in order to follow the law of conservation of matter or mass.

Note : Word equations do not require balancing.

- It has been observed that the total mass of the reactants together is equal to the total mass of the products, *i.e.* mass of the substances before and after a chemical change remains the same.
- This is possible only when the total number of atoms of each kind remains the same before and after a chemical reaction.
- This is supported by the **law of conservation of matter** which states that,

"Matter can neither be created nor be destroyed. It can only be transformed from one form to another."

The need for balancing a chemical equation

A chemical equation needs to be balanced so as to make the number of atoms of the reactants equal to the number of atoms of the products. *This is because a chemical reaction is just a rearrangement of atoms. The atoms themselves are neither created nor destroyed during the course of a chemical reaction. [Law of conservation of matter].*

Therefore, the balanced chemical equation for the reaction between hydrogen and chlorine is written as :

$$H_2 + Cl_2 \rightarrow 2HCl.$$

You will learn more about balancing chemical equations in higher classes.

***A balanced chemical reaction** is the one in which the number of atoms of each element on the reactant side is equal to the number of atoms of that element on the product side.*

RECAPITULATION

- A chemical reaction involves the transformation of an original substance into an altogether new substance(s).
- A chemical reaction can be represented with the help of the symbols or the formulae of the elements and the compounds taking part in that reaction. This gives a chemical equation.
- A chemical reaction can also be represented in the form of a word equation in which the names of the reactants and products are written.

- The substances that react with each other are called reactants and they are represented on the left hand side of the equation. The substances that are formed as a result of the reaction are called products. They are represented on the right hand side of the equation.
- Certain necessary conditions for a chemical reaction to happen are — close contact, solution form, heat, light and catalyst.
- Characteristics of chemical reactions are — change of colour, evolution of a gas, formation of a precipitate, change of state, change of smell and evolution/absorption of heat.
- A chemical equation needs to be balanced to make it follow the law of conservation of mass.
- The law of conservation of mass states that mass can neither be created nor destroyed, it can only be transformed from one form to another.

EXERCISE

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(i) A chemical equation is a statement that describes a chemical change in terms of:
(a) Symbols and formulae *(b)* Energy
(c) Number of atoms *(d)* Colours

(ii) Balancing a chemical equation is based on :
(a) Law of conservation of mass
(b) Mass of reactants and products
(c) Symbols and formulae
(d) None of the above

(iii) Copper carbonate when heated, turns :
(a) Blue *(b)* Green
(c) Black *(d)* Yellow

(iv) When lead acetate solution is added to potassium iodide solution, a precipitate is formed which is :
(a) Red *(b)* Yellow
(c) White *(d)* Black

(v) This gas has a rotten egg smell :
(a) Ammonia *(b)* Hydrogen
(c) Hydrogen sulphide *(d)* Oxygen

(vi) When sodium carbonate reacts with dilute hydrochloric acid, the gas evolved is :
(a) Carbon dioxide *(b)* Nitrogen
(c) Oxygen *(d)* Hydrogen

(vii) One of the conditions necessary for a chemical reaction to occur is :
(a) Evolution of a gas
(b) Physical contact between reactants
(c) Change in colour
(d) Formation of a precipitate

(viii) What happens when a chemical reaction takes place ?
(a) A precipitate is formed
(b) A gas is evolved
(c) The substance changes its colour
(d) All of the above

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

(i) **Assertion (A) :** A chemical reaction can be written in the form of a chemical equation with the help of symbols and formulae of the substances involved.
Reason (R) : In a chemical equation, the reactants are written on the right hand side and products are written on the left hand side of the arrow.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(ii) **Assertion (A) :** There is a need to balance a chemical equation.
Reason (R) : Matter can neither be created nor be destroyed, it can only be converted from one form to another.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(iii) **Assertion (A) :** When a few drops of an aqueous solution of silver nitrate are added to the aqueous solution of sodium chloride, a white precipitate of silver chloride is formed.
Reason (R) : Some substances react only when they are heated together.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(iv) **Assertion (A) :** The correct formula of magnesium chloride is Mg_2Cl.
Reason (R) : The combining capacity of magnesium is two, while that of chlorine is one.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

3. Fill in the blanks :
(a) The substances which undergo chemical change are called ..... .
(b) The substances formed as a result of a chemical reaction are called ..... .
(c) During a chemical reaction, transfer of ..... takes place.
(d) The basic condition necessary for a chemical reaction is ..... .
(e) In some chemical reactions, an insoluble ..... is formed when two solutions are mixed.

4. Write **'true'** or **'false'** for the following statements:
(a) No new substance is formed during a chemical reaction. .....
(b) Hydrogen sulphide has rotten egg smell. .....
(c) When potassium iodide solution is added to lead acetate solution, a red precipitate is formed. .....
(d) A black residue is formed when sugar is heated. .....
(e) When iron and sulphur are heated together a grey mass is formed which is attracted by a magnet. .....

5. Match the following :

(a) The substance undergoing a chemical change	*(i)* Product
(b) The reaction in which energy is released	*(ii)* Photosynthesis
(c) The substance that increases the speed of a chemical reaction	*(iii)* Reactant
(d) A photochemical reaction	*(iv)* Exothermic reaction
(e) Represented on the right side of the arrow in a chemical equation	*(vi)* Catalyst

6. Name the gas evolved when :
(a) Calcium carbonate is heated
(b) Potassium chlorate is heated
(c) Zinc reacts with dilute hydrochloric acid
(d) Iron sulphide is heated with dilute hydrochloric acid.

7. Name the type of energy involved in the given reactions from the following list :
[Light energy, heat energy and electrical energy]
(a) Decomposition of water *(b)* Photosynthesis
(c) Magnesium and oxygen

B. SHORT ANSWER TYPE QUESTIONS

1. Define :
(a) Chemical reaction
(b) Catalyst.

2. Differentiate between reactants and products.

3. State two ways in which a chemical reaction can be represented by showing the reactants and products.

4. What happens to the atoms of the reacting molecules during a reaction ? Do they get destroyed ?

5. Write your observations when the following reactions take place :
   (a) Sugar is heated.
   (b) Dilute acetic acid is poured on baking soda.
   (c) An aqueous solution of sodium chloride is mixed with an aqueous solution of silver nitrate.
   (d) Ammonium chloride is heated with sodium hydroxide.
   (e) Water is added to quicklime.
   (f) A glass rod dipped in ammonia solution is brought near a test tube containing concentrated hydrochloric acid.

6. Write word equations for the following reactions:
   (a) Magnesium is burnt in oxygen.
   (b) Copper carbonate is heated.
   (c) Hydrogen gas is burnt in oxygen.
   (d) Sodium reacts with water.
   (e) Iron is heated with sulphur.

7. Identify the reactants and products in the following equations and represent them by word equations.
   (a) $Zn + HCl \rightarrow ZnCl_2 + H_2$
   (b) $CO + O_2 \rightarrow CO_2$
   (c) $Ca + O_2 \rightarrow CaO$
   (d) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + NaCl$
   (e) $FeS + HCl \rightarrow FeCl_2 + H_2O$

8. Write chemical equations for the following word equations using the symbols and formulae of the reactants and products.
   (a) Calcium carbonate → Calcium oxide + Carbon dioxide
   (b) Zinc oxide + Carbon → Zinc + Carbon monoxide
   (c) Iron + Sulphur → Iron sulphide
   (d) Calcium oxide + Water → Calcium hydroxide
   (e) Carbon + Oxygen → Carbon dioxide
   (f) Sulphur + Oxygen → Sulphur dioxide

9. State the law of conservation of matter.

C. LONG ANSWER TYPE QUESTIONS

1. State four conditions necessary for chemical reactions to take place.

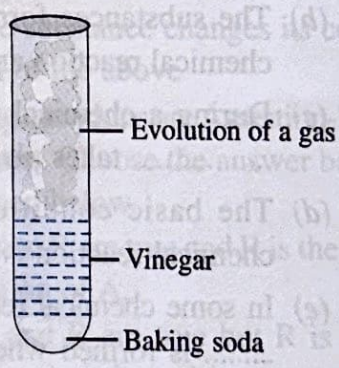
2. (a) What is a chemical equation ?
   (b) Why do we need to balance chemical equations?

3. Differentiate between the following by giving suitable examples :
   (a) Chemical reaction and chemical equation.
   (b) Exothermic and endothermic reaction.
   (c) Skeletal and balanced chemical equation.

4. A chemical reaction is represented as follows :
   $$H_2 + Cl_2 \rightarrow 2HCl$$
   (a) What information do you get from this equation ?
   (b) Write word equation for the above equation.

D. CASE STUDY BASED QUESTION

A teacher demonstrated an experiment in the class to show the evolution of a gas during a chemical reaction. She took some baking soda in a test tube and added some vinegar to it.



(a) What do you observe when vinegar is added to baking soda ?
(b) Name the gas evolved.
(c) Give the chemical names of baking soda and vinegar.
(d) Why is baking soda used while baking cakes ?
(e) In which food item can vinegar be used in place of lemon ?

Answers

Exercise : 5 — Objective type questions

1. *(i)–(a), (ii)–(a), (iii)–(c), (iv)–(b), (v)–(c), (vi)–(a), (vii)–(b), (viii)–(d)*
2. *(i)–(c)* : Assertion is true but the reason is false as reactants are written on the left hand side and products on the right hand side of the arrow in an equation.

   (ii)–(a) : The reason correctly explains the assertion. Since matter can neither be created, nor be destroyed, it is necessary to balance a chemical equation in order to follow the law of conservation of mass.

   (iii)–(b) : Both assertion and reason are true but they are independent statements. The formation of white precipate of silver chloride does not need heating, it only requires contact between two solutions.

   (iv)–(d) : Assertion is false as the correct formula of magnesium chloride is $MgCl_2$. The reason is true.

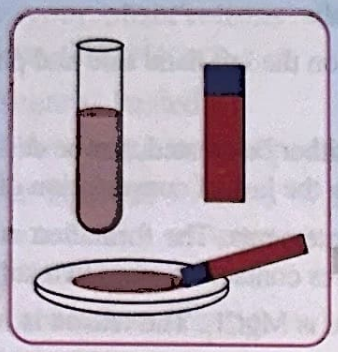
Project

Invisible writing

- Take some pure sodium hydrogen carbonate (baking soda). Make a thick paste using water in a container. Insert an ear bud in the paste and write your name on a plane sheet of paper. Do you manage to read your name written on the paper ? **No.**
  Now expose the paper with name written, to sunlight for a few minutes and watch.
  You will be able to read your name now. Discuss it with your teacher.

Metals and Non-metals

6



THEME

In day- to- day life many elements are commonly found such as iron, aluminium, zinc, lead, chlorine, carbon, sulphur etc. and their compounds. The elements have been classified in two classes, namely metals and non- metals. In this theme children will learn the classification of elements as metals and non-metals on the basis of their properties.

LEARNING OBJECTIVES

The children will be able to :

- differentiate between metals and non-metals on the basis of their physical properties such as
- lustre, conduction of electricity and heat, malleability, ductility, sonority, melting point, boiling point, density, strength;
- describe common uses of some of the metals and non-metals;
- describe the cause of corrosion of iron and other metals;
- list different ways of preventing corrosion of metallic articles used in daily life;
- list some properties and uses of metalloids.

INTRODUCTION

There are millions of substances in this world, but they are all made up of a limited number of basic substances called **elements**. You have already studied about elements as pure substances which are made up of one kind of atoms only. These elements can be represented by symbols and formulae.

You must be familiar with certain elements like gold, silver, iron, aluminium, etc. They are commonly used in making jewellery, utensils, machines and various other tools and are called metals. You have also studied about the elements oxygen, hydrogen, nitrogen, carbon, etc. They are essential for the growth and survival of all living beings and are called non-metals. Metals and non-metals find various applications in our daily life.

Activity 1

Make a list of five commonly used metals in our daily life along with their uses.

The **earth's crust** is a major source of these elements. Some of them are in abundance while some are present only in little amounts. Certain elements are found in free state, such as gold, while most of them are found in combined states. Air contains some of the non-metallic gaseous elements in free state.

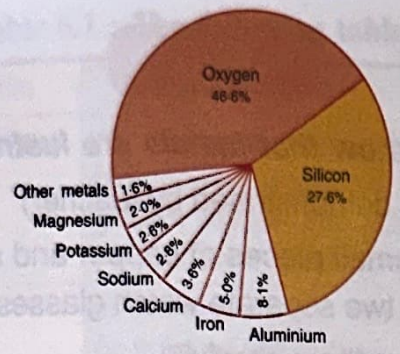


Fig. 6.1 : Showing relative percentage of elements in earth's crust

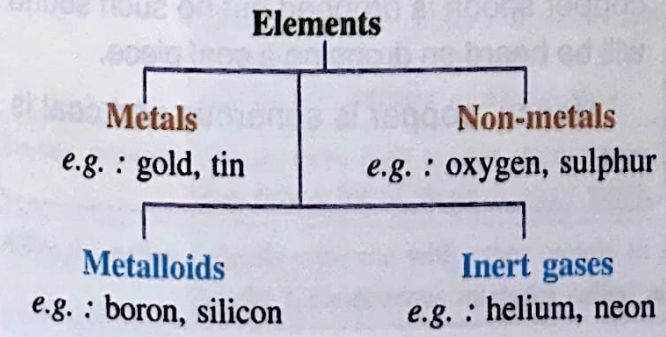
Do You Know ?

- The most abundant non-metals in earth's crust is oxygen.
- The most abundant metal in earth's crust is aluminium.

CLASSIFICATION OF ELEMENTS

For convenience of study, elements are divided into two broad classes : **metals and non-metals**. This division of elements is based on the fact that there are certain properties which are found only in metals and certain other properties found only in non-metals.

But there are also some elements that show the properties of both metals and non-metals. They are known as **metalloids**. Some common metalloids are boron, silicon, arsenic and antimony. The **noble (inert) gases** form the fourth category of elements.



METALS

Most of the elements known to us are metals.

For example : Gold, silver, platinum, aluminium, iron, zinc, tin, nickel, calcium, magnesium, etc.

The knowledge of metals is very old to mankind. Copper was perhaps the first metal to be used by man for making utensils, weapons and other objects. Most metals have a significant role in our daily life. They constitute the mineral wealth of a country.

OCCURRENCE OF METALS

Metals occur in nature in both free and combined states.

Gold and platinum are exclusively found in free state because they are not reactive. Copper, silver and mercury are also found in free state to some extent due to their low reactivity.

But most metals occur in a combined state as oxides, sulphides, carbonates, nitrates, chlorides, etc. since they are very reactive.

Sea water is also a source of metals such as sodium, potassium, magnesium and calcium in the form of their salts.

NON-METALS

Non-metals are very less in number in comparison to metals. Excluding inert gases, there are only eleven non-metals known to us. They are hydrogen, oxygen, nitrogen, carbon, fluorine, chlorine, bromine, iodine, sulphur, phosphorus and astatine.

Non-metals too play a vital role in everyday life. We need oxygen to breathe and

for respiration. Plants need carbon dioxide for photosynthesis. Proteins, carbohydrates and fats, which are important food nutrients, are made mostly of non-metals carbon, hydrogen, oxygen and nitrogen.

OCCURRENCE OF NON-METALS

- Non-metals occur in earth's crust in a combined state as oxides, carbonates, sulphates, nitrates, etc.
- In atmosphere, non-metals occur both in free state as oxygen and nitrogen and in combined state as carbon dioxide and water.
- The most abundant non-metal in earth's crust is oxygen.
- Nitrogen forms 78% of air in free state and is the most abundant element of air. It also forms an important food nutrient, protein.

Now Answer

1. Which is the second most abundant metal in the earth's crust ?
2. Which is the second most abundant element in the earth's crust ?
3. Which are the gaseous non-metals present in free state in air ?

COMPARISON OF GENERAL PROPERTIES OF METALS AND NON-METALS TO DISTINGUISH THEM

To recognise metals and non-metals, it is necessary to study and compare their general properties.

But first, let us do the following activities with some metals and non-metals to observe their properties.

Activity 2

To show that metals are lustrous.

(To be demonstrated by teacher)

Take small pieces of copper and sulphur powder in two separate watch glasses.

Observe them carefully.

You will notice that copper shines but sulphur does not shine. That means copper is **lustrous** but sulphur is **not lustrous**. Copper is a metal and sulphur is a non-metal.

Activity 3

To show that metals are hard.

Take small pieces of iron, copper, aluminium and magnesium. Try to cut these metals with a sharp knife.

What do you observe?

You will find that metals are hard and cannot be cut with a knife.

Activity 4

To compare the sonority of metals and non-metals.

Take a copper spoon and a coal piece.

Drop them one by one on the floor.

You will hear a specific sound when copper spoon is dropped but no such sound will be heard on dropping a coal piece.

Hence, copper is sonorous and coal is not.

Table 6.1 : The following table shows the general properties of metals and non-metals

Properties	*Metals*	*Non-metals*
Physical state	All metals are solids at room temperature. (*Exception :* Mercury is a liquid while gallium changes into liquid above 30°C, *i.e.* even when placed on palm of the hand.)	Non-metals exist in all the three states. Oxygen, nitrogen, chlorine, etc. are gases. Bromine is a liquid while phosphorus, sulphur, and carbon are solids.
Lustre	Metals are lustrous, *i.e.* they shine and can be polished (*Exception :* Lead is a dull metal.)	Non-metals are non-lustrous, *i.e.* they have dull surfaces (except diamond, graphite and iodine which are lustrous solids.)
Hardness	Most of the metals are hard. (*Exception :* Sodium and potassium are soft metals and can be cut with a knife.)	Solid non-metals sulphur and phosphorus are soft, (whereas carbon in the form of diamond is the naturally occuring hardest substance.)
Sonority	Metals are sonorous, *i.e.* they produce a specific sound when struck with a hard object.	Non-metals are non-sonorous. They do not produce any sound when struck.
Density	Metals have high density (sodium and potassium have low density, they are lighter than water.)	Non-metals have low density. (Exception : Iodine, graphite and diamond have high density).
Solubility	Metals are insoluble in water and other organic solvents.	Non-metals can be soluble or insoluble in water or other solvents.
Melting and boiling points	Metals have high melting and boiling points. (Sodium, potassium, gallium, caesium and mercury have low melting and boiling points.)	Non-metals have low melting and boiling points. (Exception : Graphite and diamond are forms of carbon that have high melting points).
Ductility	Metals are ductile, *i.e.* they can be drawn into wires. Gold, silver, copper, aluminium and tungsten are the most ductile metals.	Non-metals are not ductile. (*Exception :* Carbon fibre)
Malleability	Metals are mostly malleable, *i.e.* they can be beaten into thin sheets without breaking. Gold is the most malleable metal. (Zinc is brittle, on beating it breaks into pieces.)	Non-metals are brittle. They are not malleable.
Thermal and electrical conductivity	Metals are good conductors of heat and electricity. Silver is the best conductor of heat and electricity followed by copper and aluminium. (*Exception :* Bismuth and mercury are poor conductors of heat. Lead is a poor conductor of electricity.)	Non-metals are bad conductors of heat and electricity. (Carbon in the form of graphite and gas carbon conducts electricity).
Tensile strength	Metals have high tensile strength, *i.e.* they can bear a lot of strain (*except* zinc).	Non-metals have low tensile strength. (*except* carbon fibre).
Alloy formation	Metals combine with other metals to form useful homogeneous mixtures called alloys.	Non-metals do not form alloys. (Carbon forms an alloy with iron called steel).

Activity 5

To show that metals are malleable while non-metals are brittle.

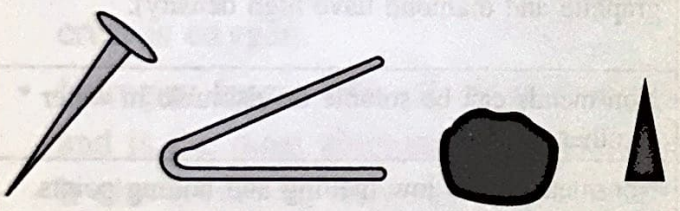
Take a small iron nail, an aluminium wire, a coal piece and a pencil lead.

Now hammer each of them separately for sometime.

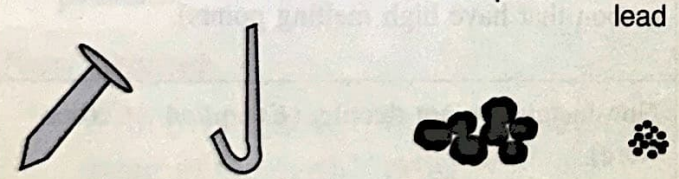
What do you observe ?

You will notice that iron nail and aluminium wire have slightly become flat, their shapes have changed, while coal piece and pencil lead are broken into pieces.

This proves that metals are malleable while non-metals are brittle.



Iron nail | Aluminium wire | Coal piece | Pencil lead



Changed shapes of iron nail and aluminium wire | Coal and pencil lead broken into pieces

Note : Have you noticed the silvery foil coated over sweets ? What is it made of ? Is it really silver ? Yes, it is a very thin sheet of silver as silver is highly malleable.

Activity 6

To show that metals are good conductors of electricity.

(To be demonstrated by teacher)

Take a battery, a bulb, a switch, aluminium wires, an iron nail and a piece of coal.

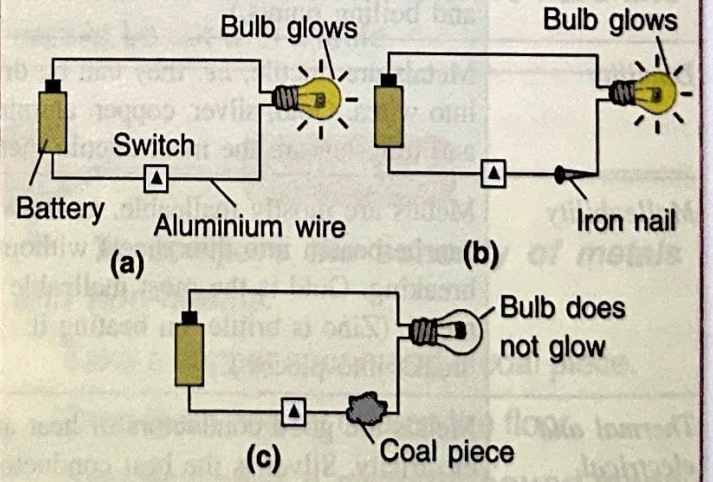
Make electric circuits as shown in Fig. (a), (b) and (c).

In the first case (a), battery, bulb and switch are connected with the aluminium wires. When the switch is on, the bulb glows. This indicates that aluminium, a metal, is a good conductor of electricity.

In the second case (b), an iron nail is connected to the wire and placed in the circuit. When the switch is on the bulb glows indicating that iron being a metal is a good conductor of electricity.

In the third case (c), a coal piece is placed in the circuit connected to the wires. You will observe that the bulb does not glow when the switch is on indicating that no current is passing through the circuit. This is because coal, a form of non-metal carbon, is a bad conductor of electricity.

Hence, it is proved that **metals are good conductors** while **non-metals are bad conductors** of electricity.



Activity 7

To show that metals are good conductors of heat.

Procedure : On a cold day, touch a metal rod and a wooden rod. Metal rod appears to be much more colder than the wooden rod

though both are at the same temperature. This is due to the fact that metals are good conductors of heat and conduct heat from our body quickly, while wood being bad conductor of heat, does not conduct heat so quickly.

Activity 8

Collect a few iron nails, copper wire, aluminium lid, zinc pieces, pencil lead, charcoal powder and sulphur powder.

Now list the properties of these substances mentioned in the following table, on the basis of the previous activities done.

Article (object)	Appearance (lustrous or dull)	Hardness (hard or soft)	Sonority (sonorous or non-sonorous	Malleable or Brittle	Conductor of electricity or non conductor of electricity
Iron nail					
Copper wire					
Aluminium lid					
Zinc pieces					
Pencil lead					
Charcoal powder					
Sulphur powder					

USES OF SOME METALS

Metals are very useful to us. They are used on the basis of their properties. A brief description of some important metals and their uses are given below.

1. **Gold (Au) :** Gold is a shiny yellow metal. It is considered as a very precious substance for its beauty, scarcity and chemical stability. Due to its chemical stability, it does not react easily with other substances. Therefore, gold is a durable substance. Gold is highly malleable and resistant to corrosion, and it is also a very good conductor of electricity. Therefore, it is used :
   (i) for making jewellery and coins.
   (ii) in the manufacture of electronic devices, like computers, telephones, home appliances, *etc.*
   (iii) in dentistry, to fill in teeth cavities (in the form of an alloy, a mixture of gold and silver).

   Pure gold is a very soft metal. In fact, it is the most malleable metal known to man. It is variously mixed with copper, nickel, silver, *etc.*, to make it harder and cheaper. 100% pure gold is 24 carat*. Generally, in India, 22 carat gold is used for making ornaments. It means that 22 parts of pure gold is alloyed with 2 parts of either copper or silver.

2. **Silver (Ag) :** Silver is a greyish white, lustrous metal. It is one of the highly malleable and ductile metals and also the best known conductor of heat and electricity. Its chemical reactivity is more than that of gold. It is used :
   (i) for making jewellery and coins.
   (ii) for filling in teeth cavities (in the form of a gold-mixed alloy).
   (iii) as a purifier of water.
   (iv) as an electrode metal, and in electroplating.
   (v) in photography in the form of silver bromide and silver nitrate.
   (vi) for making thin sheets to decorate sweets. It increases the immunity of the body.

* *CARAT is the measure of the purity of precious substances.*

Silver is not used for making electric wires, because it is a costly metal.

3. **Iron (Fe) :** Iron is the most useful and one of the cheapest metals of all. It is known to man since ancient times. It is a reactive metal and is found in combined states. Iron is used :
   (i) to make pipes, tanks, cylinders, agricultural tools, nails, wire meshes, railings, furniture, *etc.*
   (ii) to make bridges, ships, machine parts, buildings, *etc.* in the form of *steel,* an alloy containing carbon, due to its high tensile strength.
   (iii) in the construction of power transmission towers.
   (iv) for making utensils, cutlery and surgical instruments from its alloy stainless steel, which is steel with chromium or nickel.

4. **Copper (Cu) :** Copper is a reddish brown metal. It is highly ductile and a very good conductor of heat and electricity. It finds applications in :
   (i) making electric wires, coils and cables, utensils, and even semi-precious ornaments.
   (ii) making coins and statues in the form of its alloys called bronze and brass.
   (iii) electronic devices.

Activity 9

Consider some metals such as iron, copper, aluminium, lead, etc.

Which of the above metals are available in the form of wires ?

Do You Know ?

The ability of metals to be drawn into thin wires is called ductility. Gold is the most ductile metal.

You will be surprised to know that a wire of about 2 km length can be drawn from one gram of gold.

5. **Aluminium (Al) :** Aluminium is the most abundant metal found in the earth's crust, and the second most useful metal after iron. It is silvery white in colour. It is malleable and ductile, and it is a very good conductor of heat and electricity. Therefore, it is used :
   (i) as a substitute for copper to make electric cables and wires, because it is cheaper.
   (ii) to make utensils, cans for drinks, furniture, window frames, *etc.*
   (iii) for packaging foodstuffs and wrappers of medicines in the form of aluminium foil.
   (iv) as a paint to prevent rust in the form of a powder foil on telephone poles.

Aluminium is a light and shiny metal but it is a very soft metal. To make it stronger, it is mixed with other metals and then it is used to make alloys such as duralumin (Al, Cu, Mg, Mn) which is used to make the bodies of aircrafts, automobiles, machine parts, tools, etc.

6. **Zinc (Zn) :** Zinc is a bluish-white metal. It is neither malleable nor ductile; rather it is brittle. It is used :
   (i) as an electrode metal.
   (ii) for making alloys like brass and bronze.
   (iii) to make dry cells.

(iv) to coat iron sheets through the process of *galvanization*, which protects iron from rusting.

7. Mercury (Hg) : Mercury is a silvery white, liquid metal. It does not moisten glass, and it expands or contracts even on little heating or cooling. Therefore, it is used in :

(i) thermometers as thermometric liquid.

(ii) barometers and other scientific apparatus.

(iii) dentistry, for filling up cavities in teeth in the form of alloys known as *silver amalgam* and *gold amalgam*. (An alloy of metal with mercury is called amalgam).

8. Tin (Sn) : It is a silvery white metal. It is highly malleable but not ductile. It does not rust. It is used :

(i) for making cans to keep foodstuffs, especially grains.

(ii) for tinning of food cans and utensils made of other metals.

(iii) Tin coated iron sheets are used to make boxes and big containers.

9. Magnesium (Mg) : It is available in the form of ribbons. It is used :

(i) in fireworks, because it burns with a dazzling light.

(ii) for making alloys called magnalium and duralumin.

Note : Magnesium is very vital for life. The green pigment chlorophyll, present in the plants contains magnesium compounds which help to capture the solar energy for photosynthesis.

Do You Know ?

The various colours we see in fireworks are mainly produced by compounds of metals like magnesium, strontium and barium. Magnesium produces a brilliant white light, strontium compounds produce crimson and barium compounds produce green colour.

10. Lead (Pb) : Lead is a very heavy, silvery grey metal. It is used for :

(i) making storage batteries.

(ii) making sanitary pipes, tips of bullets, tin roofs, and fusible alloy (solder — a mixture of 50% tin and 50% lead).

Do You Know ?

Gold and platinum are known as noble metals because they are chemically inert. They are unaffected by air, water, acid or alkali. They are found in free state and they retain metallic lustre for a long time.

11. Platinum (Pt) : Platinum is lustrous and a very precious metal. Like gold, platinum too does not react easily with other substances. Platinum is also an excellent catalyst. It is used :

(i) for making electrodes and electrolytic cells.

(ii) for making expensive jewellery and watches.

(iii) as a catalyst in hydrogenation of unsaturated oil to prepare vanaspati *ghee*, in the manufacture of acids, *etc.*

12. Tungsten (W) : It is a shiny grey metal, occurring in solid state at room

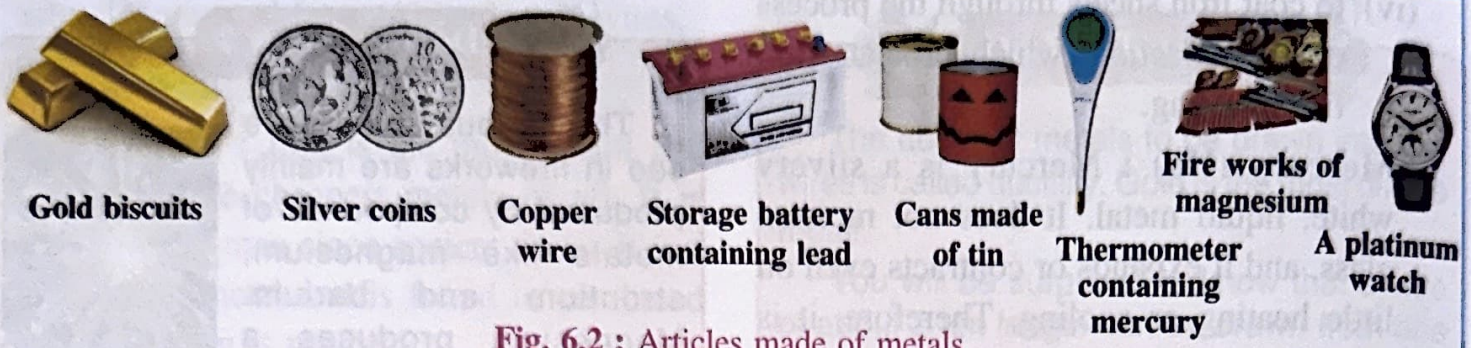


Fig. 6.2 : Articles made of metals

temperature. It can withstand high temperature because it has highest melting point among metals. It is highly resistant to acids. It is used in :

- making electrodes.
- heating elements.
- electric bulbs as filaments and cathode ray tubes.
- making heavy metal alloys used to make cutting tools.
- mining industries, etc.

Activity 10

Identify the metals from which following articles are made :

(i) Nail (ii) Electric wire
(iii) Ornaments (iv) Food cans
(v) Chocolate wrapper (vi) Sanitary pipes

USES OF SOME NON-METALS

The non-metals important to us are oxygen, nitrogen, hydrogen, carbon, iodine, chlorine, etc. They give us many useful products. Some of the non-metals and their uses are discussed below :

1. Oxygen (O_2) :

(i) Human beings and animals use oxygen for breathing and respiration.

(ii) In hospitals, oxygen cylinders are available for serious patients.

(iii) It has many industrial uses.

(iv) It is important for combustion.

2. Nitrogen (N_2) :

(i) It is used in food canning such as in packaging of potato chips and in electrical bulbs due to its non-reactive nature.

(ii) It is used for producing many compounds which are used as fertilizers like urea.

(iii) Liquid nitrogen is used as a coolant in refrigeration industry.

3. Carbon (C) : Carbon is one of the most important and widely distributed elements. It mainly occurs as coal, graphite and diamond in *free state*, while in a *combined state* it is present in all living organisms as food nutrients like carbohydrates, proteins, vitamins, *etc*. In the atmosphere, it is present as carbon dioxide.

(a) Coal : Coal is mostly used as a fuel in homes and industries. Many trains still run on coal. It is used in the pharmaceutical and textile sectors as a source of synthetic chemicals and textiles.

(b) Graphite : It is a crystalline, black, lustrous substance. It is soft and slippery, and marks paper black. Graphite is used :

(i) in the leads of pencils.

(ii) as electrode material in electrolytic cells because it is a good conductor of electricity.

(iii) as a solid lubricant (*e.g.* in grease) for machine parts.

(iv) for making heat resistant crucibles for keeping molten metals because of its high melting point and good conductivity of heat.

(v) in nuclear reactors to control the speed of nuclear reactions.

(c) Diamond : It is one of the crystalline forms of carbon. It shines brilliantly and is the hardest naturally occurring substance known to us.

(i) Pure diamond is used as a precious gem in jewellery.

(ii) Impure diamond is used for cutting glass and grinding hard substances, and in drilling heads.

(d) Charcoal : (i) It is used as a decolourising agent in sugar industry.

(ii) It is used to purify water as it removes the colour and smell if present in water.

4. **Iodine (I_2) :** Iodine is a dark grey, crystalline solid, with a metallic lustre. It is insoluble in water but soluble in alcohol and potassium iodide solutions. It is used :

(i) in a very small quantity in the form of iodized salt for the healthy growth of the human body. Iodine deficiency causes goitre.

(ii) in photographic films in the form of potassium iodide.

(iii) to make *tincture of iodine* and *iodex*, which are used as an antiseptic and pain reliever (analgesic), respectively. Tincture of iodine is solution of iodine in alcohol.

5. **Chlorine (Cl_2) :** It is a greenish yellow gas with a pungent, suffocating smell. It is fairly soluble in water, forming a pale yellow solution, called chlorine water. Chlorine is used :

(i) as a bleaching agent for rough and hard fibres like jute and cotton.

(ii) as a disinfectant for sterilizing both drinking water and swimming pool water.

(iii) in the manufacture of chemicals like D.D.T., B.H.C. and bleaching powder, which are used as insecticide, pesticide and disinfectant, respectively.

1. **DDT**—Dichlorodiphenyl trichloro ethane.
2. **BHC** — Benzene hexachloride.

(iv) in the manufacture of mineral acids like hydrochloric acid.

(v) For preparing polyvinyl chloride (P.V.C.), a type of heat resistant plastic used for making pipe and frames.

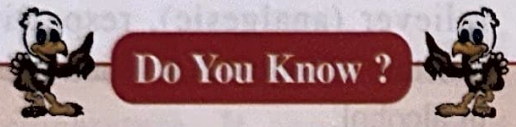
6. **Fluorine :** It is a yellow gas with an irritating (offensive) smell. Fluorine is used:

(i) as sodium fluoride in toothpastes and mouthwashes.

(ii) as fluorides to prevent dental problems like tooth decay and cavities.

(iii) as chlorofluorocarbons (CFCs) in refrigerants (used in air conditioners refrigerators, etc.) and aerosol sprays.

(iv) in making Teflon (a fluorocarbon), which is used as a coating in non-stick pans and other utensils.



CFCs are harmful for the environment as they are potent greenhouse gases. They play a significant role in depleting the ozone layer. They are being phased out and replaced by Hydrochlorofluorocarbons (HCFCs) and Hydrofluorocarbons (HFCs), which have far less detrimental effects on the ozone layer.

7. Hydrogen (H_2) :

(i) It is used as a reducing agent.

(ii) It is used for converting vegetable oils into ghee (hydrogenation of oils).

(iii) It is used to make ammonia which is used to prepare fertilizers.

8. Sulphur (S_8) :

(i) It is used as fungicide and insecticide.

(ii) It is used in the manufacture of sulphuric acid.

(iii) It is widely used in the manufacture of sulphur drugs used as antibiotics.

(iv) It is used in match box industry for making striking heads of the match sticks (Phosphorous sulphide).

(v) It is used to vulcanize rubber (a process of converting rubber latex to hard rubber).

METALLOIDS

Metalloids are the elements which show some properties of metals and some properties of non-metals. They all are solids. They are silicon, boron, arsenic, antimony, germanium, tellurium and polonium.

USES OF SOME METALLOIDS :

1. Silicon (Si) : Silicon is one of the most useful metalloids available to mankind. It does not occur in free state in nature. In combined state as **sand**, it is the second most abundant element of the earth's crust (next only to oxygen). Sand is chemically known as silicon dioxide (SiO_2), also called as silica.

- Highly pure silicon is used in making microchips for computers, transistors, solar cells, rectifiers and other solid state devices that are used extensively in the electronic and present age space industries.
- It is used in the manufacture of a waterproof material called "silicone". **Silicone** is used to make bags, umbrellas, raincoats, etc.
- It is used to prepare a substance called **silicon carbide** which is the hardest compound prepared by man. It is used in various kinds of grinding tools.
- In the form of **sand** (SiO_2), it is used to prepare concrete, bricks and cement. Sand is the principal ingredient of glass.
- In the form of silicates, silicon is used in enamel, pottery, etc. and for preserving eggs.
- It is also used to make electrical steel (silicon steel), an alloy of iron in which magnetic properties are retained.

2. Germanium (Ge) : It is a hard, lustrous, greyish white metalloid. It is chemically similar to tin and silicon.

- Germanium is used as a semiconductor.
- It is used as a transistor in many electronic

appliances when mixed with arsenic, gallium, antimony, etc.

- Germanium is also used to form alloys and as a phosphor in fluorescent lamps.

3. Antimony (Sb) : It is a lustrous grey metalloid.

- Antimony is used in electric industry to make semiconductor devices.
- It is alloyed with lead to improve its hardness and strength and is used in batteries.
- It is also used in printing presses as type metal.
- Antimony compounds are used to make paints, enamels, glass pottery, etc.

INERT OR NOBLE GASES

They are non-metallic, colourless, gaseous elements. They are chemically inert, *i.e.* they do not combine with other elements to form compounds, hence they are called as inert or noble gases. They are found only in traces. They are six in number : Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe) and Radon (Rn).

USES OF INERT GASES :

(i) **Helium (He) :** It is the second lightest element known to man. It is used for filling up weather observation balloons.

(ii) **Argon (Ar) :** It is used for filling into electric bulbs so as to provide an inert atmosphere to the heating filament. It improves the quality and life of the bulb (but commonly nitrogen is used because of its easy availability).

(iii) **Neon (Ne) :** It is used in making advertising signboards and tubelights. It emits a coloured light when a high voltage current is passed through it.

(iv) **Radon (Rn) :** It is the only radioactive inert gas used for cancer treatment.

(v) **Xenon (Xe) and Krypton (Kr) :** They find application in photography.

CORROSION IN METALS

Most of the reactive metals, when exposed to the atmosphere, react with moist air and get corroded due to which their properties are destroyed and their use as metals diminishes. By reacting with moist air, they change into more stable compounds.

Rusting of iron

Iron is a metal which gets easily corroded. When exposed to moist air, it forms a reddish brown layer known as **rust** on its surface. The process of formation of rust is called as **rusting**.

Rust is hydrated ferric oxide ($Fe_2O_3 \cdot x^*H_2O$), which is flaky and easily crumbles off from the surface of the metal, thus exposing the free iron surface for further rusting.

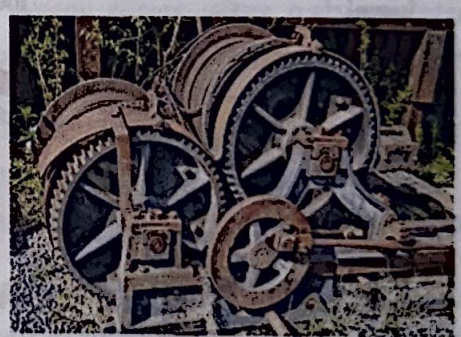


Fig. 6.3

$$\text{Iron} + \text{Oxygen} \xrightarrow[\text{slow process}]{\text{Atmospheric moisture}} \text{Hydrated ferric oxide (rust)}$$

$$4Fe + 3O_2 \xrightarrow[\text{slow process}]{x\,H_2O} 2Fe_2O_3 \cdot x\,H_2O$$

* *x may vary in number*

***Rusting** is a slow oxidation process in which iron slowly reacts with oxygen (of the air) in presence of moisture and produces a flaky brown substance, called **rust**.*

Rusting corrodes iron, weakens iron structures and slowly destroys the metal, thus causing heavy economic loss.

Activity 11

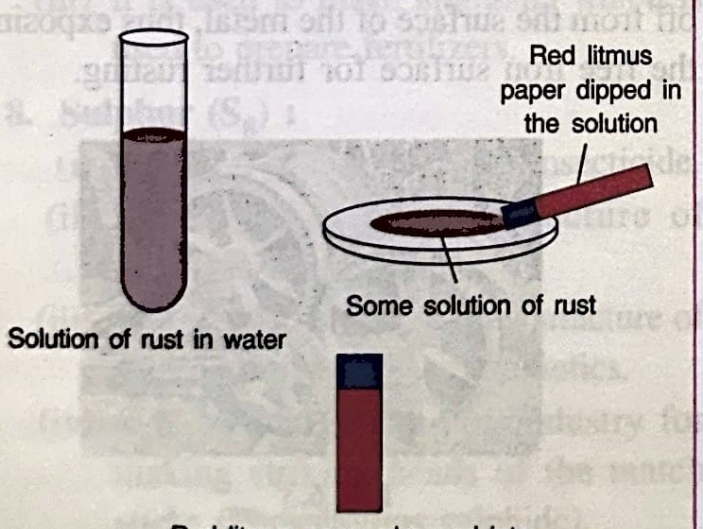
To show the basic nature of rust.

Take some rust from the surface of a rusted iron nail. Crush it into powder. Take this powder in a test tube, add some water and shake the test tube properly.

Now introduce blue litmus paper and red litmus paper in the mixture of water and rust, one by one.

You will observe that red litmus turns blue.

This proves that rust ($Fe_2O_3 \cdot xH_2O$) is basic in nature.



Conditions for rusting

Air (oxygen) and **moisture** are the two conditions necessary for rusting. This can be shown by the following activity.

Activity 12

Take three small test tubes marked A, B and C. Put some bright iron nails in each of them.

In test tube A, put some boiled water which expels all the air (oxygen) from it. Close it with a rubber stopper so as to make it air tight.

In test tube B, put a few pieces of anhydrous calcium chloride to absorb the moisture inside. Tightly close the test tube with an airtight rubber stopper. In test tube C, put some tap water and leave it open.

After a few days, it is observed that only the nails in test tube C get rusted. This is because they are in contact with both air and water. The nails in test tubes A and B are not rusted because, in test tube A, there is water but no air and in test tube B, there is air but no water. This proves that both air and water are necessary for rusting.

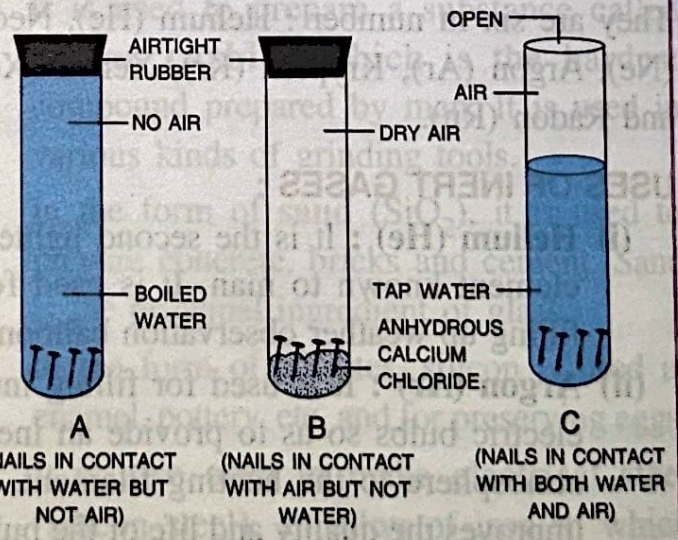


Fig. 6.4

The presence of acidic gases such as CO_2, SO_2, SO_3 and NO_2 in the air increases the rate of rusting of iron.

Look at a silver object at your home. It could be the silver trophy you have won in a school competition or a silver coin. Compare its shine with the shine of silver objects you watch in a jewellery shop. Is the shine same ? If not, try to find the reason for the decline in shine.

This is because a coating of black silver sulphide is formed on the surface of silver due to pollutant gases present in atmospheric air.

Metals like copper, aluminium, lead, etc. also react with moist air, when exposed to the atmosphere.

Copper forms a green deposit of basic **copper carbonate** when exposed to moist air. If you have copper articles at home, you can notice the green deposit on their surfaces after some days.

Aluminium has a strong affinity for oxygen, hence it reacts with air to produce **aluminium oxide** (Al_2O_3).

When **lead** gets exposed to moist air, it forms a white deposit on its surface which is basic **lead carbonate**.

Prevention from Rusting

Iron is one the most useful metals to mankind. It is used for making heavy machines, bridges, vehicles, railway equipments, utensils, *etc.*

But iron gets rusted easily in the presence of air and moisture. If rust is allowed to form unchecked, iron gets corroded, resulting in the wastage of metal. Thus, the prevention of rusting of iron is essential.

Iron can be protected from rusting if the surface of iron is prevented from coming in contact with moist air. This is done by the following methods :

By painting

1. (a) ***Painting with red lead*** : This coating is put on the heavy iron parts of vehicles, machinery and materials such as grills, gates, etc used in buildings.

   (b) ***Oil paint*** : This paint is applied on doors and windows made up of iron which already have a base coating of red lead.

   (c) ***Enamel coating*** : Enamel is a mixture of iron or steel with silicates. It is used to coat cooking appliances, refrigerators, *etc.*

   (d) ***Coal tar*** : It is used for coating the lower parts of ships and bridges.

   (e) ***Plastic coating*** : It is used to cover steel pipelines, hangers, stands, etc.

2. ***Grease or oil*** : Metallic tools and machine parts are protected from rusting by applying a thin film of grease or oil that also lubricates them.

3. **Metallic coating**

   (i) ***Galvanization*** : In this process, iron articles are dipped in molten zinc. Galvanization is used for those iron articles which are exposed to heavy moisture such as roof sheds, buckets, tubs, water pipes, *etc.* Water pipes after galvanization are known as G.I. (galvanized iron) pipes.

Galvanized steel is not used for food containers, since acids present in food react with zinc to form poisonous compounds.

   (ii) ***Tin plating*** : In this case, iron sheets are dipped in molten tin. These sheets are used to make boxes and cans for packaging vegetables, meat, beverages, *etc.* Tin is a corrosion-resistant and non-toxic metal.

   (iii) ***Aluminium painting*** : This process protects iron bridges, iron furniture, car bodies, telephone poles, *etc.* from rusting.

   (iv) ***Alloy formation*** : By this process, iron is fused with nickel, chromium, carbon,

etc. to form stainless steel, which is a lustrous, corrosion-resistant substance. It is used for making cutlery, surgical instruments, *etc.*

(v) ***Electroplating*** : Iron articles are electroplated with copper, chromium, tin, nickel, *etc.* for decorative purposes in jewellery, and for the durability of cutlery, cooking gas stoves, motor parts, etc. But it is a costly process.

The iron pillar near Qutub Minar is notable for its rust-resistant composition. It is made up of iron mixed with high percentage of phosphorous which has prevented it from corrosion.

It has not rusted for the last eight hundred years. It highlights the achievements of ancient India in chemical engineering.

RECAPITULATION

- Elements are classified into metals, non-metals, metalloids and inert gases.
- Metals are lustrous, hard, mostly solids, have high density, high melting point, high boiling point, malleable, ductile, good thermal and electrical conductors, have high tensile strength and are sonorous while non-metals are not.
- Metals and non-metals are very important and useful to mankind.
- Metalloids are the elements showing some properties of metals and some properties of non-metals.
- Inert gases are non-reactive non-metals like helium, neon, argon, etc.
- The properties of reactive metals are degraded due to exposure to moist air. The most easily corroded metal is iron due to the formation of rust ($Fe_2O_3 \cdot xH_2O$).
- Rusting of iron can be prevented by different methods, such as painting, greasing, galvanization, electroplating, alloy formation, etc.

EXERCISE

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(*i*) The most abundant metal present in the earth's crust is :
(*a*) Iron (*b*) Calcium
(*c*) Aluminium (*d*) Potassium

(*ii*) The most abundant non-metal present in earth's crust is :
(*a*) Oxygen (*b*) Nitrogen
(*c*) Carbon (*d*) Hydrogen

(*iii*) A metal found in a free state is :
(*a*) Copper (*b*) Zinc
(*c*) Sodium (*d*) Gold

(*iv*) A non-metal occurring in the solid state is:
(*a*) Bromine (*b*) Phosphorus
(*c*) Chlorine (*d*) Nitrogen

(*v*) Which of the following metals exists as a liquid under normal conditions ?
(*a*) Mercury (*b*) Bromine
(*c*) Magnesium (*d*) Lithium

(*vi*) A metal with melting point less than 50°C is :

(a) Gallium (b) Iron
(c) Gold (d) Aluminium

(vii) A metal which is neither ductile nor malleable is :
(a) Copper (b) Silver
(c) Zinc (d) Aluminium

(viii) The colour of rust which is a hydrated oxide of iron is :
(a) Reddish-brown (b) Green
(c) White (d) Black

(ix) Aluminium is not used to make :
(a) Foils (b) Wires
(c) Fireworks (d) Utensils

(x) A metalloid used in the manufacture of computer microchips is :
(a) Antimony (b) Germanium
(c) Silicon (d) Arsenic

(xi) A metalloid used to make glass is :
(a) Sulphur (b) Germanium
(c) Silicon (d) Antimony

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

(i) **Assertion (A) :** Most of the metals can be beaten into sheets without breaking.
Reason (R) : Most of the metals are malleable.
(a) (1) (b) (2) (c) (3) (d) (4)

(ii) **Assertion (A) :** Mercury is used as a thermometric liquid.
Reason (R) : Mercury can expand and contract only at high temperatures.
(a) (1) (b) (2) (c) (3) (d) (4)

(iii) **Assertion (A) :** Non-metals have low melting and boiling points.
Reason (R) : Diamond and graphite are lustrous solids.
(a) (1) (b) (2) (c) (3) (d) (4)

(iv) **Assertion (A) :** The second most abundant non-metal in the earth's crust is silicon in the form of sand.
Reason (R) : A compound of silicon called silicon carbide is the hardest substance prepared by man.
(a) (1) (b) (2) (c) (3) (d) (4)

(v) **Assertion (A) :** Helium is used for filling up weather observation balloons.
Reason (R) : The inert gas helium is the second lightest element known to humans.
(a) (1) (b) (2) (c) (3) (d) (4)

3. Fill in the blanks :

(a) The most ductile metal is ..... .

(b) A metal stored in kerosene oil is .....

(c) Tungsten metal is a good conductor of ..... .

(d) ..... is a soft metal.

(e) ..... is the hardest compound known to us.

(f) A non-metal used to purify water is ..... .

(g) A metal that gives dazzling effect to crackers when they explode is .....

(h) A chemical compound that makes up the striking heads of match sticks is ..... .

4. Write *'true'* or *'false'* for the following statements :

(a) Silver is used to make electric cables.

(b) Iodine acts as an antiseptic in the form of tincture of iodine.

(c) Sodium can be cut with a knife.

(d) Antimony is a metal.

(e) Sand is an oxide of silicon.

5. Match the following :

Column A	Column B
(a) Helium	(1) Electric bulb
(b) Neon	(2) Thermometer
(c) Argon	(3) Semiconductor
(d) Germanium	(4) Weather observation balloons
(e) Mercury	(5) Advertising signboards

6. Name a metal which is :
   (a) the most malleable,
   (b) brittle,
   (c) as precious as gold,
   (d) easily cut with a knife,
   (e) used in making electric cables,
   (f) used as a thermometric liquid,
   (g) the best conductor of electricity.

7. Name a non-metal which is :
   (a) a good conductor of heat and electricity,
   (b) the hardest naturally occurring substance,
   (c) used to kill germs in water,
   (d) lustrous,
   (e) used for filling into electric bulbs,
   (f) used for cancer therapy,
   (g) liquid at room temperature.

B. SHORT ANSWER TYPE QUESTIONS

1. Choose the odd one out and give reason for your answer.
   (a) Gold, silver, copper, sodium
   (b) Iron, aluminium, silicon, tin
   (c) Silver, copper, aluminium, zinc
   (d) Sulphur, phosphorus, carbon, bromine
   (e) Brass, bronze, steel, iron

2. Name the metals present in the following alloys:
   (a) Brass *(b)* Bronze
   (c) Duralumin *(d)* Stainless steel

3. Mention *two* uses of the following metals and non-metals :
   (a) Iron *(b)* Aluminium
   (c) Gold *(d)* Oxygen
   (e) Iodine *(f)* Germanium
   (g) Tungsten

4. Give one use of the following inert gases :
   (a) Helium *(b)* Neon
   (c) Argon *(d)* Radon

5. Give reasons :
   (a) Magnesium is used in fireworks.
   (b) Aluminium is used in making aircrafts.
   (c) Copper is used in making electric cables.
   (d) Graphite is used in the leads of pencils.
   (e) Impure diamond is used to cut glass.
   (f) Gold is mixed with copper and nickel.
   (g) Tungsten is used in electric bulbs.

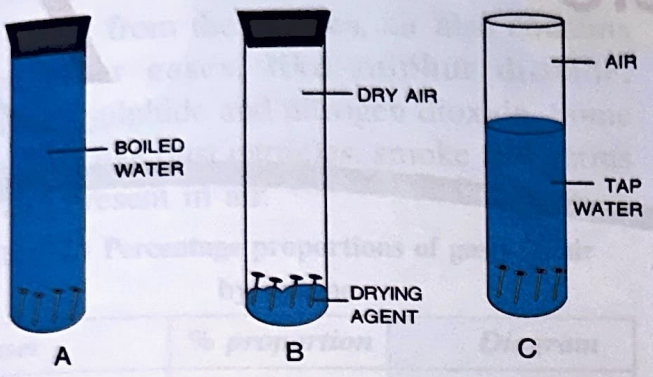
C. LONG ANSWER TYPE QUESTIONS

1. Give *four* differences between metals and non-metals with reference to their :
   (a) Melting point and boiling point,
   (b) Conductivity of heat and electricity,
   (c) Malleability
   (d) Solubility in water

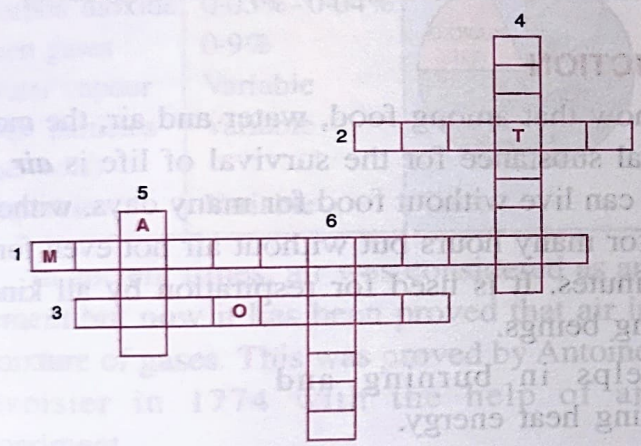
2. *(a)* What are metalloids ?
   (b) Give *two* uses of :
   (i) Silicon *(ii)* Antimony

3. *(a)* What is rusting ?
   (b) What is the chemical formula of rust ?
   (c) What happens to iron due to rusting ?
   (d) State two methods by which rusting of iron can be prevented.
   (e) Three test tubes A, B and C are shown ahead containing clean iron nails.
   In test tube A iron nails are kept in boiled water.
   In test tube B, iron nails are in contact with dry air only.
   In test tube C nails are kept in tap water.
   (i) Nails in which of the test tubes will get rusted ? Why ?

(ii) Why is there no rusting of iron nails in the other two test tubes ? Explain.



4. Solve the following crossword puzzle with the help of the given clues.



Across :

1. The property due to which a metal can be beaten into sheets.
2. This is the property due to which the metals shine.
3. The property due to which metals produce sound.

Down :

4. The property due to which a metal can be drawn into wires.
5. The substance formed when two or more metals are mixed together.
6. Non-metals are given this term because they can be easily broken down into pieces.

D. CASE STUDY BASED QUESTION

Pratham was looking at the silver trophy he had won in a school competition last year. He noticed that its shine had diminished. He went and asked his chemistry teacher why this would happen and he was told about corrosion of metals.

(a) Why does the shine of the silver trophy diminish after sometime ?

(b) Name the metal on which a green deposit is formed when exposed to air. What is that green deposit ?

(c) Why is it necessary to paint bridges, car bodies, telephone poles, etc. ?

Answers

Exercise : 6 — Objective type questions

1. *(i)*–*(c)*, *(ii)*–*(a)*, *(iii)*–*(d)*, *(iv)*–*(b)*, *(v)*–*(a)*, *(vi)*–*(a)*, *(vii)*–*(c)*, *(viii)*–*(a)*, *(ix)*–*(c)*, *(x)*–*(c)*, *(xi)*–*(c)*
2. *(i)*–*(a)* : The reason correctly explains the assertion. Metals are malleable, that is why they can be beaten into sheets without breaking.

   (ii)–*(c)* : Assertion is true but the reason is false as mercury can expand and contract even at very low temperatures, that is why it is used as a thermometric liquid.

   (iii)–*(b)* : Both assertion and reason are true but they are two independent statements.

   (iv)–*(b)* : Both assertion and reason are true about silicon but they are two independent statements.

   (v)–*(a)* : The reason correctly explains the assertion. Helium is used in weather balloons because it is light and does not react.

Air and Atmosphere

7

THEME

Air is a mixture of some gaseous components which have wide use in daily life. For example, nitrogen is an important constituent of fertilizers and oxygen is essential for our body for sustenance of life. These gases have important physical and chemical properties and uses.

LEARNING OBJECTIVES

The children will be able to :

- review that air is a mixture of gases.
- recall the components of air.
- discuss the use of oxygen and nitrogen in different life processes.
- explain from an activity that mass change takes place during combustion.
- express the reaction in the form of word equation.
- describe the preparation of oxygen in the laboratory using potassium chlorate/hydrogen peroxide and manganese dioxide as a calalyst.
- understand the concept of catalyst.

A. AIR : A Mixture of Gases

INTRODUCTION

- You know that among food, water and air, the most essential substance for the survival of life is ***air***. A person can live without food for many days, without water for many hours but without air not even for a few minutes. It is used for respiration by all kinds of living beings.
- Air helps in burning and producing heat energy.
- Air is invisible and transparent because it is a mixture of colourless gases.
- Although we cannot see air, we can feel its presence.

OCCURRENCE

- Air occurs in the atmosphere which surrounds the earth and extends to about 300 km above its surface.
- Air also occurs in water in a dissolved state which helps aquatic plants and animals to survive.

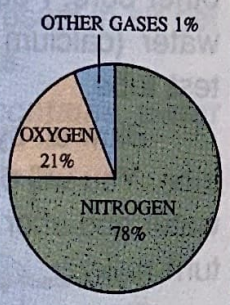
CONSTITUENTS OF AIR

The main constituents of air are nitrogen and oxygen. It also contains carbon dioxide and water

vapour in small amounts. Inert gases and ozone are also present in small amounts in air.

Apart from these gases, air also contains some other gases, like sulphur dioxide, hydrogen sulphide and nitrogen dioxide. Some impurities like dust particles, smoke and germs are also present in air.

Table 7.1 Percentage proportions of gases in air by volume

Gases	*% proportion*	*Diagram*
Nitrogen	78%	
Oxygen	21%	
Carbon dioxide	0·03% - 0·04%	
Inert gases	0·9%	
Water vapour	Variable	
Dust particles (aerosols)	Variable	
Impurities	Variable	

In ancient times, air was considered as an element but now it has been proved that air is a mixture of gases. This was proved by Antoine Lavoisier in 1774 with the help of an experiment.

Lavoisier's experiment to show that nitrogen and oxygen are the main constituents of air :

Lavoisier took some mercury in a special glass apparatus called a retort and heated it. The other end of the retort was connected to a bell jar containing air. The bell jar was kept in an inverted position over mercury contained in a trough as shown in the figure below.

The following were his observations :

1. A red layer of mercuric oxide was formed on the hot surface of mercury in the retort.

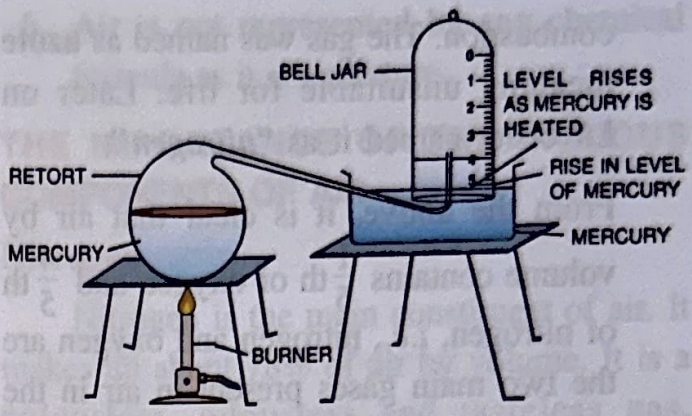


Fig. 7.1 : Lavoisier's Experiment

2. The level of mercury in the bell jar rose by $\frac{1}{5}$th of the total volume of the bell jar.

The following were his conclusions :

1. $\frac{1}{5}$th portion of air in the bell jar was used up by mercury in the retort to form the red substance. This air was ***active air***. This could be re-obtained on strongly re-heating the red substance. This gas supported burning better than air and also supported life. Lavoisier named this active air as *"oxygen"*.

$$\text{Mercury} + \text{Oxygen} \xrightarrow{\text{heat}} \text{Mercury (II) oxide (Red powder)}$$

$$2\,Hg + O_2 \xrightarrow{\text{heat}} 2\,HgO$$

$$\text{Mercury (II) oxide} \xrightarrow{\text{heat}} \text{Mercury} + \text{Oxygen}$$

$$2\,HgO \xrightarrow{\text{heat}} 2\,Hg + O_2$$

2. The remaining $\frac{4}{5}$th portion of air in the bell jar was ***inactive air*** as it did not support burning of air. It was found that this inactive air neither supports combustion (burning) nor life. The $\frac{4}{5}$th air in the bell jar was tested by putting a burning candle into the gas. The flame was extinguished. It did not support

combustion. The gas was named as **azote** meaning unsuitable for life. Later on Lavoisier named it as *"nitrogen"*.

From the above, it is clear that air by volume contains $\frac{1}{5}$th of oxygen and $\frac{4}{5}$th of nitrogen, *i.e.*, nitrogen and oxygen are the two main gases present in air in the ratio of 4 : 1 by volume. The rest of the gases are present in very small amounts. This can also be shown by a simple activity given below.

Activity 1

To show that air contains oxygen (an active part) and nitrogen (an inactive part).

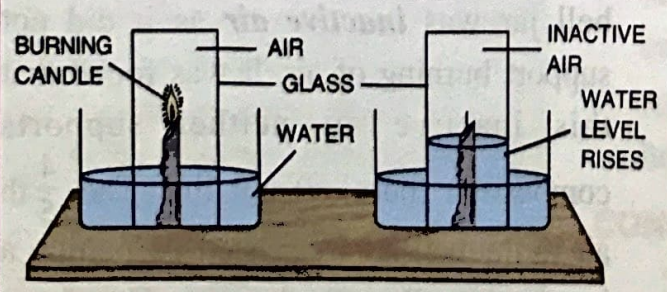
Fix a candle in the middle of a shallow container. Fill the container with some water. Cover it with an empty inverted glass jar and mark the level of water inside the jar.

Lift the jar and light the candle, then cover it with the inverted glass jar.

Observe carefully.

Does the candle continue to burn or gets extinguished ? Does the level of water inside remains the same or not ?

You will notice that the candle continues to burn for sometime and then gets extinguished. The water level rises slightly *i.e.*, upto 1/5th part of the jar containing air. This part is **active air** *i.e.* oxygen which helps candle to burn. When it is used up, candle stops burning. This 4/5th part of air still present in the jar is **inactive air** that does not support burning and it is nitrogen.



Air contains oxygen and nitrogen

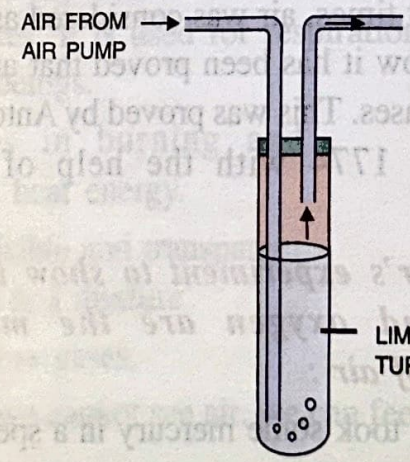
Activity 2

To show that air contains carbon dioxide.

Take a test tube fitted with a two-bored rubber-cork. Fit a long bent tube through one bore and fit a short bent tube through the other bore. Pour some freshly prepared lime water (calcium hydroxide solution) into the test tube.

Blow air by an air pump through the long tube. You will observe air bubbles in the lime water and after some time the lime water will turn milky.

This shows that *air contains carbon-dioxide.*



Air contains carbon dioxide

Note : Lime water is used in the chemical test to identify carbon dioxide gas because the two substances react to form an insoluble white solid calcium carbonate due to which lime water turns milky.

Lime water + Carbon dioxide → Calcium carbonate + Water

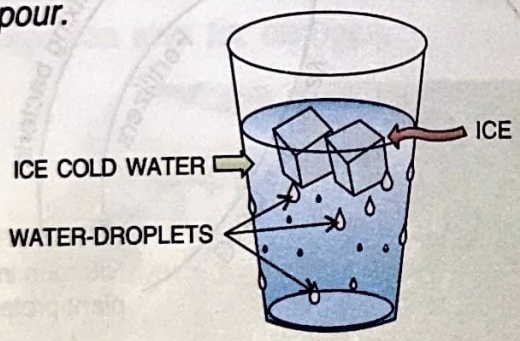
Activity 3

To show that water vapour is present in air.

Take a glass tumbler. Fill it half with ice cold water. You will observe that fine water droplets get deposited on the outer wall of the glass tumbler. These droplets have certainly not passed through the material of the glass tumbler from inside.

These water droplets must have come from air. Due to the cold surface of the glass tumbler, the water vapours present in air get condensed into water droplets.

This proves that *air contains water vapour.*



Air contains water vapour

There are enough evidences to show that air is a mixture and not a compound.

1. The composition of air is not fixed. It varies from place to place and from time to time.
2. The components of air retain their individual properties.
3. Liquid air has no definite boiling point.
4. No energy exchange occurs when the components of air are mixed with each other.
5. Components of air can be separated by simple physical methods. It involves their liquefaction followed by fractional distillation.
6. Air is not represented by any chemical formula as it is a mixture.

THE IMPORTANCE OF THE VARIOUS COMPONENTS OF AIR

Nitrogen

Nitrogen is the main constituent of air. It makes up about 78% of air by volume. It is a colourless, odourless and tasteless gas, which is slightly lighter than air as a whole. It is neither combustible nor a supporter of combustion, but controls it. It is slightly soluble in water. Its solubility in water is less than that of oxygen.

It is non-poisonous but animals die because of suffocation created due to the lack of oxygen in pure nitrogen gas.

Uses of nitrogen

1. **In the synthesis of proteins :** It is an important constituent of proteins that are necessary for the growth of plants, animals and human beings. Plants use nitrogen present in the air to make proteins .
2. **In diluting the effect of oxygen in air :** It controls the rate of burning or combustion, as it does not support burning. If nitrogen had not been present in the atmosphere, then even small sparks would have become large fires.
3. **In the manufacture of fertilizers :** It is used to make fertilizers like potassium nitrate, ammonium sulphate, urea, *etc.*
4. **In the manufacture of explosives and other chemical substances :** It is used to prepare explosives, which release huge

volumes of gases quickly with lots of energy. *e.g.* : trinitrotoluene (T.N.T.), nitroglycerine, etc.

It is also used in the manufacture of nitrogenous compounds like ammonia, nitric acid, etc.

Do You Know ?

- Most of the chemical explosives contain nitrogen.
- Explosives are used to make bombs.

5. **In the food industry for flushing food packages :** It is used for the preservation of foods, since it does not easily react. The containers used for storing foodstuffs are flushed with nitrogen to remove oxygen before they are sealed. The absence of oxygen does not allow bacterial growth. Thus, food remains fresh for a long time. The packaged potato chips that you purchase from market are filled with nitrogen.
6. **As a preservator in the hospitals :** Liquid nitrogen is used for storing and preserving blood, tissues, etc.

IMPORTANCE OF NITROGEN FOR PLANTS, ANIMALS AND HUMAN BEINGS

Nitrogen is of vital importance to the plants, animals and human beings as it provides a vital nutrient **protein** to all living beings essential for their growth and development. When we breathe in air, nitrogen is exhaled out as such. We cannot use the nitrogen present in the air directly to make protein.

It is the plants which convert nitrogen into protein while human beings and animals get it from plants. But nitrogen also cannot be absorbed directly in free form by the plants. It is first fixed up in the soil as nitrogenous compounds, nitrates and nitrites, and then absorbed by plants in soluble form. This is done by the phenomenon called **nitrogen fixation**.

When plants and animals die, they decay and nitrogen is returned to the soil. In this way, the amount of nitrogen is balanced in soil and the gas can be used again and again by plants.

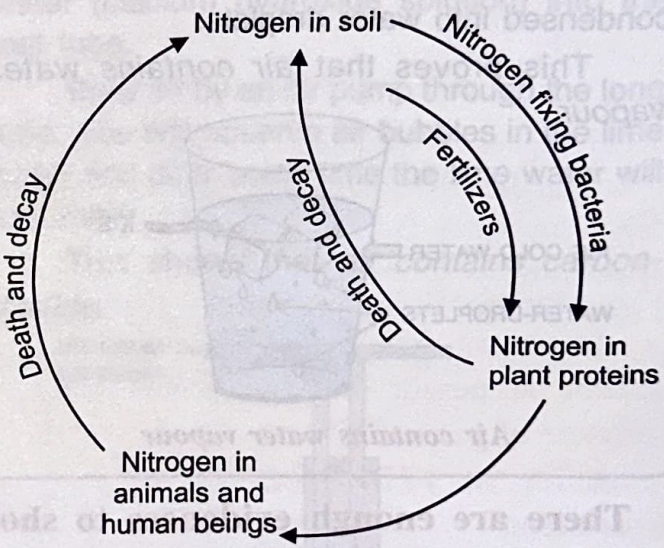


Fig. 7.2 : Nitrogen cycle.

Do You Know ?

The circulation of nitrogen through the living and non-living compounds of the biosphere (air, water, soil, plants and animals) is called nitrogen cycle.

Nitrogen cycle can also be defined as the process of converting nitrogen present in the air into nitrates and nitrites salts and then their decomposition back into nitrogen gas which mixes in the air.

Nitrogen fixation : The phenomenon by which nitrogen is converted into nitrates and nitrites and get fixed in the soil or directly due to some bacterial action is known as nitrogen fixation. Following are the two natural ways of nitrogen fixation.

1. Biological fixation :

Symbiotic bacteria (Rhizobium) living in the root nodules of leguminous plants (plants that have nodules on their roots) like peas, grams, beans, *etc.*, absorb nitrogen directly from the air and convert it into nitrates. Thereafter, the plants convert them into proteins. Some non-leguminous plants like *Ginkgo* can also fix nitrogen.

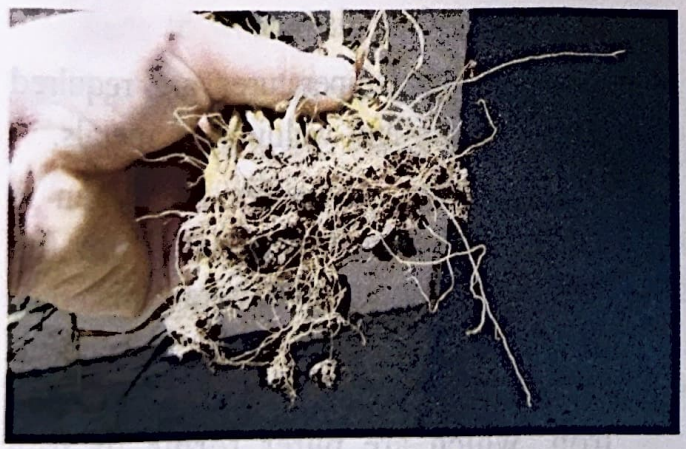


Fig. 7.3 : Root nodules in leguminous plants

Leguminous plants

1. Root nodules (containing symbiotic bacteria) —convert→ 2. Free nitrogen (in air and soil) ↓ to 3. Soluble nitrates [in soil] —converted by plants into→ 4. Proteins

2. Non-biological fixation :

During lightning, temperatures often reach as high as 3000°C. At such high temperatures, nitrogen and oxygen present in the air combine to form oxides of nitrogen.

$$\text{Nitrogen} + \text{Oxygen} \xrightarrow[3000^\circ C]{\text{Heat}} \text{Nitric oxide}$$

$$N_2 + O_2 \xrightarrow[3000^\circ C]{\text{Heat}} 2NO$$

$$\text{Nitric oxide} + \text{Oxygen} \longrightarrow \text{Nitrogen dioxide}$$

$$2NO + O_2 \longrightarrow 2NO_2$$

Nitrogen dioxide so formed then reacts with the water vapour present in the air to form nitrous and nitric acids.

During a thunderstorm, when lightning occurs, nitrogen and oxygen (present in the air) combine to form oxides of nitrogen, which are washed away by rain into the soil in the form of **nitric acid.**

These acids, so formed, reach the earth along with rainwater, and react with metal carbonates present in the soil to form metal nitrates which are absorbed by the plants from the soil and get converted into proteins.

$$\text{Calcium carbonate} + \text{Nitric acid} \longrightarrow \text{Calcium nitrate} + \text{Water} + \text{Carbon dioxide}$$

$$CaCO_3 + 2HNO_3 \longrightarrow Ca(NO_3)_2 + H_2O + CO_2$$

Lightning discharge

Nitrogen + O_2 (air) —electric spark→ Nitric oxide —O_2 (air)→ Nitrogen dioxide —H_2O (air)→ Nitric acid and nitrous acid —Minerals in Soil→ Soluble nitrate

Note : Acid rain is a bad effect of this natural process.

Oxygen

Oxygen constitutes about 21% of air by volume. It is the active part of air.

Uses :

1. **In Respiration** : Oxygen supports all life on earth. Living beings need oxygen for respiration*.
2. **In Combustion (Burning)** : Oxygen is essential for burning. Fuels burn in oxygen to produce large amount of energy along with carbon dioxde and water vapour. This energy is used by human beings for multiple purposes.

   (For details, see Chapter 7B).

Fuels : They are the substances which burn in oxygen to produce a large amount of energy in the form of heat and light.

$$\text{Fuel} + \text{Oxygen} \xrightarrow{\text{heat}} \text{Carbon dioxide} + \text{Water} + \text{Energy}$$

Some common fuels are coal, wood, LPG, CNG, petrol, diesel, *etc.*

LPG is liquefied petroleum gas used as a household fuel (contains butane and isobutane compounds).

CNG is compressed natural gas. It contains methane gas.

3. **In the field of medicines :**
   - Oxygen cylinders are provided for patients suffering from breathing problems, so as to facilitate artificial respiration.
   - Oxygen is a major constituent of *carbogen* (95% O_2 and 5% CO_2) that is given to the patients to stimulate natural breathing.
   - A mixture of oxygen and nitrous oxide is used in dentistry as local anaesthesia.
4. **In industries :**
   - *For welding and cutting of metals* :

     (a) Oxygen mixed with hydrogen as fuel produces a flame with a very high temperature (about 2800°C), known as oxy-hydrogen flame.

     (b) With acetylene, oxygen produces a flame with an even higher temperature (about 3300°C), commonly known as oxy-acetylene flame.

     Such high temperatures are required for cutting and welding of metals.
   - *For removing impurities* : Oxygen is used to remove impurities from the substances that are used as raw materials in the steel industry. Thus, it helps to convert pig iron into steel and wrought iron, which are purer forms of iron metal.
   - *In chemical industries* : Pure oxygen is used as the oxidising agent in the manufacture of nitric acid from ammonia and of sulphuric acid from sulphur dioxide.
   - *As a propellant fuel in spacecrafts* : Liquid oxygen (LOX) is carried by rockets that burn liquid hydrogen as fuel to propel the rocket at high speed.

* Respiration is a chemical process in which oxygen in the air is used by all living organisms to oxidise their food and gain energy.

$C_6H_{12}O_6$ (glucose) + $6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy

Carbon dioxide

Carbon dioxide is present in air in a very small quantity. The percentage proportion of this gas varies from 0·03 to 0·04, but it has gone up slightly because of human activities. It is a colourless, odourless gas. It is neither combustible, nor supports combustion. It is highly soluble in water.

Uses :

1. **In photosynthesis :** It is essential for the process of photosynthesis, by which green plants prepare their food. Carbon dioxide dissolved in water helps in the photosynthesis of aquatic plants.

   Carbon dioxide dissolved in water adds taste to water.

2. **In prevention of heat loss :** Carbon dioxide minimizes heat loss by radiation, by reflecting heat back to the earth's surface, particularly at night. Thus, it acts as a green house gas and balances the temperature of the earth.

   Greenhouse effect : It is a natural process that occurs when certain gases in the earth's atmosphere trap the sun's heat and keep the earth's surface warm. Such gases are called greenhouse gases. *Examples :* Carbon dioxide, methane, ozone, water vapour, nitrous oxides, etc.

3. **In fire extinguishers :** Carbon dioxide gas is used in fire extinguishers as it does not support burning.

   Carbon dioxide is heavier than air, hence it settles like a blanket over the fire and cuts off its oxygen supply due to which the fire gets extinguished.

4. **In food industry :** It is used to prepare baking soda (sodium bicarbonate), an important ingredient of baking powder. It is used as a refrigerant in the form of dry ice for preserving fruits, vegetables, meals, etc. Dry ice is solid carbon dioxide formed on cooling carbon dioxide gas.

   $$\text{Carbon dioxide (gas)} \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \text{Dry ice (solid carbon dioxide)}$$

   Carbon dioxide is also used in the manufacture of carbonated drinks.

5. **In the manufacture of chemicals and fertilizers :** It is used to prepare chemicals such as washing soda (sodium carbonate). It is also used to prepare fertilizers like urea.

6. **In oxygen cylinders for artificial respiration :** Carbon dioxide is mixed with 95% of oxygen to form **carbogen**, which is used to stimulate breathing in patients.

Water vapour

Water vapour in air is also known as moisture. The amount of moisture present in the air is known as **humidity**. But the percentage proportion of water vapour in air varies. The more is the amount of water vapour present in the air, the higher is the humidity.

Uses :

1. It determines the earth's climatic conditions. The presence of water vapour in air causes rain. At low temperatures, water vapour condenses to form snow, sleet, mist, frost, hail, dew, fog, *etc.*

2. It controls the rate of evaporation from the bodies of plants and animals.
3. It is essential for the growth of plants and for the good health of animals and human beings.

Water cycle : The change of water from one form to another in nature, which results in continuous circulation of water from earth's surface to the atmosphere and from the atmosphere back to the earth's surface is called water cycle.

Inert (noble) gases

About 0·96% of air, by volume, is made up of inert gases. Helium, Neon, Argon, Krypton, Xenon and Radon. Argon is the most abundant inert gas.

The most abundent inert gas in air is Argon.

Uses :

Helium is used in weather observation balloons.

Neon is used in advertising signboards and tubelights.

Argon is used in electric bulbs to provide an inert atmosphere.

Radon is used for cancer treatment as it is radioactive in nature.

Xenon and **Krypton** are used in photography.

Dust particles

These are solid, minute soil particles present in the air. Water vapours condense around dust particles to form clouds and cause rains. An excess of dust particles in the air also causes serious respiratory problems. Dust particles also trap heat and help in its scattering.

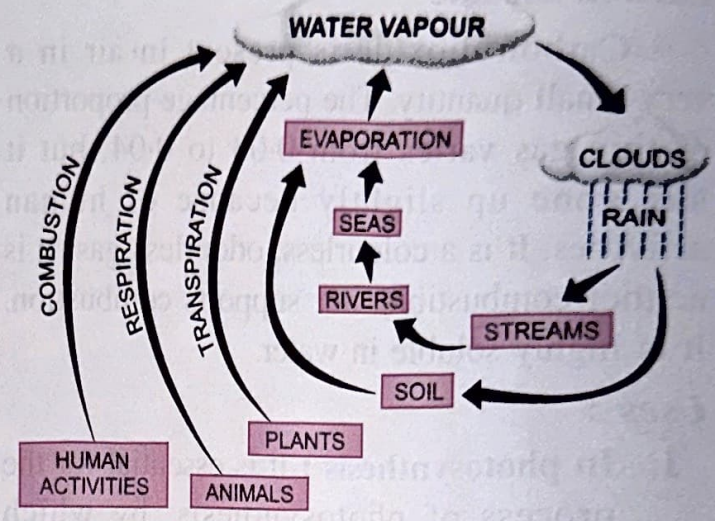


Fig. 7.4 : Water Cycle

Do You Know ?

The smell of cooking food in the kitchen spreads out because the molecules of gases present in air are in constant motion and diffuse easily in the available space.

AIR QUALITY

Air is important for the survival of human beings but it should be free from impurities.

When air contains undesirable substances that are harmful to plants and animals, it is said to be polluted. The harmful substances in air are called *air pollutants*. Air pollution is a major problem created by humans.

Some common pollutants in air are smoke, dust, soot, oxides of carbon, sulphur and nitrogen such as carbon monoxide (CO), sulphur dioxide (SO_2), sulphur trioxide (SO_3), nitrogen dioxide, etc. Other pollutants are chlorofluoro-carbons (CFCs), methane and ammonia (evolved by decaying dead plants and animals).

Though air also gets polluted by natural processes like volcanic eruptions, crop pollination, *etc.*, but mostly it is polluted by human activities like,

- burning of fuels and fibres,
- cutting of forests (deforestation),
- emission of harmful gases from vehicles,
- use of chemicals like fertilizers, insecticides and pesticides in agricultural activities,
- an increase in number of industries,
- use of chemical weapons in wars (tear gas, bombs, nuclear explosions, etc.).

TYPES OF POLLUTANTS IN AIR AND THEIR HARMFUL EFFECTS :

1. Suspended particles : **Smoke** is present in the air in a large quantity, due to the burning of fossil fuels. It carries minute suspended particulate matter* (SPM) that cause serious respiratory problems, like *asthma* and *bronchitis*. Smoke makes the atmosphere dense and reduces the visibility leading to many road accidents.

(i) Particles of lead oxide present in the automobile exhaust can cause brain damage in children. Lead compounds formed due to burning of leaded petrol obstruct development of blood cells.

(ii) Particles of dust, cement, ash and soot (carbon particles) in smoke may cause bronchitis, while asbestos fibres can cause a disease called silicosis.

(iii) Pollen grains too are suspended particles in air which act as a pollutant. They can cause allergic reactions in human beings.

Do You Know ?

Burning of coal in insufficient (or limited) supply of air (oxygen) produces carbon monoxide. This gas causes suffocation and death. That is why one should not sleep in a room with all windows and doors closed with a burning coal stove.

2. Oxides of carbon : Due to incomplete combustion of fossil fuels, carbon monoxide is produced.

Carbon monoxide is a highly poisonous gas which combines with the haemoglobin present in the blood to form a stable compound *carboxyhaemoglobin* from which haemoglobin cannot be recovered back by any method. Thus it prevents the blood to carry oxygen for respiration. Breathing in carbon monoxide rich air causes headache, dizziness and might even lead to death.

Carbon dioxide is non-poisonous but excess of it in the atmosphere adds to global warming.

3. Oxides of sulphur and nitrogen : Fossil fuels like coal, petrol, *etc.*, contain sulphur and nitrogen as impurities. Therefore, when these fuels are burnt, gases like sulphur dioxide, sulphur trioxide, hydrogen sulphide and nitrogen dioxide are produced. These gases are present in variable amounts in the air around big cities and industries. SO_2 and SO_3 are highly poisonous and cause serious respiratory problems. H_2S is highly obnoxious and causes headache. Oxides of nitrogen form

* *Particulate matter consists of minute particles of solid and liquid matter suspended in air.*

a mixture of smoke and fog known as **smog** which affects our eyes too. They also cause acid rain which damages crops and buildings and natural resources like water bodies and patches of land.

4. Other pollutants (CFCs) : Pollutants like chlorofluorocarbons (used as refrigerants and solvents), when mixed in air, react with the ozone layer causing its depletion. As a result, the density of air is reduced and ultra-violet (UV) rays from the sun reach directly to the earth. The UV rays of the sun cause skin tanning and skin cancer.

Table 7.2 : Various pollutants and the health problems caused by them

Pollutants	Health problems
A. Particles in air	
1. Smoke	1. Asthma, bronchitis (respiratory problems)
2. Lead oxide	2. Brain damage in children
3. Soot, dust, cement, ash	3. Bronchitis
4. Asbestos fibre	4. Silicosis (lung disease)
5. Pollen grains	5. Allergic reactions
B. Pollutant gases	
1. Carbon monoxide	1. Headache, dizziness and even death
2. Sulphur dioxide, sulphur trioxide	2. Respiratory problems, headache
3. Oxides of nitrogen	3. Eyes, nose, throat problems, cancer, etc.

Today, the number of people suffering from cataract and snow blindness has increased because of an increase in the percentage of UV-rays reaching the earth's surface.

Acid rain : Gases like sulphur dioxide, nitrogen dioxide, *etc.* react with the water present in the air to form sulphurous acid, sulphuric acid, nitrous acid and nitric acid respectively. These acids fall on the earth along with rain making it acidic and become a part of the water cycle.

A rain which is acidic in nature is called acid rain.

Equations for the chemical reactions to form acid rain :

sulphur dioxide + water → sulphurous acid

$SO_2 + H_2O \rightarrow H_2SO_3$

sulphur trioxide + water → sulphuric acid

$SO_3 + H_2O \rightarrow H_2SO_4$

nitrogen dioxide + water → nitrous acid + nitric acid

$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$

Acid rain has many harmful effects.

- It damages wildlife in the ecosystem where it falls. It can kill fishes and other aquatic animals and plants. Acid rain damages the nutrition level of leaves of the plants.
- It makes the soil more acidic and less fertile, thereby thus seriously damaging the growth of plants, *i.e.* it leads to the loss of soil fertility.
- It corrodes metals and damages buildings and historical monuments especially those made of marble and limestone (calcium carbonate) because acids can easily breakdown carbonates.
- Acid rain promotes corrosion of metallic structures, such as railway tracks, bridges, etc.

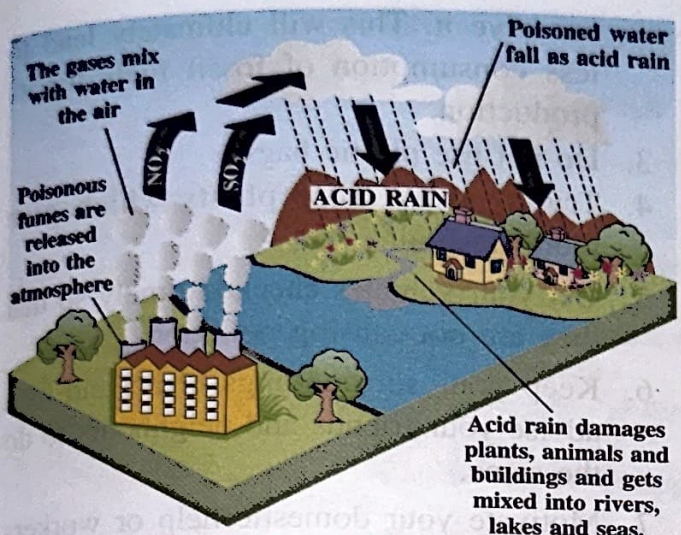


Fig. 7.5 : Acid rain

The Taj Mahal at Agra is said to be suffering from marble cancer. Why ? This is because Taj Mahal is completely made of marble and acid rain reacts with marble to form another powdery substance. Hence Taj Mahal is slowly changing its colour from white to yellow.

Do You Know ?

Rain water has always been slightly acidic as carbon dioxide in the air dissolves in rain water to form carbonic acid. $H_2O + CO_2 \rightarrow H_2CO_3$.

Global warming : An increase in the percentage of carbon dioxide, methane, nitrous oxide and chlorofluorocarbons traps the heat causing the temperature of the earth and its surroundings to rise. This is known as *global warming*.

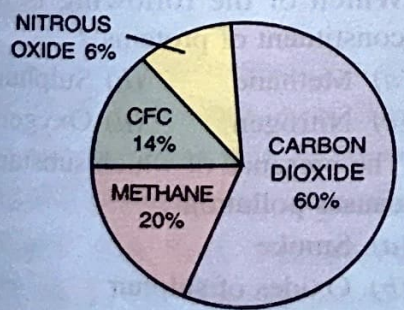


Fig. 7.6 : Contribution of different gases to global warming

You can understand the effect of global warming by comparing the temperatures of the inside of a parked car and the air outside it.

HOW TO PREVENT AIR POLLUTION

The whole world is facing serious problems due to air pollution. It has adverse effects on human health, vegetation and the atmosphere. Thus, it is necessary to control the levels of air pollutants released into the air to protect life on earth. Air pollution can be prevented by taking the following steps :

1. *By growing more trees and plants.* Trees absorb carbon dioxide for photosynthesis and release oxygen formed during this process. Thus, they help in balancing the amount of carbon dioxde and oxygen in air. They also help in bringing rain due to which other harmful gases are removed from air. [Do you know about Van Mahotsav, when lakhs of trees are planted in July every year ?].
2. *By installing tall chimneys with filters in factories and power plants* so that the smoke coming out of them is filtered and released high up into the air and with more solid particles depositing on the filter inside the chimney walls.
3. *By using internal combustion engines in vehicles* for complete and efficient burning of fuel and to reduce the amount of unburnt fuel and carbon monoxide.

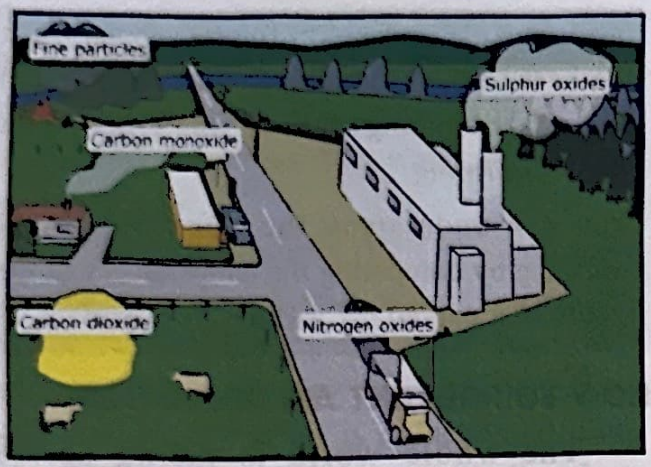


Fig. 7.7 : The common agents of air pollution in most cities.

The installation of catalytic convertors has been recommended in vehicles to reduce air pollution.

HOW CAN YOU CONTRIBUTE TO CONTROL AIR POLLUTION ?

1. Do not burn paper, garbage, dry leaves and other waste materials in open because they produce smoke and some poisonous gases which pollute the air. It would be a better option to put them in a compost pit rather than burning them.
2. Do not misuse electricity and try to conserve it. This will ultimately lead to less consumption of fossil fuels for its production.
3. Do not use plastic bags.
4. Try to grow more plants which give more fresh air.
5. Get your vehicles checked timely so that they are not causing pollution.
6. Keep your surroundings clean and also advise your friends and neighbours to do the same.
7. Motivate your domestic help or workers in your surroundings to use LPG for cooking instead of other materials which pollute the environment.

Activity 4

You have various options to reach your school such as walking, going by bicycle, travelling by bus or other public transport, using a car individually or travelling by car pool. Discuss in your class the impact of each of these options on the quality of air.

EXERCISE — I

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(i) The most abundant gas present in air is:
(a) Nitrogen (b) Oxygen
(c) Carbon dioxide (d) Water vapour

(ii) The gas used to prepare baking soda is :
(a) Nitrogen oxide (b) Carbon dioxide
(c) Sulphur dioxide
(d) Nitrogen dioxide

(iii) Which of the following is the main constituent of proteins ?
(a) Methane (b) Sulphur dioxide
(c) Nitrogen (d) Oxygen

(iv) The presence of which substance in air causes pollution ?
(a) Smoke
(b) Oxides of sulphur
(c) Chloroflurocarbons
(d) All of the above

(v) The gas which protects the earth from ultraviolet rays is :

(a) oxygen (b) carbon dioxide
(c) inert gases (d) ozone

2. Name the following :

(a) The process by which atmospheric nitrogen is converted into nitrate and nitrite salts and gets fixed in the soil ..... .

(b) Substances which burn in air to produce a large amount of heat and light energy .....

(c) The amount of moisture present in the air .....

(d) Tiny droplets of water which remain suspended in air ..... .

(e) Harmful substances present in air which cuase pollution ..... .

B. SHORT ANSWER TYPE QUESTIONS

1. Define the following terms :

(a) Pollutants (b) Acid rain
(c) Global warming (d) Smog

2. Give one use for each of the following inert gases:

(a) Argon (b) Helium
(c) Neon (d) Radon
(e) Krypton (f) Xenon

C. LONG ANSWER TYPE QUESTIONS

1. Answer the questions put against each of the following constituents of air :

(a) *Nitrogen :* Explain its significance for plants and animals.

(b) *Oxygen :* What is the percentage proportion of oxygen in air ? Why is oxygen called active air ?

(c) *Carbon dioxide :* "Although carbon dioxide plays no role in respiration, all life would come to an end if there is no carbon dioxide in air." Support this statement with relevant facts.

(d) *Water vapour :* Explain its role in modifying the earth's climate.

2. "Air is a mixture." Support this statement citing at least *three* evidences.

3. What is air pollution ? Give one harmful effect each of sulphur dioxide, nitrogen dioxide and hydrogen sulphide present in the air ?

4. (a) What are the causes of air pollution ?

(b) Suggest *five* measures to prevent air pollution.

B. Oxygen

Symbol : O; *molecular formula* : O_2; *atomicity* : 2; *valency* : 2.

DISCOVERY

Oxygen is a non-metallic diatomic gas. It is the most important constituent of air as it is vital for the existence of life on earth and also supports combustion to produce energy.

Oxygen was first discovered by Joseph Priestley in 1774. He obtained the gas by heating mercury (II) oxide. The gas was first named as "active air" due to its properties.

Later, in 1789, *Antoine Lavoisier* studied the properties of the gas and proved that it is an element. He named it "*oxygen*" which means *acid producer*, because oxides of some non-metals when dissolved in water produce acids.

OCCURRENCE

Oxygen is the most abundant element on the earth. It is available both in : (i) free state and (ii) combined state.

(i) In free state, oxygen occurs in the atmosphere to the extent of 21% by

volume and 23.2% by mass. The atmosphere also contains oxygen in the form of ozone [O_3].

(ii) In the combined state, oxygen is present in air, water, animal and plant tissues, and in minerals.

(a) About 89% (by mass) of water is oxygen.

(b) The human body has 65% oxygen (by mass).

(c) Plants contain 60% oxygen (by mass).

(d) About 50% of the earth's crust is made up of oxygen in the form of oxides, carbonates, silicates, *etc.*

(e) It is also an essential component of carbohydrates, proteins, fats and nucleic acids.

Some common compounds containing oxygen

1. Oxides : Silica (sand : SiO_2)
   Haematite (iron ore : Fe_2O_3)
   Bauxite (aluminium ore : $Al_2O_3 \cdot 2H_2O$)
2. Carbonate : Limestone (calcium carbonate : $CaCO_3$)
3. Sulphate : Gypsum (calcium sulphate : $CaSO_4 \cdot 2H_2O$)

Laboratory preparation of oxygen

1. *By the action of heat on potassium chlorate:*

Chemicals required :

(i) Solid potassium chlorate

(ii) Manganese dioxide powder (a catalyst)

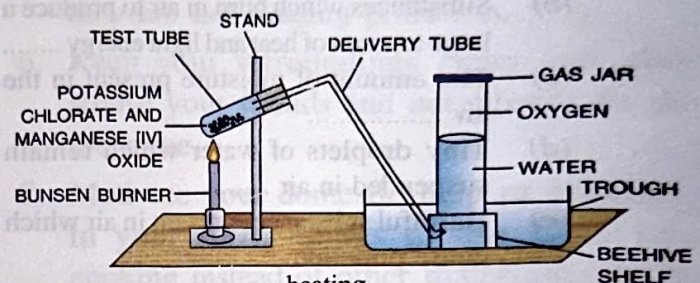
Chemical equation for the reaction :

Potassium chlorate $\xrightarrow[\text{Manganese dioxide}]{\text{heating}}$ Potassium chloride + Oxygen (g)

$$2KClO_3 \xrightarrow[MnO_2]{\text{heating}} 2KCl + 3O_2$$

Procedure : Apparatus is set up as shown in fig. 7.8(a). Potassium chlorate is a white solid, which is taken in the test tube. When it is heated strongly, it decomposes to give oxygen gas.

But potassium chlorate need heating to a high temperature before it decomposes.



Equation : $2KClO_3 \xrightarrow[MnO_2]{\text{heating}} 2KCl + 3O_2[g]$

Fig. 7.8(a) : Preparation of oxygen from potassium chlorate, using manganese dioxide as catalyst

The rate of decomposition of potassium chlorate can be increased by using manganese dioxide as a catalyst due to which the decomposition takes place quickly at a lower temperature. The gas is collected by downward displacement of water.

CATALYST : A catalyst is a substance that increases or decreases the rate of a chemical reaction without itself undergoing any chemical change. Different substances are used as catalysts in different chemical reactions.

2. *From hydrogen peroxide :*

Chemicals required :

(i) A dilute solution of hydrogen peroxide

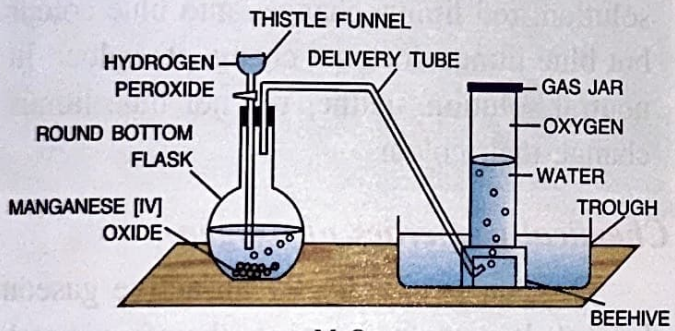
(ii) Manganese dioxide powder, which acts as a catalyst.

Chemical equation for the reaction :

$$\text{Hydrogen peroxide} \xrightarrow{\text{Manganese dioxide}} \text{Water} + \text{Oxygen}$$

$$2H_2O_2 \xrightarrow[\text{as catalyst}]{MnO_2} 2H_2O + O_2$$

Procedure : A small quantity of manganese dioxide which is a black powder, is taken in a flat bottom flask. A two-bore stopper is fixed into the mouth of the flask. Into one of the bores a thistle funnel is fitted, and through the other, a delivery tube is introduced into the flask. The other end of the glass tube (delivery tube) passes into a beehive shelf placed in a trough containing water as shown in Fig. 7.8(b).



Equation : $2H_2O_2 \xrightarrow[\text{as catalyst}]{MnO_2} 2H_2O + O_2[g]$

Fig. 7.8(b) : Preparation of oxygen from hydrogen peroxide, using manganese dioxide as catalyst

Now, hydrogen peroxide is added drop by drop into the flask with the help of the thistle funnel (slow addition controls the rate of production of oxygen). Manganese dioxide decomposes hydrogen peroxide very fast, and liberates oxygen, which passes through the delivery tube and bubbles out through the water in the trough.

Now, a gas jar filled with water is inverted over the beehive shelf. Oxygen is collected in the jar by downward displacement of water. As the gas keeps collecting in the gas jar, the water level in the jar keeps decreasing. When all the water has been displaced, the jar becomes full of oxygen. The jar is now covered with a greased lid and removed from the beehive shelf. In this way, a number of gas jars can be filled up with oxygen gas.

Since water is displaced downward by the gas collecting in the jar, the process is called *downward displacement of water. Why is oxygen collected by downward displacement of water? The reasons are:*

1. Oxygen is only slightly soluble in water (0·7% of only). Therefore, it can be collected over water without fear of excessive dissolution.
2. Oxygen is slightly heavier than air, so it cannot be collected over air.

Why hydrogen peroxide is preferred to potassium chlorate for the preparation of oxygen in the laboratory ?

In the laboratory, oxygen can be prepared by heating potassium chlorate with manganese dioxide but hydrogen peroxide is preferred for the following reasons :

1. No heating is required.
2. The rate of evolution of O_2 is moderate and therefore under control.
3. H_2O_2 is a safe chemical, also called oxygenated water.

To prepare oxygen gas from potassium chlorate, heating to a high temperature is required which may cause

(i) cracking of glass apparatus being used.

(ii) accidents and burn injuries.

Therefore, the use of potassium chlorate is avoided for safety reasons.

PROPERTIES OF OXYGEN

Physical properties of oxygen

1. Oxygen is a colourless, odourless and tasteless gas.
2. It is non-poisonous in nature and supports life.
3. It is slightly heavier than air.
4. It is only slightly soluble in water (0·7%). Oxygen dissolved in water is used by aquatic plants and animals for respiration.
5. It is liquefied into a bluish liquid at –183°C by applying a high pressure. Liquid oxygen has its boiling point at –183°C and its freezing point is –218.4°C.

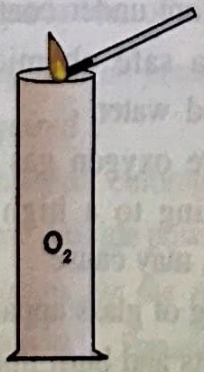
Activity 5

To show that oxygen supports burning and is neutral to litmus test.

Take a gas jar filled with oxygen. Introduce a glowing splinter in the jar. You will notice that the glowing splinter rekindles, but the gas does not catch fire.

Now, keep moist blue and red litmus papers in the jar. There is no change in the colours of either of the litmus papers.

From this, the following properties of oxygen are deduced :



The glowing splinter relights

1. *Combustibility* : Oxygen supports burning (or combustion) but does not burn itself, which means it is non-combustible.
2. *Neutral to litmus test* : Oxygen is neutral to litmus test. It does not change the colour of either the red litmus paper or the blue litmus paper.

Litmus : Litmus is a substance used as an indicator to know the nature of a solution. There are two types of litmus papers or solutions – *red* and *blue*.

In acidic solution, blue litmus changes into red colour but red litmus does not show any change in its colour. In alkaline or basic solution, red litmus changes into blue colour but blue litmus does not change its colour. In neutral solution, neither red nor blue litmus change their colour.

Chemical properties of oxygen

Oxygen is chemically an active gaseous element. It supports burning, therefore metals, non-metals and a large number of compounds react with oxygen to produce new products which essentially contain oxygen in them along with other elements. The new compounds so formed are called **oxides** and the process is called **oxidation**.

There are two types of oxidation :

1. Fast oxidation – Burning/combustion
2. Slow oxidation – Rusting, respiration

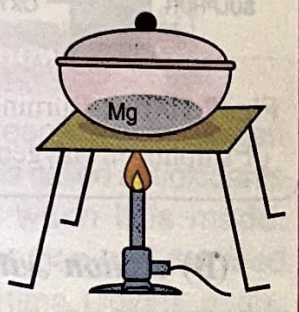
1. Fast oxidation : When a substance combines with oxygen with the release of a large amount of heat and light energy, the process is called fast oxidation.

2. Slow oxidation : When a substance combines with oxygen slowly over a period of time with the release of very less amount of heat energy, it is called slow oxidation. No light energy is released during such oxidation. When a substance burns in air, it gains weight. This can be shown by the following activity.

Activity 6

To show that magnesium gains weight on burning.

Take a piece of magnesium in a crucible with a lid. Weigh it and then heat it by opening the lid after short intervals to let the air enter the crucible. When the whole of the magnesium is burnt, cool the crucible and weigh it again.



Magnesium + Oxygen $\xrightarrow{\text{heat}}$ Magnesium oxide

$2Mg + O_2 \xrightarrow{\text{heat}} 2MgO$

What do you observe ?

Does the weight increase or decrease ?

You will find that the weight is increased after heating. This is due to the gain of oxygen from the air to form magnesium oxide. Hence, mass is increased after burning.

Oxides : Oxides are binary compounds formed by the chemical combination of a substance (metal or a non-metal) with oxygen. *Examples :* Sodium oxide (Na_2O), Sulphur dioxide (SO_2), *etc.*

Reactions of oxygen with non-metals and metals

(A) *Action with non-metals :* Non-metals like carbon, sulphur and phosphorus burn in oxygen to produce their respective oxides, with

Table 7.3 : The products and observations of the reactions between non-metals and oxygen.

Non-metal	*Reaction with oxygen*	*Observations*
1. Carbon	(a) Carbon + Oxygen → Carbon dioxide + heat $C + O_2 \xrightarrow{\text{heat}} CO_2$ + heat energy	***Flame :*** Carbon burns with bright sparks, forming carbon dioxide.
2. Sulphur	(a) Sulphur + Oxygen $\xrightarrow{\text{heat}}$ Sulphur dioxide + heat $S + O_2 \xrightarrow{\text{heat}} SO_2$ + heat energy	***Flame :*** Sulphur burns with a bright bluish flame, giving the pungent smell of sulphur dioxide.
3. Phosphorus	(a) Phosphorus + Oxygen $\xrightarrow{\text{heat}}$ Phosphorus pentoxide + heat $4P + 5O_2 \xrightarrow{\text{heat}} 2P_2O_5$ + heat energy	***Flame :*** Phosphorus burns with a dazzling flame, producing dense white fumes of P_2O_5.

Note : When oxygen supply is insufficient, carbon reacts with it to produce carbon monoxide gas.

$2C + O_2 \xrightarrow{\Delta} 2CO$

(insufficient)

the release of heat. The process of burning and the nature of the oxide formed can be better understood by the activity given ahead. (Activity 7)

Burning of hydrogen in oxygen : Hydrogen burns in oxygen at a very high temperature to produce water. Water is a neutral oxide.

$$\text{Hydrogen} + \text{Oxygen} \xrightarrow[\text{spark}]{\text{electric}} \text{Water} + \text{heat}$$

$$2H_2 + O_2 \xrightarrow[\text{spark}]{\text{electric}} 2H_2O + \text{heat}$$

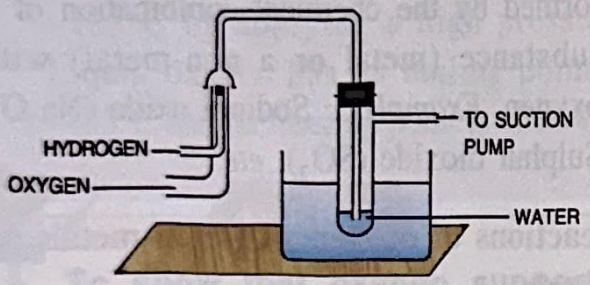
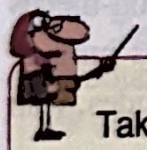


Fig. 7.9 : Burning of hydrogen and oxygen.



Activity 7

Take

(i) a piece of charcoal

(ii) a small amount of sulphur and

(iii) a piece of phosphorus each in three separate deflagrating spoons. Heat them until charcoal starts glowing and sulphur and phosphorus catch fire. Lower each of them quickly in separate gas jars containing oxygen.

When carbon, sulphur and phosphorus stop burning, remove the deflagrating spoons. Insert the moist blue litmus paper in each of them. **Blue** litmus paper turns **red**. This shows that oxides of carbon, sulphur and phosphorus are acidic in nature.

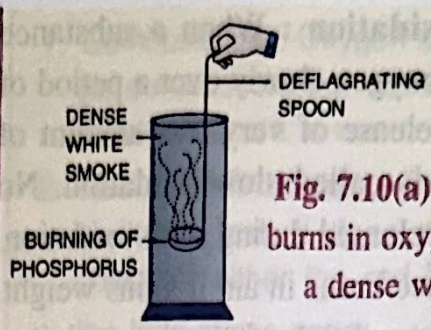


Fig. 7.10(a) : Phosphorus burns in oxygen giving off a dense white smoke

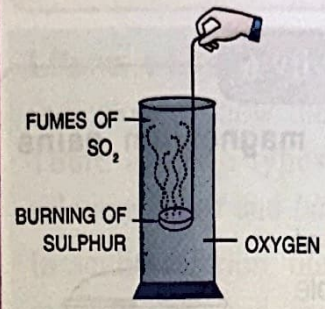


Fig. 7.10(b) : Burning of sulphur in oxygen

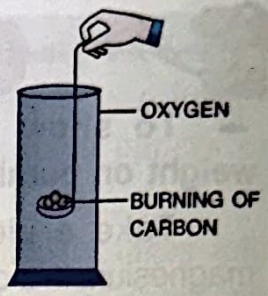


Fig. 7.10(c) : Burning of carbon in oxygen

(B) ***Action with metals :*** Metals react with oxygen to give oxides. Most of these oxides are basic in nature. Soluble basic oxides change the colour of red litmus into blue. This can be shown by the following activity.

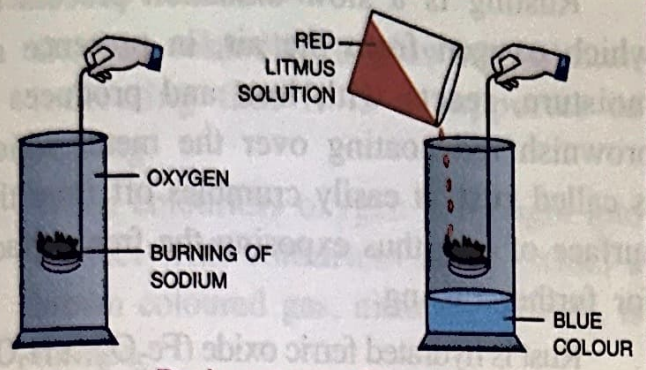
Activity 8

Take a piece of sodium in a deflagrating spoon. It starts burning on its own. Now, lower it into a jar of oxygen. It burns more brightly.

Once the metal stops burning and the jar is cooled, introduce red litmus solution and observe the change in its colour. Red litmus solution changes into blue.

Repeat the same activity with metals like calcium and potassium and observe the respective processes. Each time red litmus changes into blue.

This shows that oxides of metals are basic in nature.



Basic nature of metals

Note : This test is shown only by the metals which make soluble oxides.

Do You Know ?

Food can go bad on reacting with oxygen in the air. Hence, chemicals called antioxidants are added to the food when it is made. Antioxidants react with oxygen and the food remains fresh for a longer time. Onions, garlic, tomatoes and turmeric are good sources of antioxidants.

Burning of candle in oxygen : Candle burns in oxygen to produce carbon dioxide, water and energy in the form of heat and light.

Wax + Oxygen ⟶ Carbon dioxide + Water vapour + Energy

Respiration : Respiration is a natural chemical process in which oxygen in the air is used by living organisms to oxidise their food and to produce energy along with carbon dioxide and water vapour.

Glucose + Oxygen → Carbon dioxide + Water vapour + Energy

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy

Respiration is a continuous process taking place in living organisms.

The energy released during this process is used by living beings for their various activities.

Most living organisms use oxygen from the air present around us.

Soil also holds some air in between the spaces of its particles. This oxygen is used by soil organisms and the roots of the plants. Oxygen is also dissolved in water which supports the aquatic life.

Combustion (burning) : Combustion or burning is a chemical process in which a

Table 7.4 : The products and observations of the reactions between metals and oxygen.

Metal	*Chemical reaction with oxygen*	*Observations*
1. Sodium	(a) Sodium + Oxygen → Sodium oxide + heat $4Na + O_2 \rightarrow 2Na_2O$ + heat	***Flame :*** It burns brightly, with a brilliant yellow flame.
2. Calcium	(a) Calcium + Oxygen → Calcium oxide + heat $2Ca + O_2 \rightarrow 2CaO$ + heat	***Flame :*** It burns with a bright brick red flame, forming a white solid.
3. Magnesium	(a) Magnesium + Oxygen → Magnesium oxide + heat $2Mg + O_2 \rightarrow 2MgO$ + heat	***Flame :*** It burns with a bright dazzling light, forming the white powder of magnesium oxide.
4. Iron	Iron + Oxygen → Triferric tetraoxide + heat $3Fe + 2O_2 \rightarrow Fe_3O_4$ + heat (insoluble in water)	***Flame :*** Red hot iron burns with a bright spark, forming an oxide of iron.

substance burns in oxygen to produce a large amount of energy in the form of heat and light along with carbon dioxide and water vapour.

Example : Coal burns in air to produce heat and light energy.

$$\text{Coal} + \text{Oxygen} \xrightarrow{\text{heat}} \text{Carbon dioxide} + \text{Energy}$$

$$C + O_2 \xrightarrow{\text{heat}} CO_2 + \text{Energy}$$

Condition for burning

For a substance to burn, it needs to be heated to attain a minimum temperature known as ignition temperature. At this temperature, the substance combines chemically with oxygen to produce its respective oxide.

Ignition temperature **:** The minimum temperature to which a substance must be heated to make it burn is called its ignition temperature.

Differences between burning/combustion and respiration

Burning	*Respiration*
1. Burning is a fast oxidation process.	1. Respiration is a slow oxidation process.
2. It occurs at a high (ignition) temperature.	2. It occurs at body temperature.
3. Large amount of energy is released in the form of heat and light.	3. Small amount of energy is released in the form of heat only.
4. Burning is an artificial process and needs initiation.	4. Respiration is a natural process.

RUSTING

Rusting of iron is a case of action of oxygen on metals.

Rusting is a slow oxidation process in which oxygen from the air, in presence of moisture, reacts with iron and produces a brownish red coating over the metal which is called **rust**. It easily crumbles off from the surface of iron thus exposing the free surface for further rusting.

Rust is hydrated ferric oxide ($Fe_2O_3 . x H_2O$).

The equation of the reaction is :

$$\text{Iron} + \text{Oxygen} \xrightarrow[\text{slow process}]{\text{Moisture}} \text{Hydrated ferric oxide (rust)}$$

Air and moisture are necessary for rusting. The presence of acidic gases like CO_2, SO_2, etc. increases the rate of rusting of iron.

Since iron is a very useful metal, it is essential to prevent rusting of iron. If we protect iron from coming into contact with moisture/ moist air, rusting will not occur. This is done by painting, greasing, metallic coating, etc.

COMPARISON OF RUSTING AND BURNING

Rusting	*Burning (combustion)*
Similarities :	
1. Rusting needs oxygen.	1. Burning too needs oxygen.
2. The process forms oxides.	2. The process also forms oxides.
Differences :	
1. Rusting is slow oxidation.	1. Burning is fast oxidation.
2. Small amount of heat is released but not light.	2. Large amount of heat is released with light.
3. Both air and moisture are necessary for rusting.	3. Only air is necessary for burning.
4. Ignition is not required.	4. Ignition is required to initiate the process.

TESTS FOR OXYGEN

1. Oxygen rekindles a glowing splinter, indicating that it is a supporter of burning.
2. When colourless oxygen is brought into contact with colourless nitric oxide, a brown coloured gas, nitrogen dioxide, is formed.
3. Alkaline pyrogallol solution turns brown when oxygen is passed through it.

RENEWAL OF OXYGEN IN AIR : OXYGEN CYCLE

Living beings use atmospheric oxygen for respiration. Oxygen is also used up in the burning of fuels and in the formation of oxides of nitrogen. Yet the amount of oxygen in the air remains more or less constant.

This is because green plants return oxygen to the atmosphere by the process of **photosynthesis**. This circulation of oxygen in nature is called the oxygen cycle. The rate of photosynthesis is much faster than respiration, that is why we can say that oxygen is a '*gift from plants*' to all other living beings.

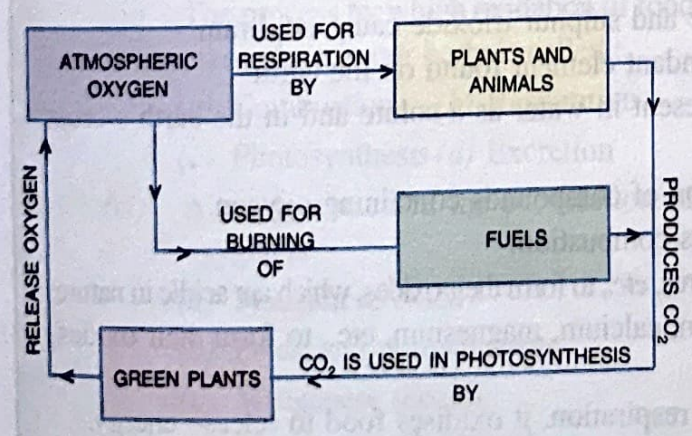


Fig. 7.11 : Oxygen cycle

Photosynthesis is the process by which carbon dioxide and water are used up by green plants in the presence of sunlight to produce glucose and oxygen gas.

Oxygen is released into the atmosphere but carbon dioxide is used up. Thus, the respective proportions of the two gases are naturally balanced.

Release of oxygen during photosynthesis

Green plants contain chlorophyll, a green pigment which traps the solar energy for photosynthesis.

$$\text{Carbon dioixde + Water} \xrightarrow[\text{sunlight}]{\text{chlorophyll}} \text{Glucose + oxygen}$$

$$6CO_2 + 6H_2O \xrightarrow[\text{sunlight}]{\text{chlorophyll}} C_6H_{12}O_6 + 6O_2$$

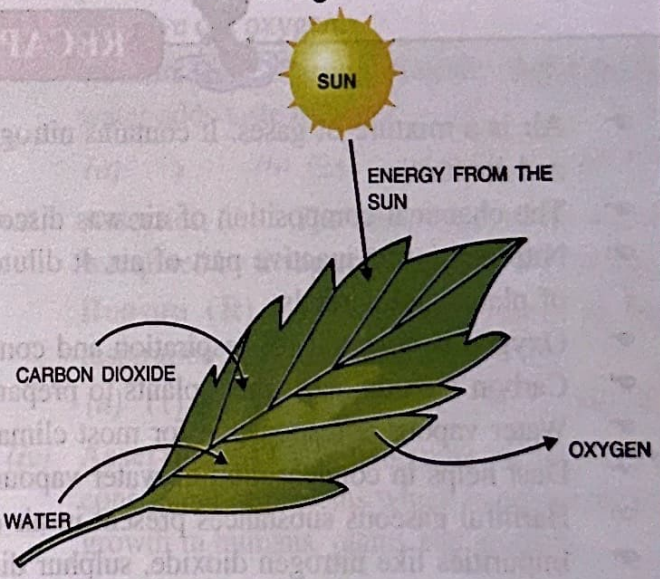
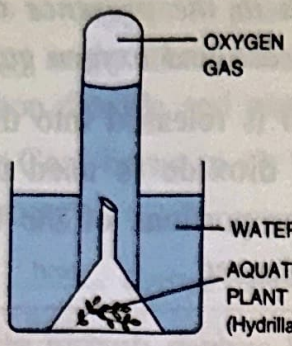


Fig. 7.12 : Release of oxygen in air by green plants during photosynthesis

Activity 9

To show that plants produce oxygen during photosynthesis.

Place a small aquatic plant like hydrilla in a beaker full of water and cover the plant with a funnel. Fill a test tube with water and invert it over the stem of the funnel. Place the apparatus in the sun.



Plants produce oxygen during photosynthesis

After some time, bubbles of a gas come out. This gas gets collected in the test tube by displacing water downward.

When the test tube has no more water left, take it out and introduce a glowing splinter in it. The splinter rekindles. This indicates the presence of oxygen.

From where has it come ? Obviously, it is produced by the plant through photosynthesis.

Note : This experiment cannot be carried out in the dark or at night, since photosynthesis cannot take place in the absence of light.

Note : The subject teacher should show burning of wood and rusting of iron objects to the students.

RECAPITULATION

- Air is a mixture of gases. It contains nitrogen, oxygen, carbon dioxide, water vapour, noble gases and impurities.
- The chemical composition of air was discovered by Lavoisier.
- Nitrogen is the inactive part of air. It dilutes the effect of oxygen. It is important for the growth of plants and animals.
- Oxygen is essential for respiration and combustion.
- Carbon dioxide helps most plants to prepare food through the process of photosynthesis.
- Water vapour is responsible for most climatic phenomena.
- Dust helps in condensation of water vapour into rain drops.
- Harmful gaseous substances present in air cause air pollution.
- Impurities like nitrogen dioxide, sulphur dioxide and sulphur trioxide cause acid rain.
- Oxygen is the most important and the most abundant element found on the earth.
- About 21% of air by volume is oxygen. It is present in water as a solute and in the earth's crust in its combined state.
- Oxygen is prepared by the thermal decomposition of compounds containing oxygen.
- Oxygen is a non-combustible gas, but it supports combustion.
- It reacts with non-metals like carbon, sulphur, phosphorus, *etc.*, to form their oxides, which are acidic in nature.
- Oxygen reacts with metals like potassium, sodium, calcium, magnesium, etc., to form their oxides, which are basic in nature.
- Oxygen is vital for the existence of life. During respiration, it oxidises food to release energy.
- In the presence of moisture and oxygen, iron forms rust, which is hydrated ferric oxide. Rusting corrodes iron. Rusting can be prevented by oiling, painting, galvanizing, electroplating, etc.
- Respiration, burning and rusting – all are oxidation processes. But burning is fast oxidation while respiration and rusting are slow oxidation processes.

EXERCISE — II

A. OBJECTIVE TYPE QUESTIONS

1. Choose the correct answer from the multiple choices given below.

(i) A fuel that releases the least amount of pollutants in the air is :

(a) sulphur dioxide
(b) chlorofluorocarbons
(c) smoke *(d)* CNG

(ii) The natural way of adding oxygen to air which involves green plants is called :

(a) photosynthesis *(b)* respiration
(c) burning *(d)* dissolution

(iii) Which one of the following is most likely to be corroded ?

(a) a stainless steel cupboard
(b) a galvanised iron bucket
(c) an iron hammer
(d) a tin-plated iron box

(iv) The substance which produces oxygen gas when heated is :

(a) Potassium chloride
(b) Potassium chlorate
(c) Potassium hydroxide
(d) Potassium carbonate

(v) The process by which oxidation of food takes place in our body is called :

(a) Combustion *(b)* Respiration
(c) Photosynthesis *(d)* Excretion

(vi) A catalyst used in the preparation of oxygen is :

(a) Manganese oxide
(b) Potassium permanganate
(c) Manganese dioxide
(d) Manganese (II) chloride

2. The following questions are Assertion-Reason based questions. Choose the answer based on the codes given below.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

(i) **Assertion (A) :** Air is invisible and transparent.
Reason (R) : Air is a mixture of colourless gases.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(ii) **Assertion (A) :** During the process of photosynthesis, plants take in carbon dioxide and give out oxygen.
Reason (R) : Carbon dioxide dissolved in water adds taste to water.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(iii) **Assertion (A) :** The component of air which is required by living organisms is only oxygen.
Reason (R) : Oxygen is essential for respiration and combustion.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

(iv) **Assertion (A) :** Nitrogen is an important constituent of proteins which is necessary for growth in humans, plants and animals.
Reason (R) : Nitrogen supports combustion.
(a) (1) *(b)* (2) *(c)* (3) *(d)* (4)

3. Fill in the blanks :

(a) ..... is the most abundant inert gas present in air.

(b) Oxides of sulphur and nitrogen combine with rain water to form ..... and ..... which cause .....

(c) ..... and ..... are the most common air pollutants.

(d) ..... discovered the oxygen gas.

(e) Oxygen occupies about ..... of air by volume.

4. Write *'true'* or *'false'* for the following statements :

(a) When magnesium is burnt, it gains weight.

(b) Oxygen reacts with metals to produce an acidic oxide.

(c) Chlorofluorocarbons cause ozone depletion.

(d) Oxygen changes moist red litmus paper to blue.

(e) Symbiotic bacteria living in the root nodules of leguminous plants absorb nitrogen directly from the air.

5. Match the following :

Column A	Column B
(a) Global warming	(i) Hydrated ferric oxide
(b) Acid rain	(ii) Manganese dioxide
(c) Rust	(iii) Carbon dioxide
(d) Catalyst	(iv) Methane
(e) Photosynthesis	(v) Nitrogen dioxide

6. Name :

(a) The most abundant element in the earth's crust.

(b) A chemical called oxygenated water.

(c) A metal highly resistant to rusting.

(d) A mixture of oxygen and carbon dioxide used for artificial respiration.

(e) Two substances from which oxygen can be obtained on a large scale.

(f) An oxide and a carbonate containing oxygen.

(g) Two substances which undergo rapid oxidation.

(h) A gas used in fire extinguishers.

B. SHORT ANSWER TYPE QUESTIONS

1. Give two examples of the following :

(a) Greenhouse gases.

(b) Substances which cause air pollution.

(c) Gases responsible for acid rain.

(d) Gases which increase the process of rusting.

(e) Fuels which produce energy without leaving any residue.

2. Define :

(a) Oxidation (b) Catalyst

(c) Respiration (d) Ignition temperature

3. What do you observe when the following substances are heated and then tested with moist blue and red litmus paper ?

(a) Sulphur (b) Phosphorus

(c) Calcium (d) Magnesium

4. What are the two types of oxidation processes? Among these, which one produces a large amount of heat and light energy ? Which type of oxidation are respiration and rusting reactions ?

C. LONG ANSWER TYPE QUESTIONS

1. What are oxides ? Give two examples of metallic and non-metallic oxides each.

2. State two differences between :

(a) Rusting and burning.

(b) Respiration and burning.

3. (a) Taking hydrogen peroxide, how would you prepare oxygen in the laboratory ?

(b) What is the role of manganese dioxide in the preparation of oxygen ?

(c) Write the word equation for the above chemical reaction.

(d) Why is hydrogen peroxide preferred in the preparation of oxygen gas ?

(e) Why is oxygen collected by downward displacement of water ?

(f) What happens when a glowing splinter is introduced in a jar containing oxygen ?

(g) What happens when oxygen gas is passed through alkaline pyrogallol solution ?

4. Give reasons :

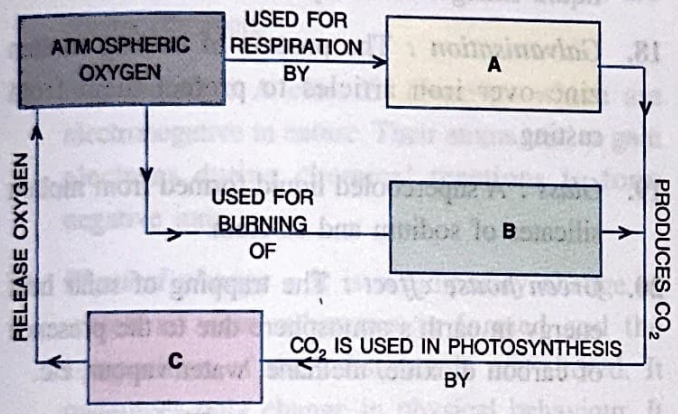
(a) Why is potassium chlorate not preferred laboratory prepration of oxygen gas ?

(b) Carbon dioxide gas is used in fire extinguisher.

5. (a) Give *four* uses of oxygen.

(b) How is oxygen naturally renewed in air ?

(c) Write the missing part/word in the following figure representing the oxygen cycle.



D. CASE STUDY BASED QUESTION

Sophia along with her classmates and teacher went on a school trip to Agra. She got an opportunity to visit Taj Mahal, the famous monument, made up of white marble. Sophia had heard a lot about its beauty but she was surprised to see the dull and pale yellow appearance of Taj Mahal.

Her teacher explained that it is due to the various factories in Agra, which emit harmful gases of sulphur and nitrogen in the atmosphere. These gases produce acid rain which has adverse effects on human health, animals, crops, buildings, etc.

(a) Name the gases which form the acid rain.

(b) What are the acids formed in the acid rain ?

(c) Give two balanced chemical equations to show the formation of acid rain.

(d) Why is Taj Mahal slowly turning yellow from white ?

(e) How can renewable sources of energy prevent air pollution ?

Answers

Exercise : 7 (I) — Objective type questions

1. *(i)–(a), (ii)–(b), (iii)–(c), (iv)–(d), (v)–(d)*

Exercise : 7 (II) — Objective type questions

1. *(i)–(d), (ii)–(a), (iii)–(c), (iv)–(b), (v)–(b), (vi)–(c)*
2. *(i)–(a)* : The reason correctly explains the assertion. Air consists of colourless gases that is why it is invisible and transparent.

   (ii)–(b) : Both assertion and reason are true but they are independent statements.

   (iii)–(d) : Assertion is false as apart from oxygen, other gases like nitrogen and carbon dioxide are also required by living organisms. The reason is true.

   (iv)–(c) : Assertion is true but the reason is false as nitrogen does not support combustion due to its inert nature.

Project

- Prepare a list of harmful effects of acid rain on bridges, cars, machines, coral reef, aquatic organisms, agriculture and discuss them in the class.

GLOSSARY

1. ***Alloy :*** A homogeneous solid mixture of two or more metals.
2. ***Anion :*** A negatively charged ion.
3. ***Atom :*** The smallest particle of an element, which can take part in a chemical reaction. It exhibits the essential properties of an element. It is electrically neutral and may or may not exist independently in nature.
4. ***Atomicity :*** The total number of atoms present in one molecule of an element or a compound.
5. ***Catalyst :*** A substance that alters the rate of a chemical reaction without itsef undergoing any chemical change.
6. ***Cation :*** A positively charged ion.
7. ***Chemical change :*** A change in which a substance loses its identity or changes its composition. Further, it is a change which is not reversible.
8. ***Chemical equation :*** A statement that describes a chemical change in terms of symbols and formulae with the reactants on left hand side and the products on right hand side of an arrow.
9. ***Chemical formula :*** Represents the composition of the molecule of a substance in terms of symbols of the atoms present in the molecule.
10. ***Chemical reaction :*** Involves the change of matter into a new substance or substances which take place during a chemical change.
11. ***Chromotography :*** The process of separating closely related but different dissolved constituents of a mxiture by their absorption on an appropriate material.
12. ***Compound :*** A substance formed by the chemical union of two or more elements, united in a definite proportion by mass.
13. ***Distillation :*** A process of getting a pure liquid from a crude liquid by boiling it and then condensing the vapours.
14. ***Ductility :*** The ability of metals to be drawn into thin wires.
15. ***Element :*** A pure substance which cannot be decomposed into simpler substances by ordinary chemical means.
16. ***Fertilizers :*** Artificially prepared chemicals which are rich in plant nutrients. They help in healthy growth of crops and increase the yield.
17. ***Freezing point :*** It is a temperature at which a liquid changes from liquid to solid state.
18. ***Galvanisation :*** The process of coating molten zinc over iron articles to protect them from rusting.
19. ***Glass :*** A supercooled liquid formed from molten silicates of sodium and calcium.
20. ***Green house effect :*** The trapping of solar heat energy in earth's atmosphere due to the presence of carbon dioxide, methane, water vapour, etc.
21. ***Ignition temperature :*** The minimum temperature to which a substance must be heated to make it burn is called its ignition temperature.
22. ***Ions :*** Electrically charged atoms or group of atoms formed by losing or gaining electrons.
23. ***Malleability :*** The property due to which metals can be beaten into sheets.
24. ***Metalloid :*** Elements which show some properties of metals and some properties of non-metals.
25. ***Metals :*** Elements which show metallic properties, a class of elements electropositive in nature. Their atoms tend to lose electrons during chemical reactions to form positive ions.
26. ***Mixture :*** Made up of two or more elements or compounds mechanically mixed together in any proportion, such that it retains the properties of those of its constituents.
27. ***Molecular formula :*** A chemical formula which gives the number of atoms of its constituent

elements in one molecule of a compound or an element.

28. ***Molecule :*** The smallest particle of a substance (an element or a compound) which can exist independently in nature and retains all the properties of that substance.

29. ***Negative valency :*** The valency of all non-metals and acid radicals.

30. ***Neutron :*** The neutral subatomic particle found in the nucleus of an atom with mass almost equal to that of a proton.

31. ***Noble gases :*** A class of elements which are chemically inert.

32. ***Non-metals :*** A class of elements which are electronegative in nature. Their atoms tend to gain electrons during chemical reactions to form negative ions.

33. ***Physical change :*** It is a temporary change in which no new substance is formed and the composition of the substance is not altered. It undergoes only change in physical behaviour. It is reversible.

34. ***Positive valency :*** The valency of all metals, hydrogen and ammonium ion.

35. ***Precipitate :*** An insoluble solid formed on reaction of two solutions.

36. ***Products :*** The substances formed as a result of a chemical change.

37. ***Proton :*** A positively charged subatomic particle found in the atoms of all elements.

38. ***Radicals :*** An atom or a group of atoms of different elements behaving as a single charged unit.

39. ***Reactants :*** The substances which take part in a chemical reaction.

40. ***Rust :*** The brown powdery mass of hydrated ferric oxide formed in the presence of moist air.

41. ***Rusting :*** The slow oxidation of iron into hydrated ferric oxide, in the presence of moist air.

42. ***Silicones :*** A compound of silicon used to prepare waterproof materials, electrical insulators and non-stick pans.

43. ***Smog :*** A noxious mixture of particulates and gases causing air pollution.

44. ***Solute :*** The dissolved substance in a liquid.

45. ***Solution :*** A homogeneous mixture containing a substance (solid, liquid or a gas) in dissolved state in a liquid.

46. ***Solvent :*** A liquid or a medium in which the solute is dissolved.

47. ***Sublimation :*** The changing of a solid substance directly into its vapours without passing through liquid stage.

48. ***Symbol :*** A short form for the full name of an element which represents an atom of that element.

49. ***Syrup :*** A supersaturated solution containing maximum amount of solute in a given volume of a solvent.

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